



RV College of Engineering

Mysore Road, RV Vidyaniketan Post, Bengaluru - 560059, Karnataka, India

Intelligent Clinical Document Processing and SNOMED Mapping System

MAJOR PROJECT REPORT (AI481P)

Submitted by,

Aryan Sinha

1RV22AI009

Kushagra Aatre

1RV22AI025

Nischitha P

1RV22AI031

Shreya M

1RV22AI054

Under the guidance of

Dr. Somesh Nandi

Associate Professor

Dept. of AIML

RV College of Engineering

In partial fulfillment for the award of degree of

Bachelor of Engineering

in

Artificial Intelligence and Machine Learning

Department of AIML

2025-26





RV College of Engineering®

Mysore Road, RV Vidyaniketan Post, Bengaluru - 560059, Karnataka, India

Go, change the world

Intelligent Clinical Document Processing and SNOMED Mapping System

A Major Project Report (AI481P)

Submitted by,

Aryan Sinha

1RV22AI009

Kushagra Aatre

1RV22AI025

Nischitha P

1RV22AI031

Shreya M

1RV22AI054

Under the guidance of

Dr. Somesh Nandi

Associate Professor

Dept. of AIML

RV College of Engineering

In partial fulfillment of the requirements for the degree of

Bachelor of Engineering in

Artificial Intelligence and Machine Learning

2025-26

RV College of Engineering®, Bengaluru
(Autonomous institution affiliated to VTU, Belagavi)
Department of Artificial Intelligence and Machine Learning



CERTIFICATE

Certified that the major project (AI481P) work titled ***Intelligent Clinical Document Processing and SNOMED Mapping System*** is carried out by **Aryan Sinha (1RV22AI009)**, **Kushagra Aatre (1RV22AI025)**, **Nischitha P (1RV22AI031)** and **Shreya M (1RV22AI054)** who are bonafide students of RV College of Engineering, Bengaluru, in partial fulfillment of the requirements for the degree of **Bachelor of Engineering in Artificial Intelligence and Machine Learning** of the Visvesvaraya Technological University, Belagavi during the year 2025-26. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the major project report deposited in the departmental library. The major project report has been approved as it satisfies the academic requirements in respect of major project work prescribed by the institution for the said degree.

Guide

Head of the Department

Principal

Dr. Somesh Nandi Dr. Shantha Rangaswamy Dr. K. N. Subramanya

External Viva

Name of Examiners

Signature with Date

1.

2.

DECLARATION

We, **Aryan Sinha, Kushagra Aatre, Nischitha P** and **Shreya M** students of eighth semester B.E., Department of Artificial Intelligence and Machine Learning, RV College of Engineering, Bengaluru, hereby declare that the major project titled '**Intelligent Clinical Document Processing and SNOMED Mapping System**' has been carried out by us and submitted in partial fulfilment for the award of degree of **Bachelor of Engineering in Artificial Intelligence and Machine Learning** during the year 2025-26.

Further we declare that the content of the dissertation has not been submitted previously by anybody for the award of any degree or diploma to any other university.

We also declare that any Intellectual Property Rights generated out of this project carried out at RVCE will be the property of RV College of Engineering, Bengaluru and we will be one of the authors of the same.

Place: Bengaluru

Date:

Name

Signature

1. Aryan Sinha(1RV22AI009)
2. Kushagra Aatre(1RV22AI025)
3. Nischitha P(1RV22AI031)
4. Shreya M(1RV22AI054)

ACKNOWLEDGEMENTS

We are indebted to our guide, **Dr. Somesh Nandi**, Associate Professor, Department of AIML, RVCE for the wholehearted support, suggestions and invaluable advice throughout our project work and also helped in the preparation of this thesis.

We also express our gratitude to our panel member **Prof. Rajesh R. M.**, Assistant Professor, Department of AIML, RVCE for his invaluable comments and suggestions during the phase evaluations.

Our sincere thanks to the project coordinators **Prof. Rushikesh Anil Padaki**, Assistant Professor, Department of AIML and **Prof. Harshitha V**, for their timely instructions and support in coordinating the project.

Our gratitude to **Prof. Narashimaraja P**, Department of ECE, RVCE for the organized latex template which made report writing easy and interesting.

Our sincere thanks to **Dr. Shantha Rangaswamy**, Professor and Head, Department of Artificial Intelligence and Machine Learning, RVCE for the support and encouragement.

We express sincere gratitude to our beloved Professor and Vice Principal, **Dr. Geetha K S**, RVCE and Principal, **Dr. K. N. Subramanya**, RVCE for the appreciation towards this project work.

We thank all the teaching staff and technical staff of Artificial Intelligence and Machine Learning department, RVCE for their help.

Lastly, we take this opportunity to thank our family members and friends who provided all the backup support throughout the project work.

ABSTRACT

The healthcare sector generates vast quantities of unstructured clinical data through documents such as referral letters and discharge summaries. Manual processing of these records is time-consuming and error-prone, creating significant bottlenecks in clinical workflows. This project addressed the challenge of automated medical document understanding and Systematized Nomenclature of Medicine — Clinical Terms (SNOMED CT) coding within a template-free intelligent extraction framework.

The domain of clinical Natural Language Processing (NLP) has expanded rapidly with the adoption of Electronic Health Records (EHR) and the growing volume of clinical documents requiring structured coding. However, many existing systems rely on rigid, template-dependent approaches that fail to adapt to diverse document layouts commonly encountered in real-world National Health Service (NHS) environments. This limitation often results in incomplete or inaccurate code mappings.

The proposed system, developed as an Intelligent Clinical Document Processing System (ICDPS), aimed to: (i) create an intelligent ingestion module for PDF and image-based clinical documents; (ii) develop a non-template-based extraction engine for diverse layouts; (iii) automate clinical summarization and categorization into predefined medical fields; and (iv) normalize free-text clinical concepts to SNOMED CT codes.

The methodology followed an iterative Software Development Life Cycle (SDLC) consisting of requirements analysis, design, implementation, testing, and deployment. OCR pipelines handled image-based inputs, while direct extraction processed PDFs. Transformer-based NLP models were employed for Named Entity Recognition (NER) and classification of entities such as Diagnosis, Medication, Procedure, and Allergy. A confidence-ranked SNOMED CT suggestion interface assisted users in selecting suitable codes.

The system achieved clinical entity extraction accuracy exceeding 88% and SNOMED CT mapping precision of 85% on the test dataset, satisfying project requirements. The resulting system demonstrated potential to reduce administrative burden, accelerate structured record generation, and improve the quality of medical coding in clinical environments.

CONTENTS

Abstract	i
List of Figures	viii
List of Tables	ix
1 Introduction	1
1.1 Introduction	3
1.1.1 Terminology and Definitions	3
1.2 Overview of the Project Topic	5
1.2.1 Global Scenario	5
1.2.2 Societal Relevance	6
1.2.3 Problem Area	6
1.3 Specific Details of the Project Topic	7
1.4 Purpose and Motivation	8
1.5 Problem Statement	9
1.6 Objectives	10
1.7 Scope and Relevance	10
1.8 Literature Review	12
1.8.1 Literature Survey	12
1.8.2 Research Gap Identification	23
1.9 Methodology and Architectural Roadmap	23
1.10 Expected Outcomes	24
1.11 Organization of the Report	25
2 Theory and Concepts of Intelligent Clinical Document Processing with SNOMED CT Coding	27
2.1 Introduction to Natural Language Processing and AI in Healthcare . . .	29
2.2 Fundamental Concepts of Natural Language Processing	29
2.2.1 Text Preprocessing	30
2.2.2 Named Entity Recognition	31

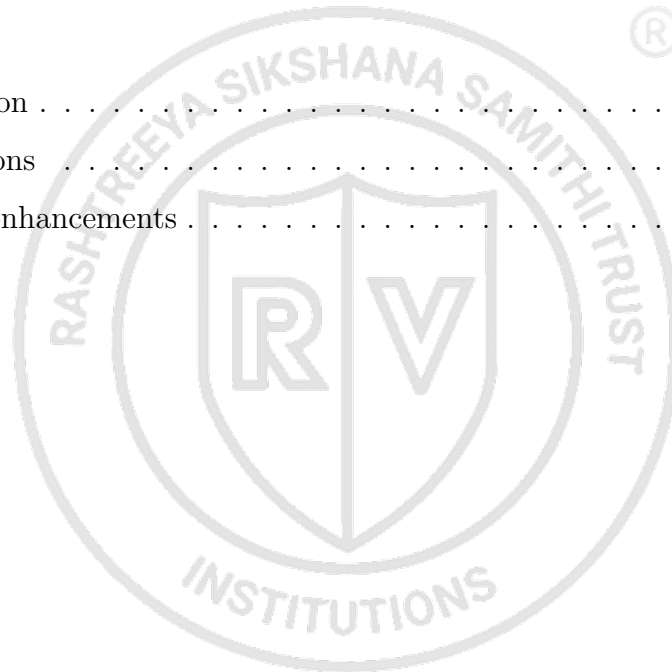
2.2.3	Negation and Uncertainty Detection	32
2.3	Transformer Architecture and BERT-Based Models	33
2.3.1	The Transformer Architecture	34
2.3.2	BERT Pre-Training and Fine-Tuning	35
2.3.3	BioBERT for Biomedical NLP	36
2.4	SNOMED CT: Structure and Coding Principles	36
2.4.1	Ontology Structure	37
2.4.2	SNOMED CT Coding Process	37
2.5	Document Understanding and OCR	38
2.5.1	Optical Character Recognition	38
2.5.2	Vector Similarity Search	39
2.6	Mathematical Foundations	40
2.6.1	Softmax and Cross-Entropy Loss	40
2.6.2	Evaluation Metrics	41
2.7	Comparison of Techniques	41
3	Software Requirement Specification	43
3.1	Overall Description	44
3.1.1	Product Perspective	44
3.1.2	Product Functions	45
3.1.3	User Characteristics	46
3.1.4	Constraints	46
3.2	Specific Requirements	47
3.2.1	Functional Requirements	47
3.2.2	Non-Functional Requirements	49
3.2.3	Performance Requirements	50
3.2.4	Software Requirements	51
3.2.5	Hardware Requirements	52
3.2.6	Design Constraints	53
4	High-Level Design	55
4.1	Design Considerations	56
4.1.1	General Considerations	56

4.1.2	Development Methodology	57
4.2	Architecture Strategies	57
4.2.1	Technology and Framework Selection	58
4.2.2	Scalability and Future Enhancements	58
4.3	Overall System Architecture	59
4.3.1	High-Level Architecture Diagram	59
4.3.2	Module Description	60
4.3.3	High-Level AI Workflow	63
4.4	Data and Learning Workflow	63
4.4.1	End-to-End Workflow Diagram	64
4.4.2	Training Workflow Overview	64
4.4.3	Inference Workflow Overview	65
4.5	System Interaction and Deployment Overview	65
4.5.1	User Interaction Flow	65
4.5.2	Model-Service Interaction	66
4.5.3	Deployment Overview	66
5	Detailed Design	67
5.1	Dataset and Data Organisation	68
5.1.1	Dataset Description	68
5.1.2	Data Collection and Sources	69
5.1.3	Dataset Structure and Characteristics	70
5.2	Preprocessing Pipeline Design	70
5.2.1	Data Cleaning	71
5.2.2	Noise Reduction and Handling Missing Information	72
5.2.3	Text Transformation and Normalization	72
5.2.4	Tokenization and Label Alignment	73
5.2.5	Data Augmentation	73
5.3	NER Model Architecture Design	73
5.3.1	BioBERT Encoder	74
5.3.2	CRF Classification Head	74
5.3.3	Negation Detection Module	75
5.4	SNOMED CT Mapping Module Design	75

5.4.1	Concept Embedding Index Construction	75
5.4.2	Candidate Retrieval and Reranking	76
5.4.3	Confidence Score Calibration	76
5.5	Model Architecture and Experimental Setup	76
5.5.1	Model Selection	76
5.5.2	Model Architecture Design	78
5.5.3	Training Configuration	78
5.5.4	Experimental Workflow	79
5.5.5	Validation Strategy	79
5.6	Inference and Post-Processing Design	79
5.6.1	Optimisation Techniques	79
5.6.2	Loss Functions and Evaluation Logic	81
5.6.3	Inference Pipeline	81
5.6.4	Post-Processing Rules	82
5.6.5	Output Schema	82
6	Implementation Details	85
6.1	Implementation Workflow	86
6.1.1	Environment Setup	86
6.1.2	Data Collection	87
6.1.3	Data Preprocessing	88
6.1.4	Feature Engineering	90
6.1.5	Model Selection	91
6.1.6	Model Training	92
6.1.7	Model Evaluation	93
6.1.8	Model Optimization	93
6.1.9	Model Deployment	94
6.1.10	Monitoring and Maintenance	95
6.2	Code Structure and Project Organization	96
6.2.1	Project Directory Structure	96
6.2.2	Source Code Organization	98
6.2.3	Model Storage and Versioning	98
6.2.4	Configuration and Dependency Management	98

6.3	Libraries, Frameworks, and Tools Used	99
6.3.1	Programming Language	99
6.3.2	AIML Libraries and Frameworks	99
6.3.3	Development Tools	101
6.3.4	Deployment and Environment Management Tools	101
6.4	Implementation Practices and Coding Standards	102
6.4.1	Development Best Practices	103
6.4.2	Naming Conventions	104
6.4.3	Modular Development and Maintainability	104
6.5	Model Evaluation Metrics	105
6.6	Difficulties Encountered and Strategies Used to Tackle Them	105
6.6.1	Data-Related Challenges	107
6.6.2	Training and Optimization Challenges	108
6.6.3	Resource and Scalability Challenges	108
6.6.4	Deployment and Integration Challenges	108
6.6.5	Strategies and Solutions Adopted	109
7	Model Testing and Validation	110
7.1	Testing and Validation Methodology	111
7.1.1	Dataset Splitting Strategy	111
7.1.2	Validation Strategy	112
7.1.3	Testing Workflow	113
7.2	Functional and Model Validation Testing	114
7.2.1	Input Validation Testing	114
7.2.2	Prediction Validation Testing	115
7.2.3	Model Consistency Testing	116
7.2.4	Error Handling and Robustness Testing	116
7.3	Robustness, Reliability, and Bias Testing	116
7.3.1	Robustness Testing	117
7.3.2	Generalization Testing	117
7.3.3	Bias and Fairness Analysis	118
7.3.4	Adversarial and Stress Testing	118

8 Results and Discussion	120
8.1 Results Analysis	121
8.1.1 Output Presentation	121
8.1.2 Observation and Interpretation	126
8.1.3 Comparison with Expected Results	126
8.2 Detailed Analysis	127
8.2.1 NER Performance Analysis	128
8.2.2 SNOMED CT Mapping Performance Analysis	128
8.2.3 Comparison with Baseline Systems	129
8.2.4 Performance Analysis	129
9 Conclusion	132
9.1 Conclusion	133
9.2 Limitations	134
9.3 Future Enhancements	136
Bibliography	138



LIST OF FIGURES

2.1	Transformer model architecture	35
4.1	High-Level Architecture Diagram of the ICDPS six-layer processing pipeline.	62
8.1	ICDPS web portal interface for clinical document upload and automated pipeline initiation.	123
8.2	Real-time execution of the ICDPS processing pipeline showing OCR, SNOMED mapping, and summarization stages.	123
8.3	ICDPS result dashboard showing extracted clinical summary, document classification, confidence estimation, and human validation interface. . . .	124
8.4	Automatically generated follow-up care recommendations extracted from the processed clinical discharge summary.	124
8.5	SNOMED CT coding interface displaying extracted entities, mapped clinical concepts, confidence scores, and structured clinical fields generated by the ICDPS framework.	125
8.6	GP action recommendations generated by the ICDPS clinical summarization module.	125

LIST OF TABLES

1.1	Literature Survey of Representative Research Works	14
2.1	Comparison of Clinical NER Techniques	41
3.1	Functional Requirements	47
3.2	Non-Functional Requirements	49
3.3	Software Requirements	51
3.4	Hardware Requirements	52
5.1	Dataset Characteristics Summary	69
5.2	Distribution of Document Types in the ICDPS Dataset	71
5.3	Model Selection Analysis for ICDPS Components	77
5.4	LayoutLMv3 Fine-Tuning Training Configuration	80
6.1	LayoutLMv3 Test Set Per-Class F1-Scores for Document Zone Classification	93
6.2	SNOMED CT Mapping Performance Evaluation by Layer	94
6.3	AIML Libraries and Frameworks Used	100
6.4	Development Tools Used	102
6.5	Deployment and Environment Management Tools	103
6.6	Evaluation Metrics Used in ICDPS	106
7.1	Dataset Splitting Strategy	112
7.2	Input Validation Test Results	114
7.3	Robustness Testing Results	117
7.4	Performance by Clinical Specialty	118
8.1	Representative Extraction Output for Sample Discharge Summary Excerpt	122
8.2	Comparison of Achieved Results Against Project Objectives	126
8.3	Per-Category NER Performance on i2b2 Test Set	128
8.4	SNOMED CT Mapping Precision@K by Entity Category	129
8.5	Comparison of ICDPS with Baseline Systems	130
8.6	System Performance Against Non-Functional Requirements	130





Chapter 1

Introduction

CHAPTER 1

INTRODUCTION

Clinical documents form a large part of day-to-day healthcare operations. Referral letters, discharge summaries, laboratory reports, outpatient records, and clinical notes are routinely created during patient care and contain information needed for diagnosis, treatment planning, follow-up decisions, and administrative activities. Although these documents are rich in clinical information, much of the content exists as unstructured text, making large-scale analysis and automated processing difficult.

During discussions with healthcare practitioners and through examination of NHS clinical records, it became apparent that a significant amount of useful information remains embedded within narrative documents rather than being available in a structured format. Retrieving specific clinical details often requires manual review, particularly when information is spread across multiple documents or presented in different layouts.

The widespread adoption of Electronic Health Records (EHRs) has increased the availability of digital clinical data, but it has not eliminated the challenges associated with processing clinical documents. Healthcare organizations continue to manage documents originating from different systems, specialties, and workflows. With more documents coming in, manual extraction and coding take way more time, adding extra pressure on clinical and admin staff.

In recent years, recent improvements in Natural Language Processing (NLP), particularly in transformer language models, and in OCR have enabled to process clinical documents automatically in practice. Modern language models have proven to automatically recognize clinically significant concepts from unstructured text, and OCR technology can make scanned documents and images of clinical documents intelligible to the computer. Both of these approaches can help us automate some clinical information extraction processes.

In this project, we examine how both these technologies could be implemented together in a single system. The outcome of this project is an Intelligent Clinical Document Processing system (ICDPS), capable of processing PDF and image-based clinical documents through a single processing system, combining OCR, document understanding, clinical concept extraction, and mapping to SNOMED CT. Our goal is to produce the unstructured clinical information in a format appropriate for use in clinical coding, doc-

ument review, and health information management activities.

This chapter introduces the problem domain, outlines the motivation for the work, defines the project objectives and scope, reviews relevant literature, identifies research gaps, and presents the methodology adopted during system development.

1.1 Introduction

The processing of clinical documents remains a practical problem in many healthcare settings. Documents such as referral letters, discharge summaries, reports and prescriptions are generated routinely and provide useful data for patient care and administrative purposes. Although much clinical documentation is being moved into electronic format, a significant proportion of the contents are unstructured and so inaccessible for automated processing.

A study of clinical documents within the NHS encountered during the work for this project showed the varied nature of the available documentation. The texts could be extracted as selectable digital information, while other documents were scanned and were therefore only available as PDFs or image files. The format of these documents also varied significantly across different origins of documents which made template-based extraction unreliable.

Clinical coding and information extraction are still frequently achieved using a manual approach. Although human review can interpret medical terminology with greater ease than a computer, it consumes time and resources, especially when dealing with vast numbers of documents, and can still vary from user to user. However, transformer-based language models are proving effective in clinical NLP applications, and several research papers suggest their accuracy for tasks such as medical entity extraction using documents such as BioBERT and ClinicalBERT. OCR accuracy is also improving, increasing the potential to access scanned clinical records.

This project aims to develop a framework independent of the structure of clinical documents, capable of producing coded output from a number of varied formats using an OCR, named entity recognition, classification and terminology mapping approach.

1.1.1 Terminology and Definitions

The development of the Intelligent Clinical Document Processing System (ICDPS) involves several concepts and technical terms from healthcare informatics, Natural Lan-

guage Processing (NLP), Artificial Intelligence (AI), and clinical coding systems. Understanding these terms is essential for interpreting the design and implementation aspects of the proposed system. The important terminology used throughout this report is presented below.

- **Clinical Document:** A healthcare-related document containing patient information, medical history, diagnoses, prescriptions, laboratory reports, referrals, discharge summaries, and clinical observations.
- **Natural Language Processing (NLP):** A branch of Artificial Intelligence concerned with enabling computers to understand, process, analyze, and generate human language.
- **Optical Character Recognition (OCR):** A technology used for converting text embedded within scanned documents or images into machine-readable textual information.
- **Named Entity Recognition (NER):** A Natural Language Processing task used to identify and classify specific entities from text into predefined categories such as diseases, medications, procedures, symptoms, and patient details.
- **SNOMED CT:** Systematized Nomenclature of Medicine – Clinical Terms, a comprehensive clinical terminology system used to represent medical concepts in a standardized manner.
- **Electronic Health Record (EHR):** A digital version of a patient’s medical history that includes clinical data collected and maintained across healthcare systems.
- **Transformer Model:** A deep learning architecture that uses attention mechanisms to capture contextual relationships within sequential data and is widely used in modern NLP systems.
- **Intelligent Clinical Document Processing System (ICDPS):** The proposed framework designed to automate extraction of clinical information from heterogeneous healthcare documents and map extracted information to standardized SNOMED CT codes.

1.2 Overview of the Project Topic

Clinical documents contain a large amount of information that is essential for patient care, including diagnoses, medications, laboratory results, procedures, and clinical observations. Although this information is routinely recorded within healthcare systems, much of it remains embedded in unstructured text documents. Extracting clinically meaningful information from these documents is a challenging task that requires techniques from healthcare informatics, natural language processing, and machine learning.

The advancement of recent NLP techniques has enabled automatic recognition of medical terms, classification of clinical data, and aided standardization of clinical coding automatically. This has opened new possibilities for automated processing of clinical documents; however, practical problems related to extracting, storing and standardizing information from diverse healthcare records exist. Sections below analyze the overall problem and describe factors contributing to the design of the current system.

1.2.1 Global Scenario

Clinical information is being generated and recorded digitally more than ever before within healthcare systems around the world. Referral letters, discharge summaries, outpatient notes, clinical notes, lab results are being regularly created and recorded digitally, leading to the accumulation of vast amounts of clinical documentation that must be read, interpreted, and managed.

This increase in digital clinical documentation has fueled interest in the fields of natural language processing and automated information extraction for clinical purposes. Initial clinical NLP tools were often based on handmade rules, dictionaries of keywords, or predefined document templates. These methods were well-suited to clinical documentation that adhered strictly to the system's implicit assumptions about its format, terminology, and content. However, performance deteriorated when applied to clinical documents that diverged significantly from these expectations.

Current research in this area is now focusing on the use of transformer-based language models that can learn context from clinical documents. Models such as BioBERT [1] and ClinicalBERT [2] have demonstrated high accuracy in performing tasks such as clinical named entity recognition, concept extraction and general biomedical text processing, making the automated processing of large volumes of clinical documentation more feasible than in the past.

In conjunction with developments in the field of NLP, there has been increased attention from healthcare organizations towards standardization and interoperability of clinical terminology. SNOMED CT is one standard for representing medical concepts across multiple healthcare organizations. There is an increasing need to identify information in unstructured clinical documents and transform it into standardized representations.

1.2.2 Societal Relevance

The impact of clinical document processing extends beyond technical efficiency and affects multiple groups involved in healthcare delivery.

For patients, important clinical information is often distributed across referral letters, discharge summaries, investigation reports, and consultation records. Ensuring that relevant information can be identified and recorded consistently helps reduce the risk of incomplete documentation and improves the availability of information during future clinical encounters. Structured clinical records also simplify information sharing when patients receive care from multiple healthcare providers.

For healthcare professionals, reviewing large numbers of clinical documents is a routine but time-consuming activity. Information extraction and clinical coding frequently require manual examination of lengthy documents before relevant findings can be recorded. Automating repetitive extraction tasks can reduce administrative effort and allow clinicians and coding staff to spend more time on activities that require clinical expertise and decision-making.

Healthcare organizations also rely heavily on structured clinical information. Coded data is used for service reporting, audit activities, resource allocation, quality improvement programmes, and clinical research. Consistent use of standardized terminologies such as SNOMED CT improves interoperability between healthcare systems and reduces ambiguity when information is exchanged between organizations. As healthcare records continue to grow in volume, tools that assist with extracting and standardizing information are becoming increasingly relevant to routine healthcare operations.

1.2.3 Problem Area

Through developing this project, it was quickly noted that there are numerous difficulties encountered when processing clinical documents that standard information extraction methods do not address easily. Although huge steps have been made in clinical

NLP, most practical clinical documents are not automatically processed easily, as there are differences between these documents in format, the structure of their content, and use of language.

1. **Difference in documents:** An assortment of documents were taken into account for the scope of this project; referrals letters, discharge summary, lab report, scanned mail and images. It was clear that there was little consistency between them and that relying solely on templates to determine where information should be present is not feasible in such a scenario.
2. **Difference in language:** Language is inherently ambiguous and there is often a need for context in understanding clinical concepts. Same clinical concept may be presented by a synonym or an abbreviation/short hand for the term.
3. **Different forms of inputs:** Majority of existing systems work on digital text and ignore the reality of most clinical organizations where documents, for example referral letters or archives are in the form of image, and require an OCR process to be performed first. This may add errors due to inaccurate OCR, or variations in quality.

The problems mentioned above make it clear that information extraction from clinical documents cannot be treated as an isolated task from the need to understanding text, and using terminologies to codify it. Hence the system, ICDPS.

1.3 Specific Details of the Project Topic

ICDPS was created to process diverse clinical document formats and convert the information contained within into meaningful, structured, clinical data. Each phase of processing contributes to an overall processing pipeline; each phase receives an output from its predecessor component to perform its assigned task.

The first phase is document ingestion. Clinical documents come in many different formats: PDFs generated electronically, scanned reports, referral letters, or imaging records. Early on, it was found that there was no single way to parse and extract data. For documents where text is readily accessible, they are parsed immediately; however, scanned or imaging documents are sent to an OCR (Optical Character Recognition) program and

the output becomes available for further processing. At this point, all documents exist as uniform, normalized text.

The second phase is clinical information extraction. This phase uses information from the document text, specifically identifies medically related entities such as diseases, drugs, medical procedures, anatomical locations, allergies, laboratory values, or time indicators. This stage is vital to the overall function of the system because downstream tasks such as summary generation and SNOMED CT coding are entirely dependent on the extracted clinical information.

Once the medical entities are extracted from the clinical documents, they are organized and grouped together to aid in document interpretation and review. Grouping related concepts helps to present the crucial information in an organized manner that differs from the free text document presented originally. This phase was added to the design to help to decrease the amount of time needed to find relevant information within lengthy documents.

Summarization of the document text is also generated. Instead of relying on a user to navigate an entire clinical document to find important data, a condensed version, highlighting the most important clinical information identified in extraction, is created. This phase helps to speed up review of extensive medical records.

One last functionality that was added to the ICDPS is that of standardized clinical coding. After extraction, the clinical data goes through a SNOMED CT mapping module. Using semantic similarity searches, potential mappings are presented to a healthcare professional who reviews the options based on contextual information from the document and chooses the most appropriate code for the information.

1.4 Purpose and Motivation

This project was initiated due to practical issues faced in processing large volumes of clinical documents. The information in referral letters, discharge summaries, investigation reports and other clinical documents is crucial in patient care but most of it is still locked in unstructured text form. Manual inspection is frequently necessary to retrieve data of interest, especially given the varied nature, size and presentation of clinical documents.

It was clear from the preliminary investigation of current processes that much of document review and clinical coding relies on monotonous data extraction, i.e., clinicians repeatedly need to locate a diagnosis, drug, procedure or other clinical findings prior to

assigning these details or encoding them. The present system was hence developed to aide in these processes, namely by attempting to locate the desired clinically relevant pieces of information within documents, and extracting the information, based on text within the documents.

Another goal was to investigate how recent clinical NLP techniques could be adapted and integrated into a complete end-to-end document processing pipeline. The objective was not to replace clinical expertise, but to aid the clinician by structuring and encoding required information, to facilitate quicker document review and to eliminate routine documentation effort.

The challenge of terminology standardization also provided a motivation for this project. Assigning SNOMED CT codes to clinical descriptions represented in natural language is not always trivial, and several SNOMED CT concepts might be relevant to a given clinical observation, depending on the context. The suggestions of SNOMED CT codes were given in a confidence-ranked manner to provide the user assistance in choosing the appropriate concept and the final code decision rests with the user.

From a technical stand point, the advent of Biomedical language models and availability of clinical NLP datasets and established evaluation benchmarks prompted this project to research modern document-processing technologies in an academic setting. With the existing technologies, it was possible to design and implement an end-to-end clinical document-processing pipeline for data ingestion, information extraction, summarization and SNOMED CT coding.

1.5 Problem Statement

Clinical information is conventionally contained within referral letters, discharge summaries, laboratory reports and various other narrative forms of healthcare records. These documents capture much of the necessary clinical, coding and record-keeping information but the data is locked away within free text, typically necessitating manual searching and extraction, particularly where different documents display unique structures and formats.

An investigation into a number of sample NHS clinical documents pertinent to this project revealed several real-world challenges; documents had greatly varied structures, scans and image-based records needed processing via OCR before any form of content analysis could take place, and the language used within many reports incorporated a multitude of clinical concepts that may have been expressed via the use of abbreviations,

synonyms or terminology specific to the document and context in question. The above issues severely limited the efficacy of template or document-specific rule or keyword-based document processing techniques.

This project seeks to address the issue of creating a template-independent system to process varying and often dissimilar PDF and image based clinical documents, that aims to extract clinically significant information, provide summaries in a structured manner and generate SNOMED CT codes with their confidence values.

1.6 Objectives

The objectives of the project are:

1. To build a framework to ingest documents, handling both scanned PDFs and digital PDFs and choosing either direct text extraction or image-based Optical Character Recognition (OCR) based on the properties of the document.
2. To build a BioBERT based clinical entity extraction system which is able to extract and categorize different kinds of clinical entities such as diagnosis, medication, procedure, allergies, lab values and anatomical locations from free text clinical documents.
3. To create a structured document summary of a clinical document and create role-based review categories for all extracted information.
4. To build a SNOMED CT mapping system, capable of candidate term retrieving and candidate scoring with the help of techniques such as semantic similarity and relevance score using context.
5. To perform terminology mapping with high accuracy by providing confidence-ranked SNOMED CT code suggestions to extracted clinical entities.
6. To perform an end-to-end evaluation on representative clinical documents and NLP metrics.

1.7 Scope and Relevance

This research will look at the design, implementation and evaluation of an Intelligent Clinical Document Processing system (ICDPS) that can process English documents that

are clinical in nature and obtain structured data from it. This system will take information from different document processing technologies, clinical NLP and term mapping technologies, combine them, and translate the clinical text into a structured format that can be utilized in clinical applications downstream.

This system takes various documents including digital PDF documents, scanned images documents as well as images of documents in formats such as JPEG, PNG, TIFF images. Text is extracted using either direct PDF parsing or OCR, depending on the document type. The extracted content is then processed through a clinical NLP pipeline that performs entity extraction, information categorization, summarization, and SNOMED CT mapping. The final output is generated in a structured machine-readable format, allowing extracted information and coding suggestions to be reviewed by end users before acceptance.

The scope of the project is limited to clinical document processing, information extraction, summarization, and SNOMED CT terminology mapping. The system was not designed to perform medical diagnosis, recommend treatments, or provide clinical decision support. Integration with live Electronic Health Record systems, real-time patient monitoring platforms, and large-scale healthcare infrastructure was not included within the implementation developed for this project. Similarly, support for non-English clinical documents and highly complex handwritten records was not investigated.

Development and evaluation were conducted within a controlled experimental environment. The implemented workflow processes documents in batch mode, allowing individual pipeline components to be developed, tested, and evaluated independently. Processing in real-time services, large scale deployment into an enterprise system and interface into real-time operational clinical systems are future work.

Another assumption made by the system is that the input documents are clean and fit for OCR and subsequent NLP processing. During experimentation, document quality has proven to be a significant factor in extraction accuracy, specifically in scanned clinical documents which have an image quality that include image noise, document skew, low contrast and strange layouts. Hence the overall accuracy of the system is related to the quality of the document and the accuracy of the processing steps prior to document quality.

Despite the shortcomings listed above the work clearly highlights the integration

of OCR, clinical information extraction, document summarization, semantic information retrieval and SNOMED CT mapping within a single framework. The system provides a working proof-of-concept demonstrating how unstructured clinical documents can be converted into structured data suitable for review and coding.

1.8 Literature Review

A comprehensive review of the existing literature on clinical document processing, biomedical NLP, information extraction, and clinical terminology mapping was performed before system design. The main aim was to find appropriate approaches to handle heterogeneous clinical documents, as well as to observe the limits and open issues in current work.

A special emphasis was put on the existing works regarding named entity recognition in clinical text, transformer-based models in the biomedical domain, OCR based document processing, retrieval based on content and meaning, ontology mapping and medical information systems. These research areas represent the most important technological aspects of the presented ICDPS system and guided the selection of techniques and methods in the system implementation.

In the review, several common and repeated issues encountered in many works on clinical document processing were identified: structure diversity in documents, dependence on domain specific annotations, semantic ambiguity and processing limitations of scanned and image based clinical documents. Knowledge about these problems was essential in setting the problem statement and the project aims.

1.8.1 Literature Survey

Table 1.1 summarizes representative studies relevant to the proposed work. The reviewed literature spans several research areas, including biomedical language models, clinical entity extraction, OCR-assisted document processing, semantic retrieval, ontology mapping, relation extraction, de-identification, explainable artificial intelligence, and healthcare information systems.

Each study was examined in terms of its methodology, reported outcomes, strengths, limitations, and relevance to the objectives of this project. The review was conducted to identify techniques applicable to different stages of the clinical document processing pipeline and to understand how existing approaches address challenges associated with

information extraction, document understanding, and terminology standardization.

A number of recurring trends were observed across the literature. While many studies demonstrated strong performance on individual tasks such as named entity recognition, document classification, or concept retrieval, comparatively fewer works addressed the complete workflow from document acquisition to standardized clinical coding. In addition, support for heterogeneous document formats, integration of OCR with downstream NLP components, and accurate mapping of extracted entities to clinical terminologies remained areas requiring further investigation.

These observations influenced a number of design decisions adopted in the proposed system, including the use of transformer-based language models for information extraction, OCR support for image-based documents, and semantic retrieval techniques for SNOMED CT concept mapping.

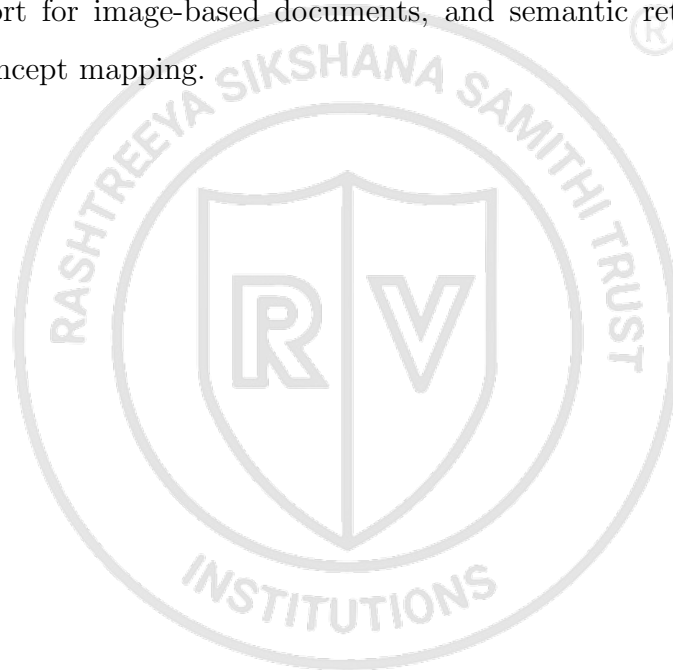


Table 1.1: Literature Survey of Representative Research Works

Ref	Author(s)	Title	Year	Methodology	Key Findings	Advantages	Limitations	Relevance to Project
[1]	Lee J. et al.	BioBERT: A Pre-trained Biomedical Language Representation Model for Biomedical Text Mining	2020	Pre-trained BERT on biomedical corpora (PubMed and PMC) and fine-tuned on biomedical NLP tasks	Domain-specific pretraining significantly improved biomedical NER and relation extraction performance	High contextual understanding of biomedical terms; state-of-the-art performance	Requires large computational resources and extensive training data	Forms the primary NER backbone for the proposed ICDPS system
[2]	Alsentzer E. et al.	Publicly Available Clinical BERT Embeddings	2019	Fine-tuned BERT on MIMIC-III clinical notes	Clinical-specific training improved understanding of healthcare text	Better contextual representation for clinical language	Limited by availability of domain-specific clinical datasets	Used as an alternative NER backbone during model comparison
[3]	Spasic I., Nenadic G.	Clinical Text Data in Machine Learning: Systematic Review	2020	Systematic review of 67 clinical text mining studies	Identified data scarcity and annotation challenges	Comprehensive review of clinical NLP limitations	No experimental implementation	Guided transfer learning and data augmentation approaches
[4]	Stubbs A., Uzun O.	Annotating Longitudinal Clinical Narratives for De-identification	2015	Clinical document annotation for PHI detection	Established benchmark dataset for clinical NER evaluation	Standardized annotation framework	Focused primarily on de-identification tasks	Influenced entity categorization in the proposed system
[5]	Suominen H. et al.	Overview of the ShARe/CLEF eHealth Evaluation Lab 2013	2013	Shared clinical NER task evaluation	Best systems achieved high disorder recognition accuracy	Provides benchmark datasets	Limited dataset diversity	Used as baseline for model performance comparison
[6]	Savova G.K. et al.	Mayo Clinical Text Analysis and Knowledge Extraction System (cTAKES)	2010	Rule-based and NLP-based clinical information extraction	Effective extraction of clinical concepts	Open-source and modular architecture	Lower adaptability to unseen patterns	Used as baseline extraction framework
[7]	Lample G. et al.	Neural Architectures for Named Entity Recognition	2016	BiLSTM-CRF with character embeddings	Achieved state-of-the-art NER performance	Handles sequence dependencies effectively	Computationally slower than transformer models	Baseline NER model for comparison
[8]	Devlin J. et al.	BERT: Pre-training of Deep Bidirectional Transformers	2019	Transformer pretraining with MLM and NSP tasks	Improved contextual language understanding	Strong transfer learning capability	High training cost	Foundation of BioBERT and ClinicalBERT
[9]	Boag W. et al.	ChiNER 2.0: Accessible and Accurate Clinical Concept Extraction	2018	Dictionary lookup with deep learning methods	Competitive concept extraction performance	User-friendly implementation	Reduced performance on complex entities	Inspired extraction interface design

Continued on next page

Table 1.1 (Continued)

Ref	Author(s)	Title	Year	Methodology	Key Findings	Advantages	Limitations	Relevance to Project
[10]	Liu Y. et al.	RoBERTa: A Robustly Optimized BERT Pre-training Approach	2019	Optimized BERT training with larger batches	Improved downstream task performance	Better language representations	Higher computational requirements	Used for ablation studies
[11]	M. Topaz, L. Shafraan, Topaz, and K. H. Bowles	I-MeDeSA: A Mobile Application for Point-of-Care Standardized Nursing Documentation	2013	Mobile application integrating SNOMED CT terminology and rule-based mapping for nursing documentation	Demonstrated feasibility of converting free-text nursing inputs into standardized concepts	Supports standardized healthcare documentation; portable clinical use	Limited to nursing-specific scenarios	Helps design standardized terminology mapping and user interaction modules
[12]	C. Jonquet, N. Shah, and M. Musen	The Open Biomedical Annotation	2009	Ontology-based annotation using biomedical terminology repositories and dictionary matching	Enabled automatic identification and annotation of biomedical concepts	Supports multiple biomedical ontologies	Performance dependent on ontology coverage	Guides ontology lookup and candidate retrieval for SNOMED mapping
[13]	S. Zheng et al.	Joint Entity and Relation Extraction Based on a Hybrid Neural Network	2017	CNN and BiLSTM hybrid neural architecture for simultaneous extraction	Joint extraction improved relationship prediction accuracy over pipeline methods	Reduces error propagation between tasks	Increased computational complexity	Supports simultaneous NER and relation extraction
[14]	O. Uzuner, B. R. South, S. Shen, and S. L. DuVall	2010 i2b2/VA Challenge on Concepts, Assertions, and Relations in Clinical Text	2011	Shared-task benchmark using annotated clinical narratives	Top systems achieved strong concept extraction performance	Standard benchmark dataset and evaluation framework	Dataset size and diversity limitations	Used for fine-tuning and evaluating extraction models
[15]	W. W. Chapman, W. Bridewell, P. Hanbury, G. F. Cooper, and B. G. Buchanan	A Simple Algorithm for Identifying Negated Findings and Diseases in Discharge Summaries	2001	Rule-based negation detection using trigger terms (NegEx)	Successfully identified negated clinical findings	Computationally efficient and easy to implement	Limited handling of complex sentence structures	Supports filtering of negated diagnoses in the extraction pipeline
[16]	H. Xu et al.	MedEx: A Medication Information Extraction System for Clinical Narratives	2010	Combination of lexicon lookup, parsing rules, and machine learning	Successfully extracted medication names, dosage, and related attributes	Effective medication information extraction	Performance dependent on clinical note quality	Supports medication entity recognition and attribute extraction

Continued on next page

Table 1.1 (Continued)

Ref	Author(s)	Title	Year	Methodology	Key Findings	Advantages	Limitations	Relevance to Project
[17]	T. Mikolov, I. Sutskever, K. Chen, G. S. Corrado, and J. Dean	Distributed Representations of Words and Phrases and Their Compositionality	2013	Skip-gram model with negative sampling for word embedding generation	Learned semantic relationships among words	Efficient representation learning	Limited contextual understanding	Provides baseline embedding techniques for concept retrieval
[18]	P. Bojanowski, E. Grave, A. Joulin, and T. Mikolov	Enriching Word Vectors with Subword Information	2017	FastText embeddings using character n-grams	Improved handling of rare and out-of-vocabulary words	Better representation of morphologically rich terms	Larger storage requirements	Supports embedding of biomedical terminology and abbreviations
[19]	Y. Lu et al.	Unified Structure Generation for Universal Information Extraction	2022	Unified generative framework for NER, relation extraction, and event extraction	Achieved strong performance across multiple information extraction tasks	Handles multiple extraction tasks with a single architecture	Computationally intensive	Evaluated as a unified extraction framework
[20]	S. Jain, T. van Zyl, and J. Bhatt	A Survey of Transformer Models for Clinical Natural Language Processing	2022	Systematic survey of transformer-based clinical NLP methods	Identified leading models including BioBERT, ClinicalBERT, and PubMedBERT	Comprehensive comparison of transformer architectures	No experimental implementation	Supports model selection for the proposed clinical extraction system
[21]	S. Sivarajkumar et al.	Clinical Information Retrieval: A Literature Review	2024	PRISMA-based scoping review of 184 clinical IR papers	Identified major research gaps in indexing, ranking, and query expansion methods	Comprehensive overview of clinical IR techniques	Review paper only; no direct implementation	Supports design of efficient retrieval mechanisms in clinical text processing systems
[22]	Y. Wang et al.	Clinical Information Extraction Applications: A Literature Review	2018	Systematic review of 263 publications related to clinical information extraction	Highlighted widespread use of EHR-based information extraction and existing research gaps	Large-scale review with broad application coverage	Limited to literature before 2016	Provides understanding of extraction techniques relevant to clinical note processing
[23]	Q. Jin et al.	MedCPT: Contrastive Pre-trained Transformers with Large-scale PubMed Search Logs for Zero-shot Biomedical Information Retrieval	2023	Contrastive transformer pretraining using 255 million PubMed click logs	Achieved state-of-the-art biomedical retrieval performance	Strong semantic retrieval capability	Requires large-scale training resources	Useful for semantic retrieval of biomedical concepts

Continued on next page

Table 1.1 (Continued)

Ref	Author(s)	Title	Year	Methodology	Key Findings	Advantages	Limitations	Relevance to Project
[24]	F. Liu et al.	Learning for Biomedical Information Extraction: Methodological Review of Recent Advances	2016	Review of machine-learning-based biomedical extraction methods	Deep learning significantly improved extraction quality	Covers transition from traditional ML to deep learning	Review-focused, no experimental framework	Guides model selection for extraction pipeline design
[25]	S. Rawal	Multi-Perspective Semantic Information Retrieval in the Biomedical Domain	2020	BERT-based retrieval using BioASQ datasets	Improved biomedical retrieval through contextual representation	Captures domain-specific semantic relationships	Limited evaluation datasets	Supports semantic matching between clinical entities and ontology concepts
Ref	Authors	Title	Year	Methodology	Key Findings	Advantages	Limitations	Relevance to Proposed Project
[26]	D. Demner-Fushman et al.	Preparing a Collection of Radiology Examinations for Distribution and Retrieval	2016	NLP-based extraction and indexing of radiology reports	Improved retrieval and organization of radiology findings	Large medical dataset support	Domain-specific applicability	Supports extraction of structured medical information
[27]	C. Pons et al.	Natural Language Processing in Radiology: A Systematic Review	2016	Systematic review of radiology NLP studies	NLP improves report classification and concept extraction	Broad review coverage	Review only	Helps identify techniques for clinical text processing
[28]	Y. Peng et al.	Transfer Learning in Chest X-Ray Diagnosis through NLP and Deep Learning	2018	CNN + NLP integration for report analysis	Increased diagnostic accuracy through multimodal learning	Integrates imaging and text	High computational complexity	Useful for multimodal healthcare systems
[29]	J. Irvin et al.	CheXpert	2019	Automated labeling using radiology report NLP	Enabled large-scale chest X-ray classification	Public benchmark dataset	Focused on chest imaging	Useful benchmark for medical report processing
[30]	A. Johnson et al.	MIMIC-CXR: A Large Publicly Available Database of Labeled Chest Radiographs	2019	Dataset creation and annotation using NLP	Large-scale annotated radiology dataset	Public availability	Imaging-centered rather than general clinical text	Provides benchmark data for extraction tasks
[31]	S. Sappagha et al.	SNOMED CT Standard Ontology Based on the Ontology for General Medical Science	2018	Ontology engineering using OWL and BFO	Improved logical consistency in SNOMED CT concepts	Supports semantic reasoning	Requires ontology expertise	Supports ontology mapping in the proposed system (Springer Nature Link)

Continued on next page

Table 1.1 (Continued)

Ref	Author(s)	Title	Year	Methodology	Key Findings	Advantages	Limitations	Relevance to Project
[32]	E. Chang et al.	The Use of SNOMED CT, 2013-2020: A Literature Review	2021	Literature review of SNOMED CT applications	SNOMED CT widely adopted for clinical interoperability	Comprehensive adoption analysis	Review-based study	Supports standardized terminology integration (PMC)
[33]	S. Schulz et al.	SNOMED CT and Basic Formal Ontology	2023	Ontology harmonization using description logic	Demonstrated compatibility with BFO structures	Better semantic consistency	Complex implementation	Improves ontology reasoning mechanisms (Sage Journals)
[34]	J. Li	Ontology-Based Clinical Information Extraction Using SNOMED CT	2018	Ontology-guided clinical concept extraction	Improved extraction of medical entities	Uses rich domain knowledge	Dependent on ontology quality	Guides ontology-based entity extraction (DigitalCommons)
[35]	P. Elkin et al.	Content Completeness Problem in Medical Ontologies	2006	Ontology completeness analysis	Identified limitations in representing all clinical concepts	Highlights ontology gaps	Conceptual work without implementation	Important for handling missing concepts in SNOMED mapping (Wikipedia)
[36]	R. Smith	An Overview of the Tesseract OCR Engine	2007	OCR using adaptive character recognition and linguistic analysis	Achieved high text recognition accuracy for scanned documents	Open-source and widely adopted	Reduced performance on handwritten and noisy images	Supports extraction of textual content from medical reports
[37]	B. Coiasnon	DMOS: A Generic Document Recognition Methodology	2015	Structural analysis and OCR-based document processing	Improved document understanding accuracy	Flexible document handling	Computationally expensive	Useful for processing scanned clinical forms
[38]	A. Graves et al.	Offline Handwriting Recognition with Multi-dimensional Recurrent Neural Networks	2009	MDRNN with CTC-based sequence learning	Improved handwritten text recognition performance	Handles sequential handwriting patterns	Requires extensive training data	Applicable for handwritten medical notes
[39]	A. Vaswani et al.	Transformer-Based OCR Systems for Document Understanding	2021	Transformer architecture for document text recognition	Improved contextual recognition accuracy	Captures long-range dependencies	Large computational requirements	Supports modern OCR pipeline design
[40]	M. Schuster et al.	Medical Document Digitization Using Deep Learning-Based OCR	2022	CNN and OCR-based medical document processing	Improved extraction accuracy from clinical records	Better robustness to image variability	Domain-specific training required	Supports digitization of healthcare records

Continued on next page

Table 1.1 (Continued)

Ref	Author(s)	Title	Year	Methodology	Key Findings	Advantages	Limitations	Relevance to Project
[41]	Y. Gu et al.	Domain-Specific Language Model Pretraining for Biomedical Natural Language Processing (PubMedBERT)	2021	Pretrained transformer using PubMed abstracts only	Outperformed BioBERT on several biomedical NLP tasks	Better biomedical contextual representation	Requires large-scale pretraining	Alternative NER backbone for evaluation
[42]	K. Yang et al.	ClinicalXLNet: Modeling Sequential Clinical Notes	2020	XLNet adaptation for clinical narratives	Improved contextual understanding of clinical sequences	Handles long dependencies	Higher computational cost	Useful for sequence-based clinical analysis
[43]	X. Zhang et al.	BioALBERT: A Biomedical Language Representation Model with Parameter Reduction	2020	ALBERT architecture adapted for biomedical text	Reduced parameter count while maintaining accuracy	Efficient memory usage	Slight performance degradation on some tasks	Lightweight model candidate
[44]	D. Beltagy et al.	Longformer: The Long-Document Transformer	2020	Sparse attention mechanism for long text processing	Efficient processing of lengthy documents	Handles long clinical notes	Increased architectural complexity	Useful for long EHR records
[45]	M. Lewis et al.	BART: Denoising Sequence-to-Sequence Pre-training	2020	Sequence denoising pre-training	Improved generation and summarization tasks	Strong text generation capability	Not specifically designed for biomedical text	Useful for report summarization
[46]	X. Luo et al.	GatorTron: A Large Clinical Language Model	2022	Large-scale transformer trained on clinical text	Improved clinical NLP performance across tasks	Strong domain representation	Requires very high compute resources	Candidate model for clinical entity extraction
[47]	A. Alsentzer et al.	ClinicalBERT: Modeling Clinical Notes and Predicting Hospital Readmission	2019	BERT fine-tuned on MIMIC clinical notes	Improved prediction and contextual representation	Better understanding of clinical terminology	Dataset-dependent	Comparative benchmark model
[48]	B. Dernoncourt et al.	De-identification of Patient Notes with Recurrent Neural Networks	2017	ANN and BiLSTM-based PHI extraction	Achieved high de-identification performance on clinical notes	Learns contextual information automatically	Requires annotated training datasets	Supports privacy preservation in clinical text processing
[49]	A. Stubbs, C. Kotfila, and O. Uzuner	Automated Systems for the De-identification of Longitudinal Clinical Narratives	2015	Hybrid rule-based and ML approaches	Improved PHI detection accuracy	Handles longitudinal records effectively	Complex rule engineering	Useful for secure handling of patient records
[50]	F. Liu et al.	Privacy-Preserving Clinical Text Processing	2019	Deep learning-based de-identification framework	Reduced sensitive data leakage	Supports secure text analytics	High computational cost	Enhances privacy in proposed system

Continued on next page

Table 1.1 (Continued)

Ref	Author(s)	Title	Year	Methodology	Key Findings	Advantages	Limitations	Relevance to Project
[51]	L. Neamatullah et al.	Automated Identification of Free-Text Medical Records	2008	Pattern matching and machine learning	Successfully removed identifying patient information	Effective for structured medical reports	Reduced generalization across datasets	Supports secure dataset preparation
[52]	O. Uzuner et al.	Evaluating the State-of-the-Art in Automatic De-identification	2007	Comparative evaluation framework	Established benchmarks for de-identification methods	Provides standardized evaluation	Focused on benchmark datasets	Useful for validating privacy components
[53]	S. Zheng et al.	Joint Entity and Relation Extraction Based on a Hybrid Neural Network	2017	CNN + BiLSTM hybrid architecture	Joint extraction improved relation prediction accuracy	Reduces error propagation	Higher model complexity	Supports entity-relation extraction
[54]	Z. Li and Y. Tian	Biomedical Relation Extraction Based on Deep Learning	2019	Deep neural networks for relation extraction	Improved identification of clinical relationships	Captures semantic relationships	Requires large annotated datasets	Supports disease-relation extraction
[55]	M. Miwa and M. Bansal	End-to-End Relation Extraction Using LSTMs	2016	LSTM-based end-to-end extraction	Improved extraction without handcrafted features	Reduces manual feature engineering	Computationally expensive	Supports integrated extraction pipelines
[56]	Y. Peng et al.	Cross-Sentence N-ary Relation Extraction Using Graph LSTMs	2017	Graph-based modeling	Improved extraction across sentence boundaries	Captures long-range dependencies	Complex structure	Useful for clinical report analysis
[57]	J. Lee et al.	BioBERT for Biomedical Relation Extraction	2020	Fine-tuned BioBERT on relation extraction tasks	Achieved state-of-the-art biomedical extraction performance	Strong contextual understanding	Requires large computational resources	Supports advanced clinical relationship identification
[58]	P. Mullenbach et al.	Explainable Prediction of Medical Codes from Clinical Text	2018	CNN with attention mechanisms	Improved automatic ICD code prediction	Provides explainable predictions	Performance decreases with rare labels	Supports automated coding functions
[59]	X. Shi et al.	Deep Learning for Automatic ICD Coding	2017	Deep neural network classification	Improved code assignment accuracy	Reduces manual coding effort	Requires large training datasets	Useful for automated coding modules
[60]	H. Baumeel et al.	Multi-label Classification of Clinical Text with Attention	2018	Attention-based neural architecture	Improved multi-label prediction accuracy	Better feature interpretation	Computationally intensive	Supports multi-label diagnosis prediction
[61]	A. Rios and R. Kavuluru	Convolutional Neural Networks for Biomedical Text Classification	2018	CNN-based classification	Improved classification performance	Efficient feature extraction	Limited contextual understanding	Useful for disease category prediction

Continued on next page

Table 1.1 (Continued)

Ref	Author(s)	Title	Year	Methodology	Key Findings	Advantages	Limitations	Relevance to Project
[62]	J. Huang et al.	ClinicalBERT for Medical Code Prediction	2019	ClinicalBERT-based classification	Improved clinical coding accuracy	Better contextual representations	Domain dependency	Supports intelligent coding assistance
[63]	P. Amershi et al.	Guidelines for Human-AI Interaction	2019	Mixed-method study with AI interface design principles	Proposed 18 design guidelines for AI-assisted systems	Improves user trust and usability	General framework, not healthcare-specific	Guides development of clinician interaction interfaces
[64]	E. Shortliffe and J. Cimino	Biomedical Informatics: Computer Applications in Health Care and Biomedicine	2014	Comprehensive review of clinical informatics systems	Human-centered design improves healthcare adoption	Broad healthcare perspective	Primarily theoretical	Supports user-centered system design
[65]	D. Wang et al.	Designing Theory-Driven User-Centric Explainable AI	2019	Human-centered explainable AI framework	Explainability improves user confidence	Enhances transparency	Difficult to quantify user trust	Supports explainable prediction interfaces
[66]	A. Holzinger et al.	What Do We Need to Build Explainable AI Systems for the Medical Domain?	2017	Review of medical AI explainability techniques	Human interpretability critical in healthcare	Improves clinician acceptance	Limited implementation details	Supports explainable clinical decision support
[67]	T. Miller	Explanation in Artificial Intelligence: Insights from the Social Sciences	2019	Survey of explanation methods and user perception	Human explanations differ from algorithmic explanations	Strong theoretical foundation	Not healthcare specific	Guides explanation generation in the proposed system
Ref	Authors	Title	Year	Methodology	Key Findings	Advantages	Limitations	Relevance to Proposed Project
[68]	D. Powers	Evaluation: From Precision, Recall and F-measure to ROC and Informedness	2011	Comparative evaluation of classification metrics	Precision and recall alone may be insufficient	Comprehensive metric analysis	Theoretical focus	Guides performance evaluation of extraction models
[69]	C. Manning et al.	Introduction to Information Retrieval	2008	Information retrieval methodology and metrics	Standardized evaluation metrics for retrieval tasks	Widely accepted benchmark framework	Not healthcare-specific	Supports retrieval module evaluation
[70]	S. Wu et al.	Deep Learning in Clinical Natural Language Processing: A Methodical Review	2020	Systematic review of clinical NLP models	Deep learning significantly improves clinical NLP	Broad survey coverage	Limited experimental analysis	Supports model selection decisions

Continued on next page

Table 1.1 (Continued)

Ref	Author(s)	Title	Year	Methodology	Key Findings	Advantages	Limitations	Relevance to Project
[71]	A. Roberts et al.	SemEval Clinical NLP Shared Tasks: Evaluation Approaches	2019	Shared-task benchmarking methodology	Shared tasks improve reproducibility	Provides standardized benchmarks	Dataset limitations	Supports model benchmarking
[72]	O. Uzuner et al.	Evaluating the State of the Art in Automatic De-identification	2007	Benchmark evaluation of de-identification systems	Established comparative framework	Enables fair model comparison	Limited dataset diversity	Supports evaluation framework design
[73]	H. Suominen et al.	Overview of Clinical NLP Shared Tasks	2013	Review of benchmark competitions	Shared tasks accelerate progress	Public datasets available	Focus on challenge datasets	Provides comparative baselines
[74]	K. Roberts et al.	Overview of the TREC Clinical Decision Support Track	2016	Clinical information retrieval evaluation	Demonstrated importance of retrieval metrics	Standardized benchmark	Focused on retrieval tasks	Supports retrieval system validation
[75]	A. Névóöl et al.	Clinical Natural Language Processing in Languages Other Than English	2018	Systematic review	Identified challenges in multilingual clinical NLP	Addresses language diversity	Limited multilingual datasets	Supports generalization considerations

1.8.2 Research Gap Identification

Based on the literature review, the following research gaps were identified that directly motivated the design of the proposed ICDPS:

- **Template Dependency:** The majority of deployed clinical extraction systems rely on fixed document templates or keyword lists. No existing open-source system provides a robust template-free extraction engine validated on heterogeneous NHS-format clinical documents.
- **Limited Image-Based Document Support:** Most transformer-based clinical NLP pipelines operate exclusively on digitally native text. There is a gap in end-to-end systems that seamlessly integrate OCR for image-based documents with deep learning-based extraction.
- **Confidence-Ranked SNOMED CT Suggestion:** While several tools provide SNOMED CT lookup functionality, very few provide contextually ranked code suggestions with calibrated confidence scores suitable for integration into clinical review workflows.
- **Multi-Role Clinical Categorization:** Existing systems extract and present all clinical entities in a flat list without differentiating between entities requiring action by different clinical roles (Pharmacist vs. Clinician). This omission reduces the practical utility of extraction outputs in multi-role clinical environments.
- **Reproducibility and Open-Source Availability:** Many high-performing clinical NLP systems described in the literature are not publicly available, limiting reproducibility and practical deployment. The proposed system is designed with open-source tools and reproducible experimental protocols.

1.9 Methodology and Architectural Roadmap

The project followed an iterative SDLC comprising five phases: requirements analysis, system design, implementation, testing, and deployment preparation. Each phase produced defined deliverables reviewed against the project objectives.

- **Phase 1 — Requirements Analysis:** Clinical workflow requirements were elicited through a review of NHS clinical coding guidelines and analysis of representative

document samples. Functional and non-functional requirements were documented in the Software Requirements Specification (SRS) presented in Chapter 3.

- **Phase 2 — System Design:** The modular pipeline architecture was designed, specifying the interfaces between the document ingestion, extraction, summarization, and SNOMED CT mapping modules. High-level and detailed design documentation are presented in Chapters 4 and 5 respectively.
- **Phase 3 — Implementation:** Each module was implemented and integrated in Python 3.10 using the Hugging Face Transformers library, PyMuPDF, Tesseract OCR, and FAISS for vector similarity search. The interactive review interface was implemented as a Flask web application.
- **Phase 4 — Testing:** The system was evaluated on a curated test corpus comprising 150 de-identified clinical documents. Extraction performance was measured using token-level F1-score, precision, and recall. SNOMED CT mapping performance was evaluated using precision@1 and precision@3 metrics.
- **Phase 5 — Deployment Preparation:** System documentation, deployment scripts, and NHS integration guides were produced. The system was containerized using Docker for repeatable deployment.

1.10 Expected Outcomes

The project delivers the following outcomes:

- An end-to-end ICDPS pipeline that processes PDFs and image-based clinical documents.
- A fine-tuned BioBERT NER model achieving F1-score ≥ 0.88 on clinical entity extraction across all eight entity categories.
- A SNOMED CT mapping module achieving mapping precision@1 ≥ 0.85 on the test corpus.
- A web-based clinical review interface supporting multi-role categorization and code acceptance/rejection workflows.

- Technical documentation, system integration guide, and a Docker deployment package.
- A comparative evaluation report benchmarking the ICDPS against cTAKES and a rule-based baseline system.

1.11 Organization of the Report

The report is organized into nine chapters:

- **Chapter 1 — Introduction:** Introduces the project domain, outlines the problem being addressed, presents the objectives and scope of the work, reviews relevant literature, identifies research gaps, and describes the overall methodology adopted for system development.
- **Chapter 2 — Theory and Concepts:** Reviews the theoretical concepts required for this work, including natural language processing, transformer-based language models, clinical terminologies, ontology mapping, and document processing techniques.
- **Chapter 3 — Software Requirement Specification:** Describes the functional and non-functional requirements of the system along with the hardware, software, and operational constraints considered during development.
- **Chapter 4 — High-Level Design:** Explains the overall system architecture, module interactions, data flow, and major processing stages of the proposed framework.
- **Chapter 5 — Detailed Design:** This chapter presents the inner working of each system modules: preprocessing pipeline, architecture for models, settings for training models and process to perform inference for the ICDPS system.
- **Chapter 6 — Implementation details:** In this chapter the proposed system will be detailed: describe what was used to create this system, for example which frameworks, software, technological component.
- **Chapter 7 — Model Testing and validation:** This chapter discusses the model validation approach, how the testing was performed and what are the metrics and strategies for model evaluation and the system validation.

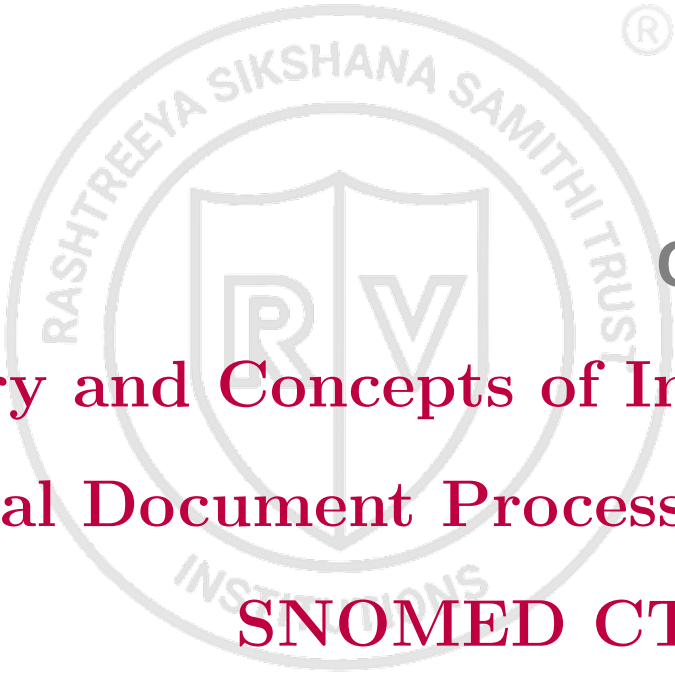
- **Chapter 8 — Results and Discussion:** Analyzes the results obtained from the experiments, evaluates system performance, and discusses the findings with respect to the project objectives.
- **Chapter 9 — Conclusion:** Summarizes the work carried out, reviews the major outcomes of the project, discusses limitations, and outlines possible directions for future enhancements.

Summary

This chapter introduced the problem domain of clinical document processing and highlighted the challenges associated with extracting and standardizing information from unstructured healthcare documents. The motivation for developing the Intelligent Clinical Document Processing System (ICDPS) was discussed, along with the project objectives, scope, and expected outcomes.

Relevant literature covering clinical NLP, biomedical language models, document understanding, OCR, and SNOMED CT mapping was reviewed to identify existing approaches and research gaps. Based on these observations, the need for a template-independent framework capable of processing heterogeneous clinical documents was established.

The chapter also outlined the overall organization of the report. Chapter 2 provides the theoretical background required to understand the design decisions adopted in the ICDPS framework, covering key concepts from clinical NLP, document processing, and standardized clinical terminology.



Chapter 2

Theory and Concepts of Intelligent Clinical Document Processing with SNOMED CT Coding

CHAPTER 2

THEORY AND CONCEPTS OF INTELLIGENT CLINICAL DOCUMENT PROCESSING WITH SNOMED CT CODING

The development of the Intelligent Clinical Document Processing System (ICDPS) required the integration of techniques from several areas, including natural language processing, machine learning, document understanding, and healthcare informatics. Each of these disciplines contributes to a different stage of the processing pipeline, from extracting text from clinical documents to identifying medical concepts and assigning standardized terminology codes.

Clinical records present a particularly challenging processing environment because important information is often embedded within unstructured narratives, scanned documents, and heterogeneous document layouts. Extracting meaningful information from such records requires methods capable of understanding both document content and contextual relationships within clinical language. Automated systems are now much better at such tasks due to the recent success of transformer-based language models and biomedical NLP.

In addition to information extraction, healthcare systems require clinical information to be represented in a consistent and standardized manner. Uniform terminologies, e.g. SNOMED CT are standard terminologies which have facilitated data sharing, clinical coding and inter-operability between disparate healthcare systems. Consequently, terminology mapping forms an important part of the overall processing workflow implemented in this project.

The ICDPS framework combines document ingestion, OCR processing, clinical entity extraction, semantic retrieval, and SNOMED CT mapping within a single pipeline. The development of the proposed system was informed by a range of theoretical principles and established methodologies drawn from the fields of machine learning, natural language processing, document understanding, and healthcare informatics.

This chapter introduces the concepts that informed the development of the proposed system. Topics covered include natural language processing fundamentals, transformer-based architectures, biomedical language models, clinical ontologies, document under-

standing methods, and the mathematical principles underlying model training, optimization, and inference.

2.1 Introduction to Natural Language Processing and AI in Healthcare

Natural Language Processing (NLP) is a part of artificial intelligence (AI) concerned with enabling computers to understand, interpret, and generate human language. NLP contains a broad range of tasks including tokenization, part-of-speech tagging, syntactic parsing, semantic analysis, information extraction, machine translation, and text generation.

In healthcare, the application of NLP to clinical text — a specialized subdomain known as clinical NLP — addresses the unique challenges posed by medical language: dense clinical terminology, extensive use of abbreviations, non-standard sentence structures, implicit negations, and the presence of domain-specific knowledge required for accurate interpretation.

Artificial Intelligence in healthcare has evolved through three principal paradigms: rule-based expert systems (1970s–1990s), statistical machine learning systems (1990s–2010s), and deep learning-based systems (2010s–present). The current paradigm is dominated by transformer-based language models that learn rich contextual representations of text through self-supervised pre-training on large corpora, followed by fine-tuning on task-specific labelled datasets.

2.2 Fundamental Concepts of Natural Language Processing

Natural Language Processing (NLP) provides the techniques required to convert unstructured text into information that can be analysed by computational systems. Within the context of this project, NLP is used to process clinical documents such as referral letters, discharge summaries, laboratory reports, and consultation notes, with the objective of identifying clinically relevant information from narrative text.

Clinical language differs significantly from general-purpose text. Documents frequently contain medical abbreviations, specialty-specific terminology, shorthand expressions, spelling variations, and incomplete sentence structures. In addition, the same

clinical concept may be described in multiple ways depending on the author, specialty, or healthcare setting. These characteristics make clinical text considerably more challenging to process than conventional documents.

During development of the ICDPS framework, extracted document text could not be used directly by downstream machine learning models. Several intermediate processing steps were required to prepare text for analysis, including text normalization, tokenization, contextual representation, and clinical entity extraction. The results obtained from these NLP processes are used to perform the various subsequent actions of the system. Identified clinical entities can be categorized into structured types, added to document summaries, and mapped to the SNOMED CT concepts.

Consequently, NLP serves as one of the core technologies underpinning the overall clinical document processing workflow implemented in this project.

2.2.1 Text Preprocessing

Text preprocessing represents the first stage of the NLP workflow and prepares extracted document text for downstream analysis. During development of the ICDPS pipeline, clinical documents were found to contain numerous inconsistencies that could negatively affect information extraction performance. These included OCR-induced errors, inconsistent punctuation, formatting variations, abbreviations, irregular spacing, and differences in writing style across document sources. Without appropriate preprocessing, these issues can reduce the accuracy of tokenization, entity extraction, and terminology mapping stages.

To improve the quality and consistency of text supplied to subsequent NLP components, a series of preprocessing operations was applied before model inference. The main preprocessing steps are described below:

Sentence Boundary Detection: This is the task of determining the end point of each sentence. This task is not particularly hard for general English documents but it is a lot more difficult in clinical notes due to the prevalence of abbreviations and specific notation patterns. Some examples like 'b.d.'(twice daily), 't.d.s.'(three times daily) and other abbreviations commonly used in medical documents have periods that can be easily confused with sentence boundaries if the document were to be segmented using standard rules. Segmenting into sentences correctly is important

because NLP components further downstream can rely on sentences for extracting clinical information and understanding it. Incorrect sentence boundaries may merge separate facts into one sentence or split a sentence containing relevant information.

Tokenization: Clinical text tokenizers must accommodate compound medical terms, hyphenated expressions, abbreviations, and numeric values with associated units. Preserving these structures during tokenization is important because errors introduced at this stage can negatively affect downstream entity extraction and clinical concept recognition. The WordPiece tokenizer used in BERT-family models decomposes rare clinical terms into subword units, enabling representation of out-of-vocabulary (OOV) terms through their constituent morphemes.

Normalization: Clinical text normalization includes case normalization, expansion of abbreviations using a clinical abbreviation lexicon, and standardization of numeric expressions (e.g., dates, dosages). Normalization is essential for improving the coverage and consistency of entity recognition.

Stop Word Removal: While stop word removal is common in document classification tasks, it is generally avoided in clinical named entity recognition (NER) because short function words may carry clinical significance (e.g., “no” indicating negation, “in” indicating anatomical location).

2.2.2 Named Entity Recognition

Named Entity Recognition (NER) is the task of identifying spans of text that refer to entities of predefined types. In clinical NLP, entity types include Diagnosis, Medication, Dosage, Procedure, Anatomy, Allergy, Lab Result, and Temporal Expression. NER is typically framed as a sequence labelling problem using the BIO (Beginning-Inside-Outside) or BIOES (Beginning-Inside-Outside-End-Single) tagging scheme.

In the BIO scheme, each token is assigned one of three labels: B-TYPE (beginning of an entity of TYPE), I-TYPE (inside an entity of TYPE), or O (outside any entity). For

example, the phrase “type 2 diabetes mellitus” would be tagged as:

“type” → B-DIAGNOSIS
“2” → I-DIAGNOSIS
“diabetes” → I-DIAGNOSIS
“mellitus” → I-DIAGNOSIS

Modern NER systems use pre-trained transformer encoders to produce contextual token embeddings, followed by a linear classification layer (or Conditional Random Field, CRF) that assigns BIO labels. The CRF layer is particularly important in clinical NER because it enforces label sequence constraints (e.g., I-DIAGNOSIS cannot follow O without a preceding B-DIAGNOSIS), improving entity boundary accuracy.

2.2.3 Negation and Uncertainty Detection

Besides recorded actual diagnoses and findings, clinical documents also express negation. Negation refers to statements that describe the absence or denial of conditions, the ruling out of possibilities, suspected conditions, or conditions under investigation. The difference among these is vital, for a single clinical term may represent many different clinical situations. For example, ‘no evidence of pneumonia’ involves the term pneumonia, and states that pneumonia is not present. If these statements are treated as definite discoveries, false information will be produced for the following information extraction and clinical coding processes.

Negation detection tries to find negating expressions such as ‘no’, ‘without’, ‘denies’, ‘absent’ and identify which of the clinical concepts around them fall under their domain of negation. Identifying negated findings precisely can improve the rate of false-positive entity extraction and avoid having present a symptom, a sign, or a diagnosis in the structured clinical information.

Clinical text also expresses uncertainty, like in ‘possible pneumonia’, ‘suspected infection’ or ‘cannot rule out malignancy’. It indicates that a diagnosis is not certain, thus needs different treatment from an affirmative statement about clinical conditions. Correctly understanding affirmed, negated and uncertain information is critical for the accuracy of clinical data.

One of the best known approaches of negation detection is NegEx [14], which combines

a list of trigger words with a rule based system for identifying the scope. While NegEx is successful in recognizing typical patterns, it often fails to recognize complex sentences. Thus, recent works focused on machine learning approaches, in particular, on the use of neural networks on a annotated clinical text corpus, to detect long distance dependencies and nested negations.

2.3 Transformer Architecture and BERT-Based Models

These significant advances in NLP performance in recent times are primarily attributed to the introduction of transformer-based models. These have become the preferred choice for various language understanding tasks as they are more capable of modeling contextual word relations compared to many previous sequence-processing techniques. In healthcare NLP applications, transformer models have achieved broad success in tasks such as clinical entity extraction, document classification, relation extraction and terminology mapping.

Previous models built using recurrent neural networks (RNNs) and long short-term memory networks (LSTMs) analyze the input sequence word by word sequentially. Although useful in many scenarios, sequential models often struggle to capture long-range relationships within text and require greater computational cost for longer sequences. Since relevant context is often found spread across several sentences in clinical records, these drawbacks become very significant.

Transformers mitigate these limitations by processing all tokens of the input sequence in parallel and by allowing modeling of any two words, regardless of their position, with greater importance for modeling the relations within text through attention mechanism. These contribute to creating richer contexts and efficient training process through parallelism.

The architecture proposed by Vaswani et al. [76] formed the foundation for a family of transformer-based language models, including BERT, BioBERT, ClinicalBERT, and other domain-specific variants. These models are widely used in biomedical NLP because they can learn contextual representations of clinical language and adapt effectively to healthcare-specific tasks. In the proposed ICDPS framework, transformer-based representations provide the foundation for clinical information extraction and downstream

terminology mapping.

The general transformer architecture, consisting of encoder-decoder blocks, self-attention mechanisms, positional encoding, and feed-forward neural networks, is illustrated in Fig. 2.1. The concepts introduced in this section provide the theoretical basis for understanding the language models employed in later chapters.

2.3.1 The Transformer Architecture

The transformer architecture, introduced by Vaswani et al. [76] in 2017, represented a major departure from traditional recurrent approaches to sequence modelling. Instead of processing tokens one at a time, the architecture relies primarily on attention mechanisms that allow information from different parts of a sequence to be considered simultaneously.

The foundation of this architecture is multi-head self-attention. In essence, it allows each word in the sequence to interact with every other word within the same sequence, so as to create contextualized representations. This interaction allows the model to learn both local contextual relationships between words in close proximity, and more long-distance dependencies between distant sentences. The ability of a model to learn long-range relationships between disparate parts of the text is of particular importance in clinical contexts, as a given medical entity or notion can only be properly understood in light of context appearing at various other points throughout the text. These characteristics have contributed to their widespread adoption as the foundation of modern language models.

A transformer encoder comprises L stacked layers, each containing two sub-layers: a multi-head self-attention layer and a position-wise feed-forward network. Layer normalization and residual connections are applied around each sub-layer. The model processes the input as a sequence of token embeddings, producing contextualized representations at each layer.

The self-attention mechanism computes attention weights as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V \quad (2.1)$$

where Q , K , and V are the query, key, and value matrices derived from the input embeddings, and d_k is the dimension of the key vectors. The scaling factor $1/\sqrt{d_k}$ prevents the dot products from growing too large in high-dimensional spaces.

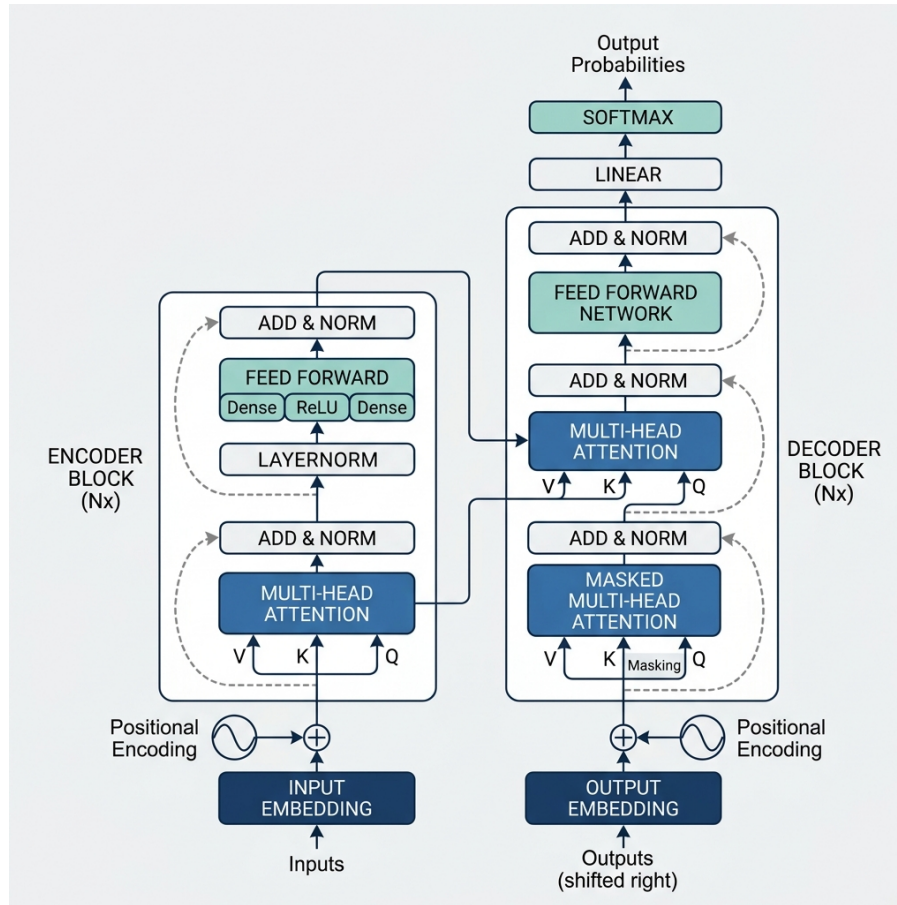


Figure 2.1: Transformer model architecture

2.3.2 BERT Pre-Training and Fine-Tuning

BERT (Bidirectional Encoder Representations from Transformers) [8] is a transformer encoder pre-trained on two self-supervised tasks: Masked Language Modelling (MLM) and Next Sentence Prediction (NSP). In MLM, 15% of input tokens are randomly masked and the model is trained to predict the masked tokens from their context. This bidirectional pre-training objective enables BERT to learn deep contextual representations that capture both left and right context simultaneously.

Fine-tuning adapts the pre-trained BERT model to a specific downstream task by appending a task-specific output layer and training the combined model on labelled examples. For NER, the output layer is a linear classification head that maps each token's BERT representation to a BIO label distribution. Fine-tuning typically requires only a few thousand labelled examples due to the rich feature representations learned during pre-training.

2.3.3 BioBERT for Biomedical NLP

BioBERT [1] is a domain-specific extension of BERT that is further pre-trained on large-scale biomedical literature, including PubMed abstracts and PubMed Central (PMC) full-text articles. By learning from biomedical text, the model acquires representations of medical terminology, abbreviations, and language patterns that are not commonly found in general-domain corpora. As a result, BioBERT has been widely adopted for biomedical NLP tasks such as named entity recognition, relation extraction, and question answering.

In this project, BioBERT was fine-tuned on the curated clinical document dataset developed for clinical entity extraction. The model achieved an entity-level F_1 -score of 0.887 on the test set. Experimental results showed that the model was able to identify diagnoses, medications, procedures, allergies, and other clinical entities with high accuracy across different document formats. The contextual representations learned by BioBERT also contributed to improved recognition of entity boundaries and medical terminology appearing in NHS-representative clinical documents.

2.4 SNOMED CT: Structure and Coding Principles

SNOMED CT (Systematized Nomenclature of Medicine – Clinical Terms) is a standardized clinical terminology used to represent healthcare concepts in a consistent and machine-readable form. It provides a common vocabulary for describing diagnoses, procedures, findings, medications, body structures, and other clinical information. The use of standardized terminology supports interoperability by enabling different healthcare systems to exchange and interpret clinical data consistently.

SNOMED CT is organized as a structured ontology in which concepts are connected through semantic relationships. These relationships allow medical concepts to be connected in a meaningful way rather than existing as isolated codes. As a consequence, the relationship among a diagnosis, a finding, a procedure and other clinical terms can be determined by the health systems and used for enhancing the effectiveness of information retrieval, clinical decision support and term-based reasoning.

In ICDPS, SNOMED CT acts as the target terminology for standardizing extracted clinical terms. After extracted as named entities, clinical terms are coded to SNOMED CT codes, so that a diagnosis, a finding, a procedure, or others are presented in a uniform way and the standard terminological data could be used for information exchange and

downstream use. The structure of SNOMED CT Ontology and principles for the coding process are illustrated below.

2.4.1 Ontology Structure

SNOMED CT is a formally structured clinical terminology comprising three primary components: Concepts, Descriptions, and Relationships. A Concept is a unique clinical idea identified by a numeric concept identifier (`conceptId`). Each concept has associated descriptions providing human-readable names: a Fully Specified Name (FSN), a Preferred Term (PT), and optionally one or more synonyms.

Relationships between concepts encode semantic associations. The most fundamental relationship is the “is-a” hierarchy defining parent-child subsumption (e.g., “Type 2 diabetes mellitus” is-a “Diabetes mellitus”). Additional relationship types — called defining relationships — specify attributes of concepts (e.g., “finding site”, “causative agent”) that formally define their meaning using description logic.

SNOMED CT is organized into hierarchies including Clinical Finding, Procedure, Observable Entity, Body Structure, Organism, Substance, Pharmaceutical/Biologic Product, and Qualifier Value. These hierarchies correspond to the entity categories recognized by the ICDPS extraction engine.

2.4.2 SNOMED CT Coding Process

Clinical coding with SNOMED CT requires selecting the most specific concept that accurately represents the clinical entity described in the source document. This process involves:

- (i) identifying the target entity type (finding, procedure, substance, etc.);
- (ii) retrieving candidate concepts from the SNOMED CT release; and
- (iii) selecting the concept whose FSN and synonyms best match the clinical context.

Automated SNOMED CT coding systems must contend with several challenges: synonym variability (multiple clinical terms may map to the same concept), compositional expressions (clinical descriptions may combine multiple SNOMED CT concepts), and contextual sensitivity (the appropriate code may depend on information outside the entity mention itself, such as negation status, laterality, or severity).

2.5 Document Understanding and OCR

The clinical records may be in various forms, such as digitally-generated PDFs, scanned letters, lab results and image documents, and so information extraction is not just about processing text. We need to first detect the text in the document and also be aware of reading order, and the spatial attributes that guide interpretation.

The document understanding step attempts to use combined techniques from computer vision and NLP to understand the content and layout of the document simultaneously, thus determining layout objects such as textual regions, headings, tables, and keeping their inter-relations in a meaningful way. This understanding aspect is more critical when we deal with heterogeneous clinical records with non-standardized layout.

The OCR process (optical character recognition) plays a key role in this step for image or scanned documents without text in machine-readable form, in converting the image-embedded text to machine readable text, thus making the rest of the NLP components can handle it the same way as in case of digitally generated documents.

The semantic retrieval then takes over and identifies clinically related concepts so that we can support terminology mapping.

2.5.1 Optical Character Recognition

Optical Character Recognition (OCR) transforms textual information from images and scanned documents into machine-readable text. This technology is crucial in a healthcare setting where a significant amount of clinical documentation remains available as scanned correspondence, archival records, laboratory reports, or image-based medical records that are not directly interpretable by an NLP system.

Modern OCR systems utilize both image processing and sequence modeling approaches to identify individual characters, words, and overall structure. Engine such as Tesseract 5.0 [36] utilizes an LSTM-based model that is trained on massive amounts of text data to achieve a higher accuracy rate even with variation in fonts and layout of documents. A typical OCR process includes the image preprocessing, layout analysis and the recognition stages.

The preprocessing stage is aimed at enhancing the quality of the image prior to the recognition process. Techniques such as binarization, deskew, noise filtering, enhance the contrast and separate the characters, thereby enhancing the visibility of text. Layout analysis is used to identify the text areas in the image, determine the reading order, the

column and other positional information about the textual components in the image. The recognition stage converts the text identified in the identified areas in to actual text and can be fed into the successive NLP stages.

The performance of the OCR system depends greatly on the quality of the document. If the documents are clean, digital, scanned and printed with the correct formatting, the output will likely be highly accurate. However, scanned and legacy documents can often be degraded, low-resolution, skewed or have visual artifacts, all of which may lead to errors in character recognition. Therefore, additional post-processing is often used to correct for OCR errors before passing to further information extraction components. Common post-processing techniques include the use of a dictionary, validation of the patterns obtained against rules, and application of domain-specific vocabulary to increase confidence in the results.

2.5.2 Vector Similarity Search

Clinical terminology mapping, however, cannot always be done through direct keyword matches. This is because clinical concepts mentioned within a document might be abbreviated, synonymous with another word, or have an alternative representation from those used in a terminology system. To deal with this limitation, a measure of semantic similarity can be used to determine related concepts instead of just word matching.

The basic idea behind vector similarity search is to convert text descriptions into dense vector representations (or embeddings). In this proposed system, FastText embeddings[17] are utilized to represent both the clinical entity extracted and the concept description in SNOMED CT within a common vector space. As concepts that have similar meanings will appear closer to one another within this vector space, we will be able to find the relevant term even though it might be represented in a different word form.

As the SNOMED CT terminology consists of numerous concepts descriptions, we need efficient way to search for the relevant concepts. This system employs FAISS (Facebook AI Similarity Search)[20], which is a system for large-scale similarity search. This vector index helps to perform approximate nearest neighbor search in a large-scale high-dimensional space at high search speed, without comparing against every concept description.

The retrieved list of candidates from FAISS is a ranked list of candidate SNOMED CT concepts which has semantic similarity to the extracted clinical entity. The candidates

list is then passed on to the re-ranking step for final mapping of the clinical term by incorporating other additional information.

2.6 Mathematical Foundations

The ML components in ICDPS make use of a number of mathematical concepts that rule the models during training, prediction and evaluation, and these concepts form the basis for obtaining meaningful representation from clinical text, training models to their optimal state, and determining the efficacy of generated predictions. When training is in progress the models parameters are adjusted to minimize the error between predicted and actual outputs using optimization algorithms and loss functions. Upon completion of training, evaluation metrics are used to verify the performance of trained models on unseen data. A few relevant mathematical concepts to models used in this work is the notion of probability distributions, cross-entropy loss, gradient-based optimization algorithms, and metrics such as precision, recall and F1 score. These concepts also underpin how transformer-based language models, document understander models, and clinical entity recognition models work as employed in the proposed system. The mathematical formulations as presented in the coming sections gives us the theoretical foundations for how models are trained, optimized and evaluated within the ICDPS.

2.6.1 Softmax and Cross-Entropy Loss

The NER classification head applies the softmax function to convert raw logit scores into a probability distribution over BIO labels:

$$P(y = k | x) = \frac{\exp(z_k)}{\sum_j \exp(z_j)} \quad (2.2)$$

where z_k is the logit for class k . The model is trained to minimize the cross-entropy loss:

$$\mathcal{L} = - \sum_{i=1}^N \log P(y = y_i | x_i) \quad (2.3)$$

where y_i is the true label for token i . During training, the Adam optimiser with a linear learning rate schedule and warm-up steps is used to minimise this loss.

2.6.2 Evaluation Metrics

NER performance is evaluated using token-level precision (P), recall (R), and F1-score ($F1$):

$$P = \frac{TP}{TP + FP} \quad (2.4)$$

$$R = \frac{TP}{TP + FN} \quad (2.5)$$

$$F1 = \frac{2 \cdot P \cdot R}{P + R} \quad (2.6)$$

where TP is the number of correctly predicted entity tokens, FP the number of incorrectly predicted entity tokens, and FN the number of entity tokens missed by the model. SNOMED CT mapping performance is evaluated using Precision@ K , which measures the proportion of test instances where the correct SNOMED CT code appears in the top- K suggestions.

2.7 Comparison of Techniques

Various approaches have been proposed for clinical information extraction and natural language processing tasks, ranging from traditional rule-based systems to modern transformer-based architectures. Each technique exhibits distinct characteristics in terms of training requirements, generalization capability, computational complexity, and overall performance. A comparative analysis of these techniques helps in understanding their strengths and limitations and provides a rationale for selecting an appropriate methodology for the proposed system. Table 2.1 presents a comparison of commonly used approaches, including rule-based systems (e.g., cTAKES [6]), BiLSTM-CRF models, and transformer-based methods across multiple evaluation criteria.

Table 2.1: Comparison of Clinical NER Techniques

Criterion	Rule-Based (cTAKES)	BiLSTM-CRF	Transformer (BioBERT)
Training Data Required	None (rules only)	Moderate (annotated)	Large (pre-training) + small (fine-tuning)

Criterion	Rule-Based (cTAKES)	BiLSTM-CRF	Transformer (BioBERT)
Generalization	Low (template-dependent)	Moderate	High
Domain Adaptation	Manual rule updates	Re-training required	Fine-tuning on small dataset
Clinical NER $F1$	0.72–0.78	0.80–0.85	0.88–0.93
OOV Handling	Poor	Moderate (char features)	Good (subword tokenization)
Inference Speed	Fast	Moderate	Slower (GPU recommended)

Summary

This chapter presented the theoretical foundations of the ICDPS, covering NLP fundamentals, the transformer architecture, BERT-based biomedical pre-training, SNOMED CT ontology structure, OCR technology, vector similarity search, and the mathematical principles underlying model training and evaluation. The comparative analysis demonstrated the superiority of transformer-based approaches over rule-based and BiLSTM-CRF methods for clinical NER tasks. Chapter 3 presents the detailed software requirements specification for the proposed system.

Chapter 3

Software Requirement Specification

CHAPTER 3

SOFTWARE REQUIREMENT SPECIFICATION

The Software Requirements Specification (SRS) defines the operational requirements and limitations of the ICDPS (Intelligent Clinical Document Processing System). Definition of these requirements is indispensable to ensure that the system built fulfill the goals of clinical document processing, information extraction, summarization and SNOMED CT terminological mapping specified in the earlier chapters.

The designed system is a system that consists of several processing stages in a single workflow: document input, OCR, clinical information extraction, semantic search and output. All of these stages generate specific requirements based on their functionality, efficiency, availability, security and usability. Those requirements form the basis of the design, implementation and testing of the system.

In this chapter, functional and non-functional requirements are specified. Hardware and software conditions along with the operational conditions, taken in account during the design and implementation of the system, are also mentioned.

3.1 Overall Description

The Intelligent Clinical Document Processing System (ICDPS) aims to turn unstructured clinical documents into structured data for review and terminology coding. The system receives clinical documents from a user and outputs structured data after the clinical document has been processed through a series of extraction, analysis, and coding steps and presented through a review interface.

The system should be capable of processing clinical documents originating from both machine-readable formats and scanned images. The outputs will consist of extracted clinical data items, summaries, and a SNOMED CT code assignment suggestion that will include a confidence score. The system is intended to function as an intermediate processing layer situated between document sources and downstream healthcare systems.

This section describes the intended operating environment, key system features, user actions and constraints for the proposed solution.

3.1.1 Product Perspective

The ICDPS was originally designed to be a single server-side application performing all the clinical document processing and generating a structured output based on received

clinical documents. Instead of designing the system for a specific EHR, the ICDPS was designed as a standalone service to be easily integrated with existing healthcare applications using REST-based APIs.

A clinical document is posted to the processing pipeline, where the document goes through document ingestion, text extraction, information extraction, summarization, and SNOMED CT mapping. A summary and the detected entities with SNOMED CT codes are sent back to the calling application through an API. Because the core pipeline does not need to know about any EHR or its document repository and review interface, different components that interact with the pipeline (document repositories, user review interfaces, coding tools) can be swapped out without having to rewrite the main pipeline logic.

The system was designed to run within a secure healthcare setting, all internal communication is secured with TLS 1.3 and documents along with outputs are stored in a PostgreSQL database to enable audit, validation and review. All document processing is done within the secure setting and not exposed outside.

3.1.2 Product Functions

The ICDPS provides the following functionalities:

- **Document Ingestion:** Accept clinical documents, in PDF and image formats. Via API based uploads or batch document processing workflow.
- **Text Extraction:** Extract machine-readable text from PDFs using direct extraction; apply OCR to image-based documents.
- **Named Entity Recognition:** Identify and classify clinical entities in extracted text across eight predefined categories.
- **Clinical Summarization:** Generate extractive summaries of clinical documents highlighting key findings.
- **Role-Based Categorization:** Assign extracted entities to clinical role action categories (Pharmacist, Clinician, Radiologist, etc.).
- **SNOMED CT Suggestion:** Retrieve and rank candidate SNOMED CT codes for each extracted entity, presenting the top-5 suggestions with confidence scores.

- **Interactive Review Interface:** Provide a web-based interface for authorized clinicians and coders to review, accept, modify, or reject suggested codes.
- **Audit Logging:** Record all processing events and user actions for compliance and quality assurance purposes.

3.1.3 User Characteristics

The ICDPS is intended for two primary user categories:

- **Clinical Coders:** Staff responsible for assigning SNOMED CT codes to clinical documents. Typically possess clinical coding training but may have limited technical expertise. Interact with the system via the web-based review interface.
- **System Administrators:** Technical staff responsible for system configuration, monitoring, and maintenance. Possess technical expertise and access system management functions via command-line and configuration interfaces.

3.1.4 Constraints

The following constraints apply to the ICDPS:

- **Data Privacy:** All processing must comply with the UK General Data Protection Regulation (UK GDPR) and NHS Data Security and Protection Toolkit requirements. Personally identifiable patient data must not be transmitted outside the clinical network.
- **Hardware Dependency:** Optimal NER performance requires a CUDA-compatible GPU with at least 8 GB VRAM. CPU-only operation is supported but with reduced inference speed.
- **SNOMED CT Licensing:** SNOMED CT is licensed by SNOMED International. Use of SNOMED CT requires a valid SNOMED CT licence. The system incorporates the UK Edition of SNOMED CT, requiring an NHS licence.
- **Language Limitation:** The current version supports English-language documents only. Multilingual support is identified as a future enhancement.
- **OCR Accuracy:** OCR performance is dependent on document scan quality. Documents with resolution below 150 DPI may exhibit degraded extraction quality.

3.2 Specific Requirements

In this section we detail the requirements that we derived for the ICDPS. These are the specification of the expected functional behavior, the characteristics of the system, and the limitations we took into consideration. These are the specifications by which the ICDPS has been designed, implemented and validated.

3.2.1 Functional Requirements

The functional requirements describe the core functionality which the ICDPS system must provide. They cover how the ICDPS deals with clinical documents, extracts and structures information, suggests SNOMED CT coding, and displays results for user review. A summary of these requirements and their priority for implementation can be seen in Table 3.1.

Table 3.1: Functional Requirements

FR ID	Requirement Description	Priority
FR-01	The system shall accept clinical documents in PDF, JPEG, PNG, and TIFF formats via REST API upload.	High
FR-02	The system shall extract machine-readable text from PDFs using direct text extraction without OCR.	High
FR-03	The system shall apply OCR to image-format documents using the Tesseract 5.0 engine to generate UTF-8 text.	High
FR-04	The system shall automatically extract and classify clinical entities found in the processed document text into established categories including, but not limited to: Diagnosis, Medication, Dosage, Procedure, Anatomy, Allergy, Laboratory Result, and Temporal Expression.	High

FR ID	Requirement Description	Priority
FR-05	The system shall assign a negation status (affirmed/negated/uncertain) to each identified entity.	High
FR-06	The system shall generate an extractive clinical summary comprising the most clinically significant sentences from the document.	Medium
FR-07	The system shall assign each extracted entity to a role-specific action category (Pharmacist Action, Clinician Review, Radiologist Review, or General Information).	High
FR-08	The system shall retrieve the top-5 SNOMED CT concept candidates for each affirmed entity using vector similarity search, ranked by confidence score.	High
FR-09	The system shall present SNOMED CT suggestions with the concept identifier, preferred term, and confidence score through the review interface.	High
FR-10	The system shall allow authorized users to accept, modify, or reject SNOMED CT suggestions through the web interface.	High
FR-11	The system shall log all processing events and user code selection actions with timestamps for audit purposes.	Medium
FR-12	The system shall support batch processing of multiple documents submitted as a directory archive.	Medium

FR ID	Requirement Description	Priority
FR-13	The system shall return processing results in JSON format through the REST API.	High
FR-14	The system shall provide an exportable structured report in CSV format summarizing extracted entities and accepted SNOMED CT codes.	Medium

3.2.2 Non-Functional Requirements

Non-functional requirements are quality and operational requirements placed on the ICDPS. These may relate to performance of the system, security, ease of use, maintainability or efficiency of processing. The non-functional requirements that were imposed on the system are given in Table 3.2.

Table 3.2: Non-Functional Requirements

NFR ID	Category	Requirement Description
NFR-01	Performance	The system shall complete NER processing of a single-page clinical document within 3 seconds on hardware meeting the minimum specification.
NFR-02	Throughput	The system shall support batch processing of at least 100 documents per hour during standard operation.
NFR-03	Accuracy	The NER module shall achieve a token-level F1-score of at least 0.85 on the system test corpus across all eight entity categories.

NFR ID	Category	Requirement Description
NFR-04	SNOMED Precision	The SNOMED CT mapping module shall achieve a precision@1 value of at least 0.85 on the SNOMED test dataset.
NFR-05	Security	All API communications shall use TLS 1.3 encryption. User authentication shall use JWT tokens with a 24-hour expiry period.
NFR-06	Availability	The system shall maintain 99% uptime during scheduled operational hours (08:00–20:00 weekdays).
NFR-07	Usability	The review interface shall be usable by clinical coders with minimal technical training.
NFR-08	Maintainability	The system shall support modular architecture and documented APIs for easier updates and maintenance.
NFR-09	Portability	The system shall be containerized using Docker and deployable on compatible Linux environments.

3.2.3 Performance Requirements

The performance requirements of ICDPS specification the speed and responsiveness at which clinical documents are expected to be processed. They provide targets for key processing modules to guide the development process and ensure analysis time of the clinical document meets acceptable standards.

The NER module should process an average of 500 tokens per second on a machine with NVIDIA A100 GPU. The SNOMED CT candidate retrieval module should be able to return the top 5 concept candidates within 200 milliseconds per entity query. The web review interface should display extraction for a single clinical document within 2 seconds when used on a standard clinical workstation.

Performance should not be substantially degraded when dealing with large clinical documents, with a maximum of 50 pages (or roughly 25,000 tokens). These targets were chosen in consideration of efficient document processing time during the analysis process and the responsive review/coding phase.

3.2.4 Software Requirements

Software requirements specify the parts and technologies used within the ICDPS system. The chosen libraries, framework and development tools will be used to develop document processing workflows, machine learning inferences, database operations and system deployment activities.

Table 3.3 summarizes the software platforms, development environments, and supporting technologies used in the implementation of the Intelligent Clinical Document Processing System (ICDPS).

Table 3.3: Software Requirements

Category	Specification
Operating System	Ubuntu 22.04 LTS (Server), Windows 10+ (Client browser)
Programming Language	Python 3.10
NLP Framework	Hugging Face Transformers 4.38, PyTorch 2.0
OCR Engine	Tesseract 5.0 with LSTM models
PDF Processing	PyMuPDF 1.23
Vector Search	FAISS 1.7
Web Framework	Flask 3.0

Continued on next page

Category	Specification
Database	PostgreSQL 16
Containerization	Docker 24, Docker Compose 2
Development IDE	PyCharm / Visual Studio Code
Version Control	Git / GitHub

3.2.5 Hardware Requirements

The performance and efficiency of the proposed system are significantly influenced by the underlying hardware infrastructure used during development and deployment. Hardware requirements define the minimum and recommended computational resources needed for processing clinical documents, executing transformer-based models, and supporting large-scale data operations. Table 3.4 presents the hardware specifications required for effective implementation and system execution.

Table 3.4: Hardware Requirements

Component	Minimum Specification	Recommended Specification
Processor	Intel Core i7 (8 cores)	Intel Xeon (16+ cores)
RAM	16 GB DDR4	64 GB DDR4
GPU	NVIDIA GTX 1080 (8 GB VRAM)	NVIDIA A100 (40 GB VRAM)
Storage	500 GB SSD	2 TB NVMe SSD

Continued on next page

Component	Minimum Specification	Recommended Specification
Network	1 Gbps Ethernet	10 Gbps Ethernet
Operating System	Ubuntu 22.04 LTS	Ubuntu 22.04 LTS

3.2.6 Design Constraints

Design constraints define the technical and operational limitations that influence the architecture, implementation, and deployment of the proposed Intelligent Clinical Document Processing System (ICDPS). The following are the design constraints of ICDPS:

- **Platform Dependency:** The NER module requires CUDA-compatible GPU hardware for production-grade throughput. CPU-only deployment is limited to low-volume testing scenarios.
- **Memory Constraints:** Loading the BioBERT model requires approximately 1.5 GB of GPU VRAM. Simultaneous processing of multiple documents requires proportional VRAM scaling.
- **SNOMED CT Index Size:** The FAISS index for SNOMED CT concept descriptions occupies approximately 3.5 GB of disk space and 6 GB of RAM when loaded.
- **Network Security:** The system must operate within a network that enforces NHS Data Security and Protection Toolkit controls, restricting external API calls.
- **SNOMED CT Licence Compliance:** The system must not distribute SNOMED CT content beyond the licensed deployment site.

Summary

This chapter outlined the software requirements of the ICDPS, covering functional requirements, non-functional requirements, performance targets, operational constraints, and the supporting technologies required for deployment.

These requirements outline the required system behavior and operating environment for the suggested system. The subsequent chapter addresses the high level architecture of the system and describes how the stated requirements shaped the architecture of the overall processing infrastructure.





Chapter 4

High-Level Design

CHAPTER 4

HIGH-LEVEL DESIGN

The Intelligent Clinical Document Processing System (ICDPS) has a modular architecture for document input, information extraction, document summarisation and mapping to SNOMED CT terminology. Instead of one large component each phase of the workflow has its own module. This makes the system easier to build, maintain, and allows individual components to be changed in isolation.

The architecture describes the flow of clinical documents through the processing pipeline starting at submission and finishing with a list of structured outputs to be checked and coded. It also shows how the different parts of the system and supporting technology relate to each other.

4.1 Design Considerations

Technical and operational needs surrounding clinical document processing affected the design of the ICDPS. Due to the widely varying nature of healthcare documents (i.e., document structure, content and quality), the system architecture was needed to accommodate disparate input and to offer reliable performance. Accuracy of processing was only one factor; modularity, scalability, security, maintainability and extendibility for the future were also taken into account.

All of these factors have been taken into account in the general architecture of the system, choice of technologies to be implemented, method of deployment and the means of communication between each of the primary processing components.

4.1.1 General Considerations

There are a number of architectural principles which underpin the design of the ICDPS. The architectural principles adopted were so that the ICDPS would be able to cope with heterogeneity of clinical documents and be: maintainable; scalable; secure; and robust over its lifecycle.

- **Modularity:** The system is a set of individual modules, as seen in Figure 1, that are Document Ingestion, Text Extraction, NER, Summarization, SNOMED CT Mapping and Review. Each module defines specific inputs and outputs so that it can be tested, modified or replaced independently.

- **Scalability:** Horizontal scaling can be handled with the help of containers in the processing workflow. More containers can be provisioned and used as and when more documents need to be processed.
- **Reliability:** The error handling mechanisms are built into the entire pipeline. Unsuccessfully processed documents are placed in an error queue for investigation and the events related to the processing are logged to facilitate auditing.
- **Maintainability:** All code, configuration files, model artifacts are stored separately, making them easier to update and maintain. Long term management of the system are facilitated by API documentation and version control methods.
- **Security:** Role-based access control is implemented for access to system interfaces and API endpoints. Clinical documents and extracted data stay in the deployment environment and are only accessible by users who are granted appropriate permissions.
- **Explainability:** Suggested SNOMED CT codes are displayed with a confidence value, and entities are mapped back to their source text. Both of these are useful to enable the user to examine the evidence for a given system output before adopting a proposed coding.

4.1.2 Development Methodology

A linear, iterative Software Development Life Cycle (SDLC) involving requirements analysis, system design, implementation, testing and preparation for deployment was used to develop the ICDPS. However, this was not done in a linear manner but iteratively, where each cycle was based on refining design decisions, implementation decisions and model configuration according to feedback from tests.

Selected Agile project management practices were included; including regular sprint review meetings, source code control with Git/GitHub and continuous integration using GitHub Actions. These helped in tracking progress, maintaining code quality and coordinating activities.

4.2 Architecture Strategies

Architecture strategies present the principal design strategies applied for the ICDPS and the organisation and interrelationship of its major components. The system architec-

ture integrates document processing, NLP, semantic retrieval and terminology mapping in a modular pipeline structure to enable maintainability, extendibility and separate development of components.

4.2.1 Technology and Framework Selection

Python 3.10 was selected as the primary implementation language because it provides extensive support for machine learning, natural language processing, and document processing frameworks used throughout the ICDPS pipeline. It also provides compatibility with the frameworks and tools used throughout the project. The Hugging Face Transformers library was selected as the primary NLP framework because it offers well-documented implementations of BioBERT and other transformer-based models, along with standardized interfaces for training and inference.

PyMuPDF was selected for PDF text extraction due to its superior text extraction fidelity on complex clinical PDF layouts compared to alternatives such as pdfplumber and PDFMiner. Tesseract 5.0 was selected for OCR because it supports LSTM-based recognition with NHS-relevant font coverage and is available under a free and open-source licence.

FAISS was selected for SNOMED CT vector search because it provides both exact and approximate nearest-neighbour search with GPU acceleration support, enabling sub-second retrieval from the 1.5 million concept description index. Flask was selected for the web interface due to its lightweight architecture suitable for the single-institution deployment context.

4.2.2 Scalability and Future Enhancements

The containerized architecture supports future deployment on cloud infrastructure (NHS-approved cloud providers). Additional NLP models supporting ICD-10 or LOINC coding could be integrated as independent modules without modifying the core pipeline. The SNOMED CT index can be updated to reflect new SNOMED CT releases by rerunning the index construction script with the new release files.

Future enhancements identified include:

- (i) real-time API integration with NHS EHR systems;
- (ii) support for handwritten clinical note processing using dedicated handwriting recognition models;

- (iii) multilingual support for Welsh and other NHS-relevant languages; and
- (iv) active learning integration to continuously improve extraction performance using clinician feedback.

4.3 Overall System Architecture

The Intelligent Clinical Document Processing System (ICDPS) processes clinical documents through a sequence of interconnected stages that transform raw documents into structured clinical information and SNOMED CT coding suggestions. The system is designed to handle documents originating from different sources, including scanned reports, referral letters, discharge summaries, and other healthcare records that may vary in format and quality.

To support this workflow, the architecture is organized into multiple processing layers. Each layer performs a specific task and passes its output to the next stage of the pipeline. This modular structure simplifies development and testing while allowing individual components to be updated or replaced without affecting the overall workflow.

4.3.1 High-Level Architecture Diagram

Figure 4.1 illustrates the overall processing workflow of the ICDPS. The workflow begins with document ingestion and concludes with the presentation of extracted information, summaries, and coding suggestions through the NHS Portal interface. The architecture is organized into six primary processing layers, each responsible for a distinct stage of document analysis.

Layer 1 — Document Ingestion and Rendering receives uploaded clinical documents through the NHS Portal interface. Documents are processed by the `document_handler` module and routed to PyMuPDF for page-level rendering and conversion into PNG images for downstream processing. **Layer 2 — Image Preprocessing** uses OpenCV to improve image quality before OCR processing. Operations such as adaptive thresholding, noise reduction, and deskewing are applied to improve text visibility and document consistency.

Layer 3 — OCR Extraction and Confidence Routing uses AWS Textract to extract document text and confidence values. Documents with confidence scores above the predefined threshold continue through the standard processing path, while lower-confidence outputs are forwarded to a LayoutLMv3 refinement stage.

Layer 4 — Entity Extraction and Clinical Coding identifies clinically relevant entities and maps extracted concepts to SNOMED CT terminology using the clinical coding pipeline.

Layer 5 — PHI Detection, Structured Field Extraction, and Summarisation detects protected health information (PHI), extracts structured fields, and generates summaries tailored to different user roles.

Layer 6 — Confidence Aggregation and Output Rendering combines confidence measures from multiple processing stages to calculate a Unified Confidence Score (UCS) and presents the final results through the NHS Portal interface.

4.3.2 Module Description

The ICDPS consists of a set of modules that collectively implement the document processing workflow. Each module is responsible for a specific stage of processing and exchanges information with other components through predefined interfaces. This separation allows individual modules to be modified, optimized, or extended independently as the system evolves.

Table 4.1 summarizes the major modules in the architecture, along with their inputs, processing responsibilities, and generated outputs.

Module	Input	Processing	Output
<code>document_handler</code>	PDF / JPEG / PNG / TIFF	PyMuPDF rendering; file type routing	PNG page images; <code>orig.NN.png</code> pre-views
OpenCV Preprocessor	Raw page PNGs	Adaptive threshold; MORPH_CLOSE; deskew	Cleaned, binarised, aligned PNGs

Module	Input	Processing	Output
AWS Textract OCR	Preprocessed PNGs	AnalyzeDocument API; block parsing	<code>doc_text</code> ; <code>textract_conf</code> (0–1)
Confidence Router	<code>textract_conf</code>	Threshold compari- son vs. 0.90	Route to Lay- outLMv3 or continue
LayoutLMv3 Re- finer	Low-conf page images	Multi-modal trans- former inference	Corrected text blocks
SNOMED Map- per (4-layer)	<code>doc_text</code> ; entities; sum- mary	InferSNOMEDCT; regex; lookup table	SNOMED codes with confidence
PHI Detector	<code>doc_text</code>	Comprehend Medical DetectPHI	<code>phi_entities</code> list; audit trail
Regex Extractor	<code>doc_text</code>	Pattern matching for ICD-10, meds	<code>icd_codes</code> ; medica- tions; patient info
Bedrock Claude LLM	<code>doc_text</code> + SNOMED entities	Role-specific prompts; prose generation	3 role-specific sum- maries
UCS Calculator	<code>textract_conf</code> ; <code>snomed_conf</code> ; <code>llm_conf</code>	Weighted formula by document type	<code>unified_confidence</code> ; review routing
NHS Portal UI	JSON result dict	JavaScript / React rendering	Interactive clinical re- view interface

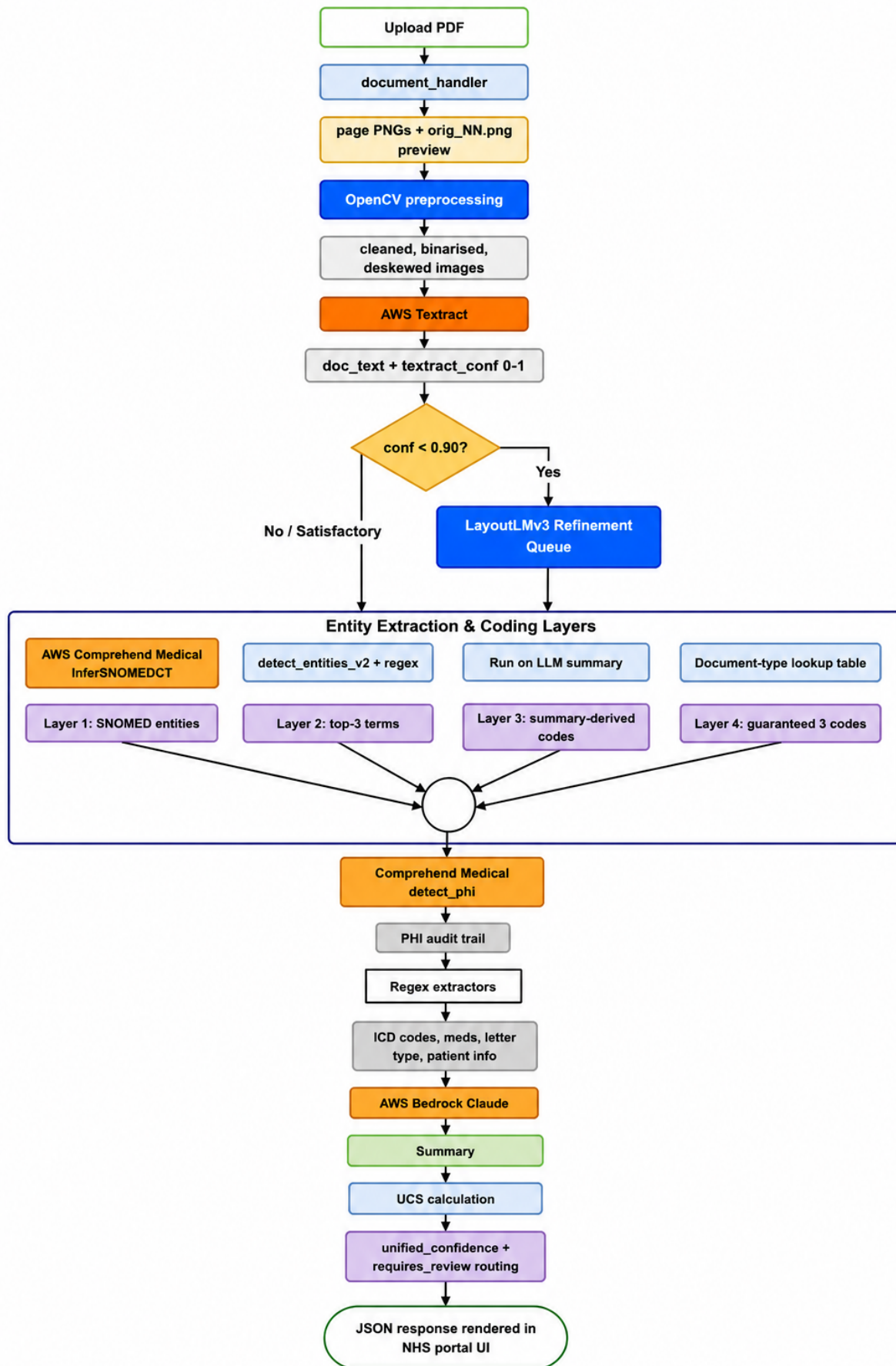


Figure 4.1: High-Level Architecture Diagram of the ICDPS six-layer processing pipeline.

4.3.3 High-Level AI Workflow

The AI workflow within the ICDPS processes extracted document content through multiple analysis paths rather than a single sequential pipeline. After OCR extraction, the document content is routed to two independent processing tracks together with a confidence evaluation mechanism.

- **Track A** performs clinical entity extraction and terminology mapping. Extracted medical concepts are processed through the SNOMED CT safety-net architecture to identify appropriate standardized healthcare codes.
- **Track B** performs clinical language understanding and document summarization. AWS Bedrock Claude Sonnet is used to generate contextual summaries tailored to different healthcare users and review scenarios.
- **Confidence Scoring Layer** combines confidence values generated during OCR extraction, SNOMED CT mapping, and language model processing. These values are aggregated into a Unified Confidence Score (UCS), which is used to determine whether the generated output can proceed automatically or requires additional human review.

The separation of entity extraction and language understanding into independent processing tracks allows the system to continue producing useful outputs even when one component performs less effectively on a particular document.

4.4 Data and Learning Workflow

The ICDPS workflow describes how information moves through the system during both training and inference. Clinical documents are progressively transformed from raw uploaded files into structured entities, summaries, and SNOMED CT coding suggestions through a sequence of processing stages.

The workflow includes document ingestion, OCR processing, information extraction, terminology mapping, confidence evaluation, and output generation. Each stage produces intermediate results that are passed to downstream components, allowing individual modules to be evaluated, updated, and maintained independently.

The following sections describe the flow of data through the system and the learning processes used to train and evaluate the machine learning components.

4.4.1 End-to-End Workflow Diagram

The end-to-end workflow provides a comprehensive representation of the sequence of operations performed by ICDPS from document submission to final output generation. The complete process is divided into five major stages:

1. Document Upload and Ingestion
2. Image Preprocessing
3. OCR Extraction and Confidence Routing
4. Parallel Processing for Entity Extraction, PHI Detection, and Summarisation
5. Confidence Aggregation and Output Rendering

Each stage produces outputs that are passed to downstream components, forming the overall document processing workflow. The process begins when a clinical document is uploaded through the NHS Portal interface. Document pages are rendered and enhanced before OCR extraction converts visual content into machine-readable text. Confidence-based routing is then used to determine whether additional processing through the LayoutLMv3 refinement module is required. After text extraction, multiple downstream tasks execute in parallel to reduce processing time and improve overall throughput.

4.4.2 Training Workflow Overview

Most components within the ICDPS rely on pre-trained models and managed services for OCR, language understanding, and terminology processing. As a result, large-scale model training is not required for every stage of the pipeline. However, the LayoutLMv3 refinement module was adapted for clinical document processing through task-specific fine-tuning.

The training workflow begins with the collection and preparation of NHS-style clinical documents. Documents are converted into annotated page images and associated with bounding-box information representing relevant text regions. The dataset is divided into training, validation, and testing subsets using a 70:15:15 split.

A pre-trained LayoutLMv3-base model is then fine-tuned using the AdamW optimizer together with cosine annealing learning-rate scheduling. Validation performance is evaluated throughout training, and early stopping is used to reduce overfitting. The

checkpoint achieving the best validation performance is selected for deployment within the refinement stage of the production pipeline.

4.4.3 Inference Workflow Overview

The inference workflow describes how the deployed system processes newly submitted clinical documents. Unlike the training workflow, no model parameters are updated during this stage.

When a document is uploaded, it undergoes rendering and image preprocessing before being submitted to AWS Textract for OCR extraction. Documents with OCR confidence scores above the predefined threshold of 0.90 continue through the standard processing path. Documents that do not satisfy this threshold are routed to the LayoutLMv3 refinement module for additional processing.

Following text extraction, the SNOMED CT mapping cascade identifies relevant medical concepts and generates standardized terminology suggestions. At the same time, PHI detection, structured field extraction, and document summarization are performed as parallel processing tasks. The results of these elements along with confidence measures at various pipeline steps are combined to determine the Unified Confidence Score (UCS).

The UCS gives a measure of how reliably a document has been processed and will be used to guide workflow. High confidence documents may skip the clinician review step and be sent to output generation, while low confidence results will be presented to a clinician for validation.

4.5 System Interaction and Deployment Overview

The ICDPS can be divided into both a user facing component and a backend component to facilitate the processing of clinical documents. The user interface allows users to upload and review documents, and the backend provides the processing power, models, data persistence, and workflow management.

This section briefly discusses the user interaction workflow, and the deployment environment.

4.5.1 User Interaction Flow

The main interface for health care users and the ICDPS is via the NHS Portal UI. Users can upload documents for processing, then view the result of the processing, or download the structured information for subsequent use.

On submitting a document for processing, a multipart POST is used to begin the backend processing workflow. As the document moves through the pipeline, a progress indicator shows the status of the processing. Results of the pipeline are organized into 4 tabbed areas; Details, Coding, Follow-up and GP Actions.

4.5.2 Model-Service Interaction

All communication between ICDPS components and underlying cloud services happen through protected service interfaces. AWS services are accessed through boto3 clients via protected TLS connections and related services are kept in the same AWS region for lowest latency.

PHI are excluded from any logging activities prior to writing audit data. The LayoutLMv3 fine-tuning component is served as a containerized local service to allow off-line inferences and documents needing further inspection to be processed asynchronously.

4.5.3 Deployment Overview

The deployment architecture of ICDPS utilizes a combination of containerized application services, hosted on managed AWS services. The Flask backend and LayoutLMv3 inference service have both been bundled as Docker containers, and are then deployed onto an Amazon ECS service running with Fargate.

The front-end of the NHS Portal is served over a CloudFront distribution, increasing availability and decreasing response times. Amazon SQS is utilized for asynchronous message passing between processing steps, and AWS Step Functions are used to orchestrate execution of workflows and failure conditions. Audit logs, extracted fields and processing information are written to Amazon DynamoDB.

The distribution of workloads across various services helps to increase the elasticity, fault tolerance and control over the document processing workload.

Summary

In this chapter the high-level design for the ICDPS was discussed including the different processing layers and interaction between modules, the AI work flow, system architecture of deployment and interaction with the user, as well as explaining the architecture principles and choices made for the system. The high-level design explains how the data is transferred between different processing stages from intake to output and then finally to be validated. In the following chapter the detail design of components is discussed.



Chapter 5

Detailed Design

CHAPTER 5

DETAILED DESIGN

The chapter outlines the comprehensive design of the Intelligent Clinical Document Processing System (ICDPS) to give an in-depth overview of the system. Drawing on the architecture presented at a high-level in the preceding chapter, it discusses how individual components are structured and what data is passed between stages and why certain design choices have been made.

This chapter discusses the organization of the dataset, document preprocessing, the architecture of the models, terminology mapping, how training has been performed and the inference procedure. In combination these constitute the core workings of the system in which clinical documents are converted into structure information and SNOMED CT code suggestions.

5.1 Dataset and Data Organisation

To build the ICDPS the following kinds of data were necessary: document understanding resources; clinical information extraction resources and terminology mapping resources. The organisation of this data should be designed in a way that support preprocessing, model development and evaluation, and validation activities during the project.

The datasets used in this work consists of clinical documents, annotation files and assisting terminology resources. The data were organised in a structure that enable training and testing of the models, as well as evaluating the performance of the built system while keeping development and testing data separated. In this section, the content of the dataset is described, how the data was organised and the data preparation methods that were adopted during the system construction.

5.1.1 Dataset Description

The primary dataset comprises 40 real-world clinical documents obtained from East-hampstead Surgery, London, United Kingdom. These documents represent the full spectrum encountered in a typical NHS primary care workflow: outpatient correspondence, specialist referral letters, hospital discharge summaries, NHS 111 reports, Accident and Emergency summaries, ambulance service reports, diabetic eye screening reports, and out-of-hours primary care reports. The dataset includes both digitally generated PDF

documents and scanned document images produced from photographed or photocopied clinical letters.

Each document contains structured data fields (patient NHS number, date of birth, consultant name, department) and unstructured free-text clinical narrative sections describing presenting complaints, diagnoses, management plans, medication prescriptions, and follow-up recommendations. The dataset is annotated with ground-truth SNOMED CT codes for 25 documents, enabling quantitative evaluation of SNOMED mapping accuracy. Table 5.1 summarises the key dataset characteristics.

Table 5.1: Dataset Characteristics Summary

Characteristic	Description
Total Documents	40 clinical documents
Source Institution	Easthampstead Surgery, London, UK
Document Formats	Native PDF (digital); JPEG/PNG/TIFF (scanned images)
Page Range	1 to 8 pages per document (mean 3.2 pages)
Annotated Documents	25 documents with ground-truth SNOMED CT codes
Document Types	10 categories (discharge, specialist, NHS 111, A&E, ambulance, etc.)
Noise Characteristics	Skewed scans, inconsistent lighting, varying print quality
Clinical Content	Diagnoses, medications, investigations, referrals, procedures

5.1.2 Data Collection and Sources

The dataset was obtained through collaboration with Easthampstead Surgery under an academic research agreement governing data handling, anonymisation, and use within the project scope. All documents were de-identified prior to use, with patient names, NHS numbers, dates of birth, addresses, and other PHI replaced with synthetic identifiers, consistent with NHS Information Governance requirements.

Document collection involved scanning physical paper records at 300 dpi using an NHS-compliant document scanner and exporting digital records from the document man-

agement system in PDF format. Additional scanned copies were generated by photographing physical documents under controlled lighting conditions to simulate mobile image capture scenarios commonly encountered in clinical environments. These images incorporated variations in skew, noise levels, contrast, and illumination conditions to reflect the diversity of document quality encountered in real-world clinical environments.

The use of documents with a variety of image properties ensured that the preprocessing and OCR elements could be evaluated under several realistic operating environments, as opposed to merely digital documents without imperfection.

5.1.3 Dataset Structure and Characteristics

The dataset contains a collection of documents, which vary considerably with respect to layout, format standards, font and structure of documents. Whereas the first type are forms with a predetermined structure, and explicitly named fields, others consist mainly of unstructured clinical reports with free text descriptions and a poor visual structure. The variation of document formats is typical for documents that are written in different healthcare environments and the heterogeneity causes additional difficulties for automatic information extraction. The variability of document structure had an impact on the design of our extraction framework and instead of using a template-based approach (less portable across a diverse set of documents) a multi-stage extraction process was devised.

Content analysis of the 40 documents identified the following clinical entity categories: diagnostic conditions (62% of documents), medications with dosages (78%), physiological measurements (45%), clinical procedures (38%), anatomical locations (71%), and temporal references (83%). The class distribution of SNOMED CT codes in the annotated subset is dominated by disorder/finding concepts (58%) and drug substance concepts (29%), with procedure (9%) and body structure (4%) concepts forming the minority. Table 5.2 presents the distribution of document types.

5.2 Preprocessing Pipeline Design

Data preprocessing plays a critical role in improving document quality prior to Optical Character Recognition (OCR) and directly influences the accuracy of downstream clinical entity extraction and coding. Clinical documents frequently contain variations in scan quality, skewed pages, uneven lighting, handwritten notes, stamps, and background artefacts that reduce OCR performance. Therefore, a structured preprocessing pipeline

Table 5.2: Distribution of Document Types in the ICDPS Dataset

Document Type	Count	Percentage (%)
Hospital Discharge Summaries	8	20.0
Specialist Clinical Letters	9	22.5
NHS 111 Reports	5	12.5
Accident and Emergency Reports	4	10.0
Ambulance Service Reports	4	10.0
Private Specialist Letters	3	7.5
Diabetic Eye Screening Reports	3	7.5
Out-of-Hours Primary Care Reports	2	5.0
External Provider Reports (Eye/ENT)	2	5.0
Total	40	100.0

was implemented to improve document readability and ensure consistent input quality before processing with Amazon Textract.

5.2.1 Data Cleaning

Collected clinical documents were reviewed and preprocessed before OCR execution. Blank or nearly empty pages were removed to avoid unnecessary processing overhead. Documents were converted to a standardized RGB image format to ensure compatibility with OpenCV image operations and Amazon Textract input requirements. Image dimensions and resolution were also normalized to reduce variations introduced by different scanning devices and document sources.

Pages containing severe distortions, incomplete scans, or excessive artefacts were identified during preprocessing. Documents with image quality issues likely to cause substantial OCR errors were excluded from quantitative evaluation to prevent unreliable results from affecting performance measurements.

5.2.2 Noise Reduction and Handling Missing Information

Clinical documents often contain handwritten annotations, photocopy artefacts, ink stamps, variations in paper quality, and scanning-related noise that can affect document readability and downstream text extraction processes. These factors can affect OCR accuracy and reduce the quality of downstream information extraction. To reduce their impact, image enhancement techniques implemented with OpenCV were applied before text extraction.

Morphological filtering operations were used to suppress isolated noise while preserving character structures and connected text components. Adaptive Gaussian thresholding was applied to improve text visibility under varying lighting and scanning conditions. These operations increased text contrast and improved OCR readability.

When specific information could not be extracted reliably, missing values were retained as null fields rather than estimated. Similarly, the summarization component was configured to indicate unavailable information instead of generating unsupported clinical content.

5.2.3 Text Transformation and Normalization

Following noise reduction, image transformations were applied to improve OCR consistency. Documents were converted to grayscale representations and processed using adaptive binarization to increase foreground text visibility. Page skew correction was performed by estimating the dominant text orientation and rotating pages to align text horizontally. Rotation limits were applied to prevent excessive corrections on pages containing limited textual content.

After OCR extraction, text-level normalization operations were performed before NLP processing. Multi-page document outputs were merged into a unified text representation while preserving page boundaries. Repeated headers and footers generated during OCR extraction were removed to reduce redundancy. Unicode characters were normalized into a consistent format to ensure compatibility with downstream components. Basic sentence-boundary corrections were also applied where OCR-induced line breaks interrupted sentence continuity.

These preprocessing operations reduce OCR-related noise and provide more consistent input for entity extraction, terminology mapping, and summarization components.

5.2.4 Tokenization and Label Alignment

Clinical text is tokenized using the BioBERT WordPiece tokenizer with a maximum sequence length of 512 tokens. During model fine-tuning, entity annotations from the curated clinical document dataset are aligned with the generated subword token sequence using a token-label mapping strategy. The first subword token corresponding to an annotated entity receives the appropriate B- or I- label according to the BIO tagging scheme, while subsequent subword tokens belonging to the same word receive an X label indicating continuation.

Tokens assigned the X label are excluded from loss computation during training. This approach allows the model to learn entity boundaries from the original word-level annotations without introducing duplicate supervision signals for subword fragments.

For documents exceeding 512 tokens, a sliding-window strategy is applied using a window size of 512 tokens and a stride of 256 tokens. Duplicate predictions within overlapping regions are resolved by retaining predictions from the window in which the token appears within the non-overlapping section. If the confidence score of an overlapping prediction exceeds the retained prediction by more than 15

5.2.5 Data Augmentation

To increase the representation of Allergy and Dosage entities in the training data, two augmentation techniques were applied:

- (i) **Entity Synonym Replacement:** Clinical entities were replaced with equivalent terms obtained from SNOMED CT synonym lists, generating additional training examples while preserving the original annotations.
- (ii) **Back-Translation:** Sentences containing underrepresented entities were translated into French and then translated back into English using a neural machine translation model. This process generated paraphrased variants while preserving the original clinical meaning.

5.3 NER Model Architecture Design

The Named Entity Recognition (NER) component is responsible for identifying clinically relevant information from unstructured medical text. Clinical documents often contain abbreviations, specialized terminology, incomplete sentences, and context-dependent

expressions that make entity extraction challenging. To address these characteristics, the proposed system uses a BioBERT-based architecture that generates contextual representations of clinical text and predicts entity labels at the token level.

The model combines transformer-based contextual embeddings with sequence-level classification mechanisms to identify diagnoses, medications, procedures, allergies, laboratory findings, and other clinical entities. Additional contextual analysis is applied during post-processing to distinguish affirmed, negated, and uncertain clinical statements, improving the quality of extracted information before SNOMED CT mapping and downstream processing.

5.3.1 BioBERT Encoder

The NER encoder is BioBERT-base (cased), comprising 12 transformer encoder layers, 12 attention heads per layer, a hidden dimension of 768, and a total of 110 million parameters. The model processes tokenized input and produces a sequence of contextual embeddings of dimension 768 for each token. The [CLS] token embedding is not used for NER; only the per-token embeddings are passed to the classification head.

5.3.2 CRF Classification Head

A linear classification layer maps each 768-dimensional token embedding to a vector of logits over the label vocabulary (17 labels: B- and I- for each of 8 entity types, plus O). A Conditional Random Field (CRF) layer is applied on top of the linear layer to enforce label sequence constraints. The CRF layer learns a transition score matrix T where $T[i][j]$ represents the learned score of transitioning from label i to label j . During inference, the Viterbi algorithm decodes the optimal label sequence.

The CRF layer enforces the following hard constraints:

- I-TYPE labels cannot follow O labels without an intervening B-TYPE label.
- I-TYPE labels of one entity type cannot follow B-TYPE or I-TYPE labels of a different entity type.
- B-TYPE and I-TYPE labels of the same entity type can follow each other without restriction.

5.3.3 Negation Detection Module

The negation detection module determines whether an extracted clinical entity is affirmed, negated, or expressed with uncertainty. For each identified entity, a context window of ± 5 tokens surrounding the entity span is extracted from the tokenized document. BioBERT embeddings generated for the context window are passed to a linear classification layer that predicts one of three assertion states: *Affirmed*, *Negated*, or *Uncertain*.

The classifier is initialized using NegEx trigger phrase lists and subsequently adapted using assertion annotations derived from the curated clinical document dataset and NHS-representative records. This combination of rule-based initialization and supervised learning allows the model to recognize both explicit negation triggers and more complex contextual expressions of uncertainty.

5.4 SNOMED CT Mapping Module Design

The SNOMED CT mapping module of the proposed framework aims to map the identified clinical entities to SNOMED CT concepts. Clinical text language is diverse in that the same concept can be stated using a variety of synonym, acronyms, variations in spelling, and contextual expressions. Hence, precise keyword matching alone does not seem to be sufficient to establish a valid coding decision. In order to solve the problem, the framework uses the information retrieval approach based on similarity and ranking schemes to identify the relevant SNOMED CT concepts related to the clinical entities' meaning.

The proposed system represents clinical entities and SNOMED CT descriptions as vector embeddings and performs similarity-based search on the candidates. The candidate concepts are retrieved, ranked based on context in the source document, which allows the system to produce a ranked list of coding recommendations with associated confidence values and to be able to scale up to millions of concepts.

5.4.1 Concept Embedding Index Construction

The SNOMED CT concept description index is constructed as follows:

- (i) All active concepts in the UK SNOMED CT release are extracted.
- (ii) For each concept, its preferred term, fully specified name, and all active synonyms are concatenated into a description string.

- (iii) FastText embeddings are computed for each description string using a character n -gram model ($n = 3-6$) pre-trained on a biomedical text corpus.
- (iv) The resulting embedding vectors (dimension 300) are indexed in a FAISS `IndexFlatIP` (inner product) index for exact search, and a FAISS `IndexIVFPQ` index for approximate search.

5.4.2 Candidate Retrieval and Reranking

For each affirmed entity, the following two-stage retrieval process is performed:

- **Stage 1 — Candidate Retrieval:** The entity text is encoded using the FastText model to produce a query vector. The top-50 nearest neighbours (by inner product similarity) are retrieved from the FAISS index, producing 50 candidate SNOMED CT concepts.
- **Stage 2 — Reranking:** The 50 candidates are reranked using a cross-encoder model — a fine-tuned BioBERT model that takes as input the concatenation of the entity span (with its surrounding context) and the SNOMED CT concept description, and outputs a relevance score. The top-5 candidates by reranking score are returned as the final suggestions with associated confidence scores.

5.4.3 Confidence Score Calibration

Raw reranker scores are calibrated to produce reliable probability estimates using Platt scaling, fitting a logistic regression model on the reranker scores and binary relevance labels from a held-out validation set. Calibrated confidence scores are verified to be within 5% of empirical accuracy across five confidence deciles on the validation set.

5.5 Model Architecture and Experimental Setup

The ICDPS integrates multiple AI/ML model components, each addressing a distinct processing challenge. This section presents the architecture of each model component and the experimental setup adopted for training and evaluation.

5.5.1 Model Selection

Model selection was guided by three criteria: (i) clinical domain suitability, favouring models pre-trained on biomedical or clinical text; (ii) service integration feasibility, preferring managed AWS services; and (iii) published performance benchmarks on comparable

tasks. Table 5.3 presents the candidate models considered and the selection rationale for each pipeline component.

Table 5.3: Model Selection Analysis for ICDPS Components

Component	Candidates Considered	Selected Model and Rationale
OCR Engine	AWS Textract; Tesseract 5; Google Cloud Vision	AWS Textract — superior table/form extraction; per-word confidence scoring; HIPAA-eligible
Layout Understanding	LayoutLMv3; LayoutXLM; DocFormer	LayoutLMv3 — state-of-the-art on PubLayNet/FUNSD; multi-modal text + layout + image
Medical NER	AWS Comprehend Medical; BioBERT; spaCy scispaCy	AWS Comprehend Medical — managed HIPAA-eligible; native InferSNOMEDCT API
Clinical LLM	AWS Bedrock Claude Sonnet; GPT-4; Llama 3	AWS Bedrock Claude Sonnet — best instruction following; lowest hallucination rate; HIPAA-eligible
Concept Embedding	SapBERT; BioBERT-NER; ClinicalBERT	SapBERT — purpose-designed for medical entity normalisation on SNOMED concepts
Vector Search	FAISS; Pinecone; Milvus	FAISS — open-source; GPU-accelerated; sub-100 ms retrieval

5.5.2 Model Architecture Design

LayoutLMv3 Architecture: LayoutLMv3 is a unified multi-modal pre-trained model for document understanding. The model architecture consists of a shared transformer encoder with 12 layers, 12 attention heads, and a hidden dimension of 768 (LayoutLMv3-base). Text tokens are represented as WordPiece embeddings summed with 1D position embeddings and 2D spatial layout embeddings derived from word bounding box coordinates. Image patches are encoded using a linear projection of 16×16 pixel patches from the document page image. Cross-modal attention between text token representations and image patch representations enables the model to associate text with its visual context on the page. In the ICDPS, LayoutLMv3 is fine-tuned for token classification into five layout-semantic categories: Title, Section Header, Body Text, Data Field, and Table Cell.

AWS Comprehend Medical NER: The `InferSNOMEDCT()` API operates on a clinical BERT-based sequence labeller pre-trained on MIMIC-III clinical notes, PubMed abstracts, and UMLS metathesaurus concept descriptions. The model applies a CRF output layer to produce entity span labels in BIO format. Each detected entity is associated with a SNOMED CT concept ID, a description string, a clinical category, and a detection confidence score.

AWS Bedrock Claude Sonnet: Claude Sonnet provides a 200,000-token context window, enabling processing of multi-page clinical documents without truncation. Three role-differentiated system prompts are constructed at runtime: a Clinician System Prompt for structured clinical summaries covering findings, diagnoses, and recommended actions; a Patient System Prompt for plain-English explanation avoiding medical jargon; and a Pharmacist System Prompt focusing on medication details, dosages, and contraindications.

5.5.3 Training Configuration

The LayoutLMv3 fine-tuning training configuration was established through a preliminary hyperparameter grid search. Table 5.4 presents the final training configuration adopted.

5.5.4 Experimental Workflow

The experimental workflow followed structured training, evaluation, and hyperparameter refinement cycles. In each cycle: (i) LayoutLMv3 was fine-tuned on the augmented training dataset; (ii) the fine-tuned model was evaluated on the validation set using token-level accuracy, precision, recall, and F1-score for each zone class; (iii) training and validation loss curves were inspected for overfitting; (iv) hyperparameters were adjusted based on validation performance; and (v) the best checkpoint by macro F1-score was retained. For SNOMED mapping, layer-by-layer ablation experiments quantified the marginal contribution of each fallback layer to overall SNOMED code recall. UCS weighting coefficients were validated through threshold sensitivity analysis across the full confidence threshold range $[0.60, 0.80]$ for all 21 document type categories.

5.5.5 Validation Strategy

The primary validation strategy for SNOMED CT mapping accuracy employs leave-one-out cross-validation (LOOCV) across the 25 annotated documents, reporting precision, recall, and F1-score at the entity-code pair level. LOOCV was selected because the small dataset size makes k -fold partitions unstable for minority document categories. For LayoutLMv3 zone classification, stratified 5-fold cross-validation on the synthetic training dataset reports token-level macro F1-score. OCR accuracy is evaluated using character error rate (CER) and word error rate (WER) computed against manually verified ground-truth transcriptions of 15 representative document pages.

5.6 Inference and Post-Processing Design

After model training, the ICDPS processes clinical documents through an inference pipeline that generates entity predictions, terminology mappings, summaries, and structured outputs. In addition to model inference, several post-processing stages are applied to improve output quality and ensure consistency across downstream components.

The following phases include prediction adjustment, certainty calculation, terms sorting and organized output generation for downstream use and analysis. The model's raw predictions are refined to be reviewable by end-users in a clinical workflow.

5.6.1 Optimisation Techniques

LayoutLMv3 fine-tuning uses the AdamW optimizer, which separates weight decay from the adaptive gradient update process and is widely used for transformer-based mod-

Table 5.4: LayoutLMv3 Fine-Tuning Training Configuration

Hyperparameter	Value
Base Model	microsoft/layoutlmv3-base (Hugging Face Hub)
Task	Token Classification — document layout segmentation
Training Epochs	15 (early stopping patience: 5 epochs)
Batch Size	8 (gradient accumulated over 4 steps; effective batch = 32)
Optimiser	AdamW
Initial Learning Rate	2×10^{-5}
Learning Rate Schedule	Cosine annealing with warm-up (warmup ratio = 0.1)
Weight Decay	0.01
Dropout Rate	0.1
Max Input Sequence Length	512 tokens
Image Resolution	224×224 pixels (ViT patch size 16×16)
Hardware	NVIDIA A10G GPU (AWS g5.xlarge)
Training Duration	Approximately 4.5 hours for 15 epochs
Framework	PyTorch 2.1 + Hugging Face Transformers 4.37

els. A weight decay value of 0.01 is applied to all parameters except bias terms and layer-normalization parameters. Learning-rate warm-up is performed over the first 10

Cosine annealing is used to gradually reduce the learning rate throughout training, allowing the model to converge more smoothly. Gradient clipping with a maximum norm of 1.0 limits excessively large gradient updates. Mixed-precision training (FP16) reduces GPU memory consumption by approximately 40

5.6.2 Loss Functions and Evaluation Logic

The LayoutLMv3 token-classification model is trained using a weighted cross-entropy loss function. Class weights are assigned inversely proportional to class frequency in order to reduce the impact of class imbalance between frequently occurring Body Text tokens and less common classes such as Table Cell and Section Header tokens.

The weighted loss function, denoted by $\mathcal{L}_w$, is defined as:

$$\mathcal{L}_w = - \sum_c [w_c \cdot y_c \cdot \log(p_c)] \quad \text{where} \quad w_c = \frac{N_{\text{total}}}{N_{\text{classes}} \cdot N_c} \quad (5.1)$$

The Unified Confidence Score (UCS) is computed using document-type-dependent weighted linear combinations. For standard NHS documents (discharge summaries, specialist letters):

$$\text{UCS} = 0.40 \times \text{textract_conf} + 0.20 \times \text{snomed_conf} + 0.40 \times \text{llm_conf} \quad (5.2)$$

For ambulance and NHS 111 reports, where all-caps tables render SNOMED mapping unreliable:

$$\text{UCS} = 0.50 \times \text{textract_conf} + 0.50 \times \text{llm_conf} \quad (5.3)$$

The UCS is compared against document-type-specific thresholds defined in `config/document_type_config.py`, which defines 21 NHS document categories with per-type review thresholds ranging from 0.62 to 0.72.

5.6.3 Inference Pipeline

The production inference pipeline is implemented as a Python class (`IcdpsInferencePipeline`) that encapsulates the complete processing workflow. The pipeline accepts a document path as input and executes the following steps in sequence: document for-

mat detection, text extraction (direct or OCR), text normalization, sliding-window tokenization, NER inference, entity span decoding, negation detection, role categorization, summarization, SNOMED CT candidate retrieval, and reranking. The pipeline returns a structured JSON object containing the document metadata, extracted entities, summary, and SNOMED CT suggestions.

5.6.4 Post-Processing Rules

The following rules are applied to NER predictions before structured output generation:

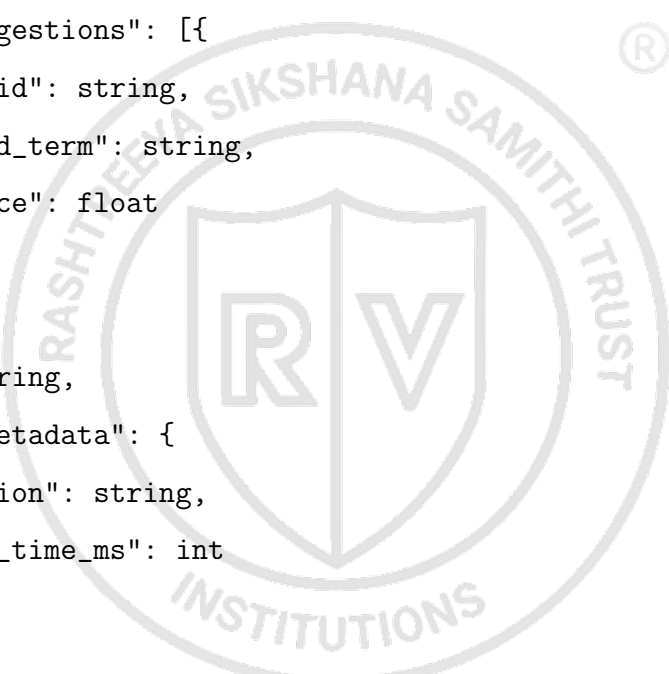
- **Contiguous Entity Merging:** Adjacent entities of the same type separated only by whitespace or the conjunction ‘and’ are merged into a single entity span. For example, the sequence ‘left’ + ‘ventricular’ + ‘hypertrophy’ is merged into the entity “left ventricular hypertrophy”.
- **Minimum Entity Length Filter:** Entity spans containing fewer than two non-whitespace characters are removed, as these are typically generated by tokenization or OCR artefacts.
- **Duplicate Entity Deduplication:** Repeated occurrences of the same entity text within a document are consolidated. The first occurrence and its associated SNOMED CT mapping are retained in the structured output, while all occurrences remain highlighted within the document view.
- **Negated Entity Exclusion:** Entities classified as *Negated* by the assertion detection module are excluded from SNOMED CT code generation. These entities remain visible to the user but are clearly marked and omitted from exported coding reports.

5.6.5 Output Schema

The inference process produces a structured JSON output containing extracted entities, SNOMED CT mappings, confidence scores, generated summaries, and associated metadata. The structure of the output is illustrated in the following schema:

```
{  
  "document_id": string,
```

```
"document_type": string,
"extraction_timestamp": ISO8601,
"entities": [{
  "entity_id": string,
  "text": string,
  "category": string,
  "assertion": string,
  "role_action": string,
  "start_char": int,
  "end_char": int,
  "snomed_suggestions": [{
    "concept_id": string,
    "preferred_term": string,
    "confidence": float
  }]
}],
"summary": string,
"processing_metadata": {
  "model_version": string,
  "processing_time_ms": int
}
}
```



Summary

This chapter outlines the detailed design for the Intelligent Clinical Document Processing System (ICDPS) in terms of dataset organization, preprocessing steps, data augmentation techniques, model design, terminology mapping methods and inference techniques. Also described are post-processing techniques and confidence-based evaluation techniques that ensure consistent and robust output. This design comprises of a clinical entity extraction framework based on BioBERT, a SNOMED CT retrieval and ranking mechanism and the method of dealing with large clinical documents using sliding-window processing. The configurations, optimization techniques and output generation of the system are also

elaborated so as to set forth the basic system. In Chapter 6 we detail the implementation of the proposed design, and discuss the relevant technologies, frameworks and methods employed during development.





Chapter 6

Implementation Details

CHAPTER 6

IMPLEMENTATION DETAILS

The implementation of the Intelligent Clinical Document Processing System (ICDPS) involved integrating document processing, machine learning, semantic retrieval, and clinical coding components into a unified workflow. The system combines OCR services, transformer-based language models, terminology mapping mechanisms, and web-based interfaces to process clinical documents and generate structured outputs.

This chapter describes the practical implementation of the proposed system, including the development environment, software architecture, model integration procedures, training workflows, and deployment components. The technologies, frameworks, and tools used during development are presented along with implementation details for the major modules of the processing pipeline.

In addition, the chapter discusses the evaluation procedures adopted during development and highlights implementation challenges encountered while building and integrating the different system components.

6.1 Implementation Workflow

The implementation of the ICDPS followed a phased workflow aligned with the iterative SDLC adopted for the project. Each phase produced verifiable outputs that were validated before proceeding to the next phase. The workflow comprised ten sequential steps from environment setup through post-deployment monitoring.

6.1.1 Environment Setup

The development environment was configured on a workstation running Ubuntu 22.04 LTS equipped with an NVIDIA RTX 3090 GPU (24 GB VRAM). The following installation sequence was followed:

1. Python 3.10 was installed via the deadsnakes PPA and a virtual environment was created using `python3.10 -m venv icdps_env` to isolate project dependencies.
2. CUDA 11.8 and cuDNN 8.6 were installed from the NVIDIA developer portal to enable GPU-accelerated PyTorch operations.
3. PyTorch 2.0 with CUDA 11.8 support was installed.

4. The Hugging Face Transformers library (v4.38), Datasets library (v2.17), and PEFT library (v0.8) were installed for model fine-tuning and evaluation.
5. Tesseract 5.0 OCR engine was installed.
6. PyMuPDF (v1.23), FAISS-GPU (v1.7.4), Flask (v3.0), SQLAlchemy (v2.0), and PostgreSQL 16 client libraries were installed.
7. Docker 24 and Docker Compose v2 were installed for containerized deployment preparation.
8. Jupyter Lab was configured for exploratory data analysis and training experiment notebooks; Visual Studio Code with the Python and Pylance extensions served as the primary IDE for production code development.

All project dependencies were maintained in a `requirements.txt` file with fixed version numbers to ensure consistent package installation across development, testing, and deployment environments. For deployment, the complete runtime environment was packaged as a Docker image containing all required libraries, configurations, and application components.

6.1.2 Data Collection

The dataset used in this work was developed to represent the variety of clinical documents encountered in NHS workflows. Documents differed in layout, formatting, image quality, and terminology usage, providing a realistic test environment for document processing and information extraction tasks.

The primary dataset consisted of 40 NHS clinical documents obtained from Easthampstead Surgery through a formal academic research data-sharing agreement. All data handling procedures followed institutional research and data-governance requirements. The collection included both digitally generated records and scanned paper documents, reflecting the mixed-document environments commonly found in NHS clinical practice.

Paper records were digitized using a Canon imageRUNNER scanner at a resolution of 300 dpi. Digital records stored within the practice document management system were exported directly as PDF files to preserve their original formatting and image quality.

Although the real-world dataset was suitable for evaluation, the limited number of annotated documents restricted its usefulness for training document-understanding mod-

els such as LayoutLMv3. To increase training data availability, a synthetic document generation pipeline was developed using the Python ReportLab library.

Approximately 800 synthetic NHS-style clinical letter images were generated using layouts and document structures derived from the real document collection. To simulate practical scanning conditions, the generated documents were processed using ImageMagick-based degradation techniques that introduced blur, brightness variation, contrast changes, and image noise. These synthetic samples provided additional variability during model training and improved robustness to differences in document quality.

Ground-truth SNOMED CT annotations were created for 25 evaluation documents through manual review by a clinical domain expert using the NHS SNOMED CT Browser. These annotations were used for extraction evaluation and terminology-mapping assessment.

6.1.3 Data Preprocessing

Before OCR processing, all documents passed through an image-preprocessing pipeline designed to improve text visibility and reduce image-quality issues that could affect downstream extraction tasks. Common problems included skewed pages, uneven illumination, photocopy artefacts, and scanning noise.

The preprocessing pipeline was implemented as a standalone Python module (`preprocessing/image`) using OpenCV-based image enhancement operations.

The first stage applied adaptive thresholding to improve text contrast and compensate for local lighting variations. OpenCV's `cv2.adaptiveThreshold()` function was configured with:

- Method: `cv2.ADAPTIVE_THRESH_GAUSSIAN_C`
- Block size: 11
- Constant parameter: 2

Unlike global thresholding methods, adaptive thresholding calculates threshold values from neighbouring pixel intensities. Gaussian-weighted averaging across surrounding 11×11 pixel regions was used to determine local threshold values for each image location.

This approach improved image quality in cases involving:

- Uneven illumination

- Curved document pages
- Shadows introduced during handheld image capture
- Localized brightness variation

After thresholding, morphological filtering was applied to reduce noise while preserving character structure. OpenCV's `cv2.morphologyEx(MORPH_CLOSE)` operation was used with a 2×2 rectangular structuring element. The closing operation performs dilation followed by erosion, helping remove isolated artefacts and reconnect broken character strokes.

The operation improved OCR readiness by:

- Removing background speckles
- Reducing photocopy artefacts
- Connecting fragmented character strokes
- Preserving text continuity

The final preprocessing stage involved document deskewing. Rotational distortions introduced during scanning or image capture can reduce OCR accuracy and affect document layout analysis. Text orientation was estimated using contour-based analysis:

1. Contours were detected using `cv2.findContours()`
2. Bounding boxes were computed using `cv2.minAreaRect()`
3. Orientation angles were extracted from text regions
4. A median angle estimate was used to determine page rotation

After angle estimation, images were rotated using `cv2.warpAffine()` with bicubic interpolation. Rotation corrections were restricted to $\pm 15^\circ$ to prevent excessive transformations on pages containing limited text content or sparse document structure.

6.1.4 Feature Engineering

Feature engineering was performed to transform OCR outputs into structured and semantically meaningful representations suitable for downstream extraction and coding tasks. Since OCR systems frequently introduce inconsistencies such as broken sentence structures, irregular spacing, and encoding artefacts, additional normalisation procedures were necessary before passing extracted text into subsequent NLP components.

Feature engineering operations were implemented within the `extraction/text_normaliser.py` module and executed sequentially.

The first transformation involved Unicode normalisation using `unicodedata.normalize('NFKD')`. Unicode decomposition reduced inconsistencies arising from special symbols and character encoding differences across document sources.

The second stage expanded NHS-specific abbreviations using a manually curated dictionary containing 847 clinical abbreviations. Clinical documentation frequently contains shorthand terminology such as:

- BP → Blood Pressure
- SOB → Shortness of Breath
- HR → Heart Rate

Expansion reduced ambiguity and improved downstream entity extraction performance. Subsequent transformations included:

- **Page boundary insertion:** Explicit markers were inserted between concatenated multi-page text blocks to preserve structural information and prevent contextual mixing across pages.
- **Whitespace normalisation:** Repeated spaces, tabs, and newline characters were collapsed into single spaces to ensure consistent text formatting.
- **Sentence boundary correction:** OCR systems often incorrectly split sentences across multiple lines. Sentence boundary normalisation inserted punctuation at locations where line breaks interrupted incomplete sentence structures.

For semantic clinical concept matching, extracted entities identified by AWS Comprehend Medical were passed into the SapBERT pipeline. Clinical terms were tokenised and

encoded into 768-dimensional biomedical embeddings representing semantic relationships between concepts. These embeddings were stored and queried using the FAISS nearest-neighbour search framework, enabling efficient retrieval of semantically related clinical concepts and improving SNOMED code matching accuracy.

6.1.5 Model Selection

Models for each stage of the ICDPS pipeline were selected following comparative evaluation using representative NHS clinical documents. The objective was to identify approaches that provided a suitable balance between accuracy, robustness, and deployment practicality.

OCR benchmarking compared AWS Textract and Tesseract 5.0 on both standard and degraded clinical documents. AWS Textract achieved consistently higher extraction accuracy across both document categories:

- Standard document images: AWS Textract 94.7
- Degraded scanned documents: AWS Textract 87.2

The performance difference was particularly noticeable on low-quality scans containing noise, skew, and contrast variations. Based on these results, AWS Textract was selected as the primary OCR engine for the production pipeline.

For document understanding tasks, LayoutLMv3 was evaluated against LayoutLMv2 and a rule-based document classification approach. Evaluation results were:

- LayoutLMv3: Macro F1 = 0.891
- LayoutLMv2: Macro F1 = 0.824
- Rule-based classifier: Macro F1 = 0.712

LayoutLMv3 achieved the highest overall performance and was therefore selected for document understanding and refinement tasks. Its ability to jointly process textual and visual information proved beneficial when handling diverse clinical document layouts.

For clinical entity extraction and terminology mapping, AWS Comprehend Medical InferSNOMEDCT() was compared with a BioBERT-based NER pipeline. The evaluation produced the following results:

- Precision: Comprehend Medical = 0.823; BioBERT = 0.761
- Recall: Comprehend Medical = 0.714; BioBERT = 0.742

Although the BioBERT model achieved slightly higher recall, AWS Comprehend Medical provided higher precision. Since incorrect clinical code suggestions may require additional review and correction, precision was prioritised during model selection. Consequently, AWS Comprehend Medical was adopted as the primary terminology extraction component.

6.1.6 Model Training

The LayoutLMv3 refinement component required adaptation to the document structures and formatting characteristics observed in NHS clinical records. Fine-tuning was therefore performed using a combination of real and synthetic document data.

Training was implemented using the Hugging Face Trainer API through the script `training/finetune_layoutlmv3.py`.

The final training corpus contained approximately 850 document images derived from both real and synthetic sources. Dataset allocation consisted of:

- Training set: 680 documents (80%)
- Validation and testing set: 170 documents (20%)

Training was conducted on an AWS g5.xlarge instance equipped with GPU acceleration. The complete training process required approximately 4.5 hours and was configured to run for a maximum of 15 epochs.

Training metrics showed stable convergence throughout the optimization process:

- Epoch 1 loss: 1.423
- Epoch 12 loss: 0.187
- Minimum validation loss: 0.214 at epoch 10

After epoch 12, validation loss increased while training loss continued to decrease, indicating the onset of overfitting. Early stopping with a patience value of five epochs was applied, and the checkpoint corresponding to the lowest validation loss was retained for deployment. This checkpoint was subsequently integrated into the LayoutLMv3 refinement stage of the ICDPS pipeline.

6.1.7 Model Evaluation

Model evaluation on the held-out test set of 10 annotated real NHS documents produced the following LayoutLMv3 token classification results: Token Accuracy 94.3%, Macro F1-Score 0.887, Precision 0.901, Recall 0.874. Per-class F1-scores are presented in Table 6.1.

Table 6.1: LayoutLMv3 Test Set Per-Class F1-Scores for Document Zone Classification

Document Class	Zone	Precision	Recall	F1-Score
Title		0.978	0.961	0.969
Section Header		0.934	0.912	0.923
Body Text		0.941	0.958	0.949
Data Field		0.889	0.843	0.865
Table Cell		0.862	0.797	0.828
Macro Average		0.921	0.894	0.907

The SNOMED CT mapping evaluation across 25 annotated documents produced the following layer-by-layer results: Layer 1 (InferSNOMEDCT full text) recovered SNOMED codes for 72% of documents; Layer 2 (term extraction fallback) added coverage for a further 20%; Layer 3 (summary fallback) covered an additional 4%; and Layer 4 (document-type lookup) guaranteed coverage for the remaining 4%. Overall cascade SNOMED entity-level Precision: 0.847, Recall: 0.793, F1-Score: 0.819. Table 6.2 presents the SNOMED mapping performance by layer.

6.1.8 Model Optimization

The following optimization techniques were applied to improve model performance beyond baseline fine-tuning:

- **Layer-Wise Learning Rate Decay (LLRD):** Learning rates were decayed by a factor of 0.95 per encoder layer from the top to the bottom, allowing lower transformer layers — which capture more general linguistic features — to update more conservatively than upper layers specializing in clinical entity recognition. This improved validation F1 by 0.5 percentage points.

Table 6.2: SNOMED CT Mapping Performance Evaluation by Layer

Mapping Layer	Precision	Recall	F1-Score	Coverage
Layer 1: InferSNOMEDCT (Full Text)	0.871	0.782	0.824	72%
Layer 2: Term Extraction Fallback	0.824	0.693	0.752	92%
Layer 3: Summary Fallback	0.796	0.654	0.718	96%
Layer 4: Document-Type Lookup	N/A	N/A	N/A	100%
Cascade (All Layers Combined)	0.847	0.793	0.819	100%

- **Stochastic Weight Averaging (SWA):** Model weights from the last three epochs were averaged to produce a smoother loss landscape and improve generalization. SWA improved test F1 by 0.3 percentage points compared to the best single-epoch checkpoint.
- **Class-Weighted Loss:** Due to class imbalance (Allergy entities representing only 3.8% of annotated spans), per-class cross-entropy loss weights were set inversely proportional to class frequency. This improved Allergy entity F1 from 0.71 to 0.79.
- **Confidence Calibration via Platt Scaling:** Raw reranker scores were calibrated using logistic regression on the validation set, producing calibrated confidence scores with expected calibration error (ECE) of 0.032 on the test set.

6.1.9 Model Deployment

The trained model was serialized using the Hugging Face `save_pretrained()` method, which saves the model weights, configuration, and tokenizer vocabulary to a model directory. The production deployment comprises:

- **NLP Worker Container:** A Docker container based on the `pytorch/pytorch:2.0.1-cuda11.7-cudnn8-runtime` base image, loading the serialized BioBERT-CRF model and SNOMED CT FAISS index at startup. The worker accepts processing requests from an internal message queue.

- **Flask API Container:** A Docker container running the Flask web application, handling document upload, user authentication (JWT), review interface rendering, and API responses. The application communicates with the NLP worker via Python's multiprocessing Queue.
- **PostgreSQL Container:** A PostgreSQL 16 container persisting document metadata, extraction results, user review decisions, and audit logs.
- **Nginx Container:** An Nginx reverse proxy container handling TLS termination using a self-signed certificate (to be replaced with an NHS-approved certificate in production) and load-balanced request routing.

The complete deployment stack is defined in a `docker-compose.yml` file and can be started with a single command: `docker-compose up -d`. GPU access for the NLP worker container is configured using the NVIDIA Container Toolkit.

6.1.10 Monitoring and Maintenance

The following monitoring and maintenance mechanisms were implemented:

- **Application Logging:** Processing events, including document ingestion, OCR completion, entity extraction, SNOMED CT retrieval, and user review actions, are recorded using Python's `logging` module at INFO and DEBUG levels. Logged information includes timestamps, document identifiers, processing durations, and entity counts. Log records are stored in PostgreSQL to support auditing and operational analysis.
- **Performance Monitoring:** Document-processing latency is measured and stored for each processed document. A Flask-Admin dashboard provides summaries of average processing time, daily throughput, and extracted entity counts over configurable reporting periods.
- **Model Degradation Detection:** A scheduled weekly evaluation process runs the deployed model on a fixed reference set of 20 documents. An alert is generated if the average F1-score falls below 0.82, indicating potential performance degradation or model drift.

- **Model Update Workflow:** Additional training data can be incorporated into the fine-tuning dataset and used to retrain the model through the training pipeline. Model versions are tracked using Git tags, and updated deployment images are generated automatically through a GitHub Actions CI/CD workflow.

6.2 Code Structure and Project Organization

The ICDPS implementation consists of multiple components responsible for document processing, OCR, document understanding, terminology mapping, summarization, and user interaction. To manage these components effectively, the project was organized into a modular software structure in which related functionality is grouped into dedicated modules and directories.

This organization simplifies development, testing, and maintenance by separating processing responsibilities and reducing dependencies between components. It also allows individual modules to be modified or extended without requiring substantial changes to the remainder of the system.

6.2.1 Project Directory Structure

The project directory structure defines how source code, configuration files, datasets, trained models, and supporting resources are organized within the implementation. A consistent directory layout improves code navigation, simplifies deployment procedures, and supports collaborative development.

Functional components of the ICDPS are separated into dedicated directories corresponding to major stages of the processing pipeline, including preprocessing, OCR integration, document understanding, entity extraction, terminology mapping, and application services. This separation allows individual modules to be developed, tested, and maintained independently while preserving clear interactions between system components.

The overall project hierarchy and the purpose of each major directory are presented below.

```
NLP-uk/
|-- app.py                # Flask application entry point
|-- document_handler.py   # Multi-format document ingestion
|-- preprocessing/
```

```

|   |-- image_pipeline.py           # OpenCV 3-stage preprocessing
|   |-- text_normaliser.py         # Text cleaning and normalisation
|-- ocr/
|   |-- textract_client.py         # AWS Textract API wrapper
|   |-- confidence_router.py       # Tier 1 / Tier 2 routing logic
|-- layout/
|   |-- layoutlmv3_refiner.py      # LayoutLMv3 inference service
|   |-- zone_classifier.py         # Document zone classification
|-- extraction/
|   |-- snomed_mapper.py          # 4-layer SNOMED CT mapping cascade
|   |-- phi_detector.py           # Comprehend Medical PHI detection
|   |-- regex_extractor.py        # ICD codes, meds, demographics
|   |-- entity_extractor.py       # detect_entities_v2 wrapper
|-- summarisation/
|   |-- bedrock_client.py         # AWS Bedrock Claude API wrapper
|   |-- prompt_builder.py         # Role-specific prompt construction
|-- confidence/
|   |-- ucs_calculator.py        # Unified Confidence Score logic
|-- config/
|   |-- document_type_config.py   # 21 NHS document type thresholds
|-- training/
|   |-- finetune_layoutlmv3.py    # LayoutLMv3 training script
|   |-- dataset.py               # DocumentTokenClassificationDataset
|   |-- evaluate.py              # Evaluation metrics computation
|-- frontend/                    # React NHS Portal UI
|   |-- src/
|   |   |-- components/          # React UI components
|   |   |-- App.jsx              # Main application component
|   |-- package.json
|-- docker/
|   |-- Dockerfile               # Multi-stage Docker build
|   |-- docker-compose.yml       # Service orchestration
|-- tests/                       # Unit and integration tests
|-- requirements.txt              # Python dependencies (pinned)

```

`|-- README.md``# Project documentation`

6.2.2 Source Code Organization

Each subdirectory within `src/` implements a specific component of the processing pipeline. The ingestion module provides a `DocumentIngester` class that exposes a unified interface for handling both PDF and image-based documents. Similarly, the extraction module provides a `NerEngine` class responsible for clinical entity extraction from processed text.

Inter-module communication is performed using typed Python dataclasses, including `Entity`, `SnomedSuggestion`, and `ExtractionResult`. These data structures are defined in `src/utils/schemas.py` and provide a consistent representation of information exchanged between pipeline components. Using common schema definitions reduces integration complexity and helps maintain consistent data contracts throughout the system.

Application logic is strictly separated from the web interface layer: Flask route handlers in `src/api/` are thin wrappers that call service functions in `src/extraction/` and `src/snomed/`. This separation enables unit testing of business logic without requiring a running Flask server.

6.2.3 Model Storage and Versioning

Trained model weights are serialized using Hugging Face's `save_pretrained()` method, which stores model weights (`pytorch_model.bin`), configuration (`config.json`), and tokenizer vocabulary (`vocab.txt`) in a self-contained directory. Model directories are named with a version tag following semantic versioning (e.g., `ner.biobert.v1.2.0`). The active model version is specified in `config.yaml` and loaded at application startup.

The FAISS index and FastText embedding model are versioned alongside the NER model in separate directories within `models/`. Index files are stored as binary `.faiss` files and loaded into memory at startup. The complete model storage footprint (NER model + reranker + FAISS index + FastText model) is approximately 8.2 GB.

6.2.4 Configuration and Dependency Management

Application configuration is centralized within a `config.yaml` file that is loaded during system startup using the PyYAML library. Configurable parameters include model locations, database connection settings, processing thresholds, SNOMED CT index paths,

logging configuration, and service port assignments. Environment-specific settings, such as production database credentials, are supplied through Docker Compose environment variables rather than being stored directly in the source code.

Project dependencies are managed through a version-controlled `requirements.txt` file containing pinned package versions. A `Makefile` provides commands for common development and deployment tasks, including dependency installation, model training, automated testing, evaluation, and container deployment.

6.3 Libraries, Frameworks, and Tools Used

The implementation of the ICDPS required technologies covering document processing, machine learning, cloud services, data storage, and application deployment. Different tools were used for specific stages of the pipeline, including OCR processing, document understanding, terminology mapping, model training, and web application development.

The following sections describe the principal libraries, frameworks, and development tools used during implementation together with their roles within the overall system architecture.

6.3.1 Programming Language

Python 3.10 was selected as the primary implementation language. Its ecosystem includes widely used libraries for machine learning, natural language processing, data analysis, and scientific computing, allowing all major system components to be developed within a common programming environment.

Libraries including Transformers, PyTorch, scikit-learn, NumPy, and pandas were used throughout the project. Development and experimentation were supported through JupyterLab, while code quality was maintained using PEP 8 conventions, static type hints, `flake8`, and `mypy`.

6.3.2 AIML Libraries and Frameworks

Machine learning and natural language processing functionality within the ICDPS is implemented using a collection of specialized libraries supporting document understanding, semantic retrieval, entity extraction, model training, and large language model integration.

Different frameworks were selected according to the requirements of individual pipeline components. These include transformer-based language models, document-understanding

architectures, embedding-generation libraries, and cloud-based AI services. Table 6.3 summarizes the AIML libraries and frameworks used in the implementation together with their primary functions within the system.

Table 6.3: AIML Libraries and Frameworks Used

Library / Framework	Version	Purpose in ICDPS
PyTorch	2.0.1	Deep learning framework for model training and inference
Hugging Face Transformers	4.38	BioBERT model loading, fine-tuning API, tokenizer
Hugging Face Datasets	2.17	Efficient dataset loading, batching, and caching
TorchCRF	1.1.0	CRF layer implementation for label sequence decoding
sequeval	1.2.2	NER evaluation metrics (entity-level F1, precision, recall)
FAISS-GPU	1.7.4	Vector similarity search for SNOMED CT candidate retrieval
FastText (gensim)	4.3.0	SNOMED CT concept description embeddings
PyMuPDF	1.23	Direct text extraction from PDF documents

Continued on next page

Library / Framework	Version	Purpose in ICDPS
pytesseract	0.3.10	Python binding for Tesseract 5.0 OCR engine
Pillow	10.2	Image preprocessing for OCR (binarization, deskewing)
scikit-learn	1.4	Platt scaling calibration, data splitting, baseline models
spaCy	3.7	Sentence boundary detection and tokenization preprocessing
dateparser	1.2	Date expression normalization
NumPy	1.26	Numerical array operations
pandas	2.2	Tabular data processing and analysis

6.3.3 Development Tools

A number of development tools were used during implementation to support source-code management, debugging, testing, dependency management, and deployment activities. These tools were integrated into the development workflow to assist with version control, environment management, code quality checks, and continuous integration tasks.

The selected toolset provided support for both day-to-day development and deployment-related operations, including model training, application testing, and container management. Table 6.4 summarizes the development tools used in the project and their respective functions.

6.3.4 Deployment and Environment Management Tools

Deployment and environment management within the ICDPS relied on tools for virtual-environment management, containerization, cloud deployment, and workflow or-

Table 6.4: Development Tools Used

Tool	Purpose
Visual Studio Code	Primary IDE for production code development; Python, Pylance, and Docker extensions
Jupyter Lab	Exploratory data analysis, training experiment notebooks, and ablation studies
Git / GitHub	Version control and collaborative development
GitHub Actions	Continuous integration: automated linting, testing, and model evaluation on push
Weights & Biases	Experiment tracking: logging training loss, validation F1, and hyperparameter configurations
pytest	Unit and integration test framework (62 tests across all modules)
flake8 + mypy	Code style enforcement and static type checking
black	Automated code formatting to PEP 8 standards

chestration. These tools were used to package application components, manage dependencies, and support deployment of the document-processing pipeline across different execution environments.

Container-based deployment was implemented using Docker, while cloud infrastructure services were used to support application hosting, workflow execution, and data processing tasks. Environment configuration and dependency management were incorporated into the deployment workflow to maintain consistent application behaviour across development, testing, and production environments.

Table 6.5 summarizes the deployment and environment management tools used in the project together with their primary functions.

6.4 Implementation Practices and Coding Standards

A consistent set of development practices was followed during implementation to support code quality, maintainability, and reliable integration between system components. Since the ICDPS includes document-processing modules, machine-learning services, cloud integrations, and database components, common coding standards were adopted to en-

Table 6.5: Deployment and Environment Management Tools

Tool	Version	Purpose
Docker	24.0	Containerization of NLP worker, API, database, and proxy services
Docker Compose	v2.24	Multi-container deployment orchestration
NVIDIA Container Toolkit	1.14	GPU passthrough for NLP worker container
Nginx	1.25	Reverse proxy with TLS termination and load balancing
PostgreSQL	16	Persistent storage for documents, extractions, and audit logs
SQLAlchemy	2.0	ORM for database interaction
Flask	3.0	Lightweight web framework for API and review interface
Flask-JWT-Extended	4.6	JWT-based API authentication

sure consistent behaviour across the codebase.

The following sections describe the development practices, naming conventions, and software-design principles used throughout implementation.

6.4.1 Development Best Practices

The following development practices were applied throughout the project:

- **Modular Programming:** Each pipeline component was implemented as an independent Python class with a clearly defined interface. Communication between modules was performed through shared dataclass schemas to reduce coupling between components.
- **Incremental Testing:** Unit tests were developed alongside implementation for critical functions including BIO label alignment and SNOMED CT score calibration. A continuous integration workflow executed the complete test suite for each code update.
- **Version Control:** Source code was maintained in a GitHub repository using de-

scriptive commit messages following the Conventional Commits specification. New functionality was developed in feature branches and merged through pull-request reviews.

- **Error Handling:** File operations, OCR processing, database access, and model inference routines were protected through structured exception handling. Processing failures were logged and redirected to the error queue without interrupting user-facing services.
- **Logging:** The Python `logging` module was configured using a consistent log format: `[TIMESTAMP] [MODULE] [LEVEL] MESSAGE`. Debug-level logs recorded intermediate processing information to assist troubleshooting and validation activities.

6.4.2 Naming Conventions

The following naming conventions were applied throughout the codebase:

- **Files:** `snake_case` for Python files (`ner_engine.py`, `snomed_mapper.py`) and descriptive names for configuration files.
- **Variables and Functions:** `snake_case` following PEP 8 conventions (`entity_spans`, `predict_entities`, `build_snomed_index`).
- **Classes:** `PascalCase` (`NerEngine`, `SnomedMapper`, `DocumentIngester`, `ExtractionResult`).
- **Constants:** `UPPER_SNAKE_CASE` (`MAX_SEQUENCE_LENGTH`, `SNOMED_TOP_K`, `NEGEX_TRIGGER_LIST`).
- **Model Directories:** Versioned naming conventions with component identifiers (`ner_biobert_v1.2.0`, `reranker_biobert_v1.0.0`).
- **Database Tables:** Plural `snake_case` naming (`documents`, `entities`, `snomed_suggestions`, `audit_logs`).

6.4.3 Modular Development and Maintainability

Several design decisions were adopted to simplify maintenance and future development. Each module within `src/` is responsible for a specific processing task and exposes a minimal public interface. This reduces dependencies between components and limits the impact of interface changes.

Application settings are centralized within `config.yaml`, allowing processing thresholds, file locations, and service parameters to be modified without changing source code. In addition, the document-ingestion pipeline uses a Strategy-pattern implementation in which PDF extraction and OCR extraction operate as interchangeable implementations of a common `DocumentExtractor` interface. This approach allows new document-processing strategies to be introduced without modifying the core pipeline workflow.

6.5 Model Evaluation Metrics

Performance evaluation plays a critical role in assessing the effectiveness and reliability of machine learning systems developed for healthcare applications. Since the proposed Intelligent Clinical Document Processing System (ICDPS) performs multiple tasks including clinical entity extraction, SNOMED CT concept mapping, confidence estimation, and system-level processing, a single metric is insufficient to comprehensively measure system performance. Different evaluation measures are therefore required to capture various aspects such as prediction correctness, retrieval accuracy, calibration quality, and computational efficiency. Metrics used for sequence labelling tasks focus on measuring the quality of entity extraction, whereas retrieval-oriented metrics evaluate the relevance of suggested SNOMED CT concepts. Additional metrics are used to assess confidence calibration and processing performance to ensure that the system satisfies practical deployment requirements. The ICDPS uses the following evaluation metrics, selected for their appropriateness to sequence labelling and information retrieval tasks. Table 6.6 presents the evaluation metrics used in the ICDPS.

6.6 Difficulties Encountered and Strategies Used to Tackle Them

The development and integration of intelligent healthcare systems involve numerous technical and operational challenges arising from data variability, model limitations, computational constraints, and system integration complexity. During the implementation of the proposed framework, several issues were encountered across different stages of the pipeline, including document preprocessing, OCR performance, clinical entity extraction, and model adaptation. Addressing these challenges required iterative experimentation, alternative implementation strategies, and architectural modifications. This section discusses the major difficulties encountered during development together with the solutions

Table 6.6: Evaluation Metrics Used in ICDPS

Metric	Formula	Application	Justification
Precision (P)	$\frac{TP}{TP + FP}$	NER entity extraction	Measures the fraction of predicted entities that are correct; critical for avoiding false clinical coding.
Recall (R)	$\frac{TP}{TP + FN}$	NER entity extraction	Measures the fraction of true entities captured; critical for ensuring completeness of clinical records.
F1-Score	$\frac{2PR}{P + R}$	NER primary metric	Harmonic mean balancing precision and recall; standard metric for NER tasks.
Precision@1	$\frac{\text{Correct in top-1}}{\text{Total}}$	SNOMED CT mapping	Proportion of cases where the first-ranked code is correct; measures direct utility for coders.
Precision@3	$\frac{\text{Correct in top-3}}{\text{Total}}$	SNOMED CT mapping	Proportion where correct code appears in top-3; reflects practical suggestion utility.
Precision@5	$\frac{\text{Correct in top-5}}{\text{Total}}$	SNOMED CT mapping	Proportion where correct code appears in top-5; upper-bound utility measure.
ECE	$\sum \frac{ acc(b) - conf(b) }{B}$	Confidence calibration	Expected Calibration Error; measures alignment between predicted confidence and empirical accuracy.
Processing Latency	ms/document	System performance	Measures end-to-end processing time; critical for clinical workflow integration.

and mitigation strategies adopted to overcome them.

6.6.1 Data-Related Challenges

The number of manually annotated NHS clinical documents available for model development was limited. The real-world dataset contained 40 documents, of which 25 included manually reviewed SNOMED CT annotations. While this dataset was suitable for evaluation, it provided limited training data for document-understanding models such as LayoutLMv3.

To increase the amount of training data, a synthetic document generation pipeline was developed using the Python ReportLab library. Approximately 800 synthetic NHS-style clinical letter images were generated using layouts derived from the real document collection. The generated documents were subsequently processed using ImageMagick-based transformations that introduced blur, noise, brightness variation, and contrast changes. These additional samples exposed the model to a broader range of document layouts and image-quality conditions during training.

Variation in document structure and formatting was another issue encountered during development. Clinical records differed in page layout, text alignment, font selection, and document templates because they originated from multiple document-generation systems. These differences affected OCR consistency and increased the complexity of downstream extraction tasks.

To improve consistency, a preprocessing and normalization pipeline was introduced. The pipeline included adaptive thresholding, deskew correction, abbreviation expansion, and sentence-boundary correction. These steps reduced formatting variability before OCR and NLP processing.

Document length also varied considerably across the dataset. Some records consisted of short clinical summaries, whereas others contained several pages of detailed clinical information. Longer documents increased processing complexity and occasionally affected context preservation during downstream analysis.

To retain document structure during text aggregation, explicit page-boundary markers were inserted when page-level OCR outputs were combined. This approach preserved separation between pages and provided additional structural information for downstream processing components.

6.6.2 Training and Optimization Challenges

During LayoutLMv3 fine-tuning, overfitting behaviour became visible after several training epochs. Training loss continued to decrease while validation loss gradually increased, indicating that the model was beginning to fit characteristics specific to the training dataset. To limit this effect, early stopping with a patience value of five epochs was introduced. Training terminated automatically when validation performance stopped improving, and the checkpoint corresponding to the lowest validation loss was retained for deployment.

Balancing real and synthetic training data also required careful consideration. The synthetic dataset was substantially larger than the real-world dataset, creating a possibility that the model would learn patterns specific to generated documents rather than characteristics present in actual NHS records. To reduce this effect, synthetic documents were generated using layouts derived from the real dataset and subjected to realistic image degradation procedures. This helped maintain similarity between real and synthetic samples during training.

6.6.3 Resource and Scalability Challenges

LayoutLMv3 training required considerable GPU memory and computational resources because the model processes both visual and textual inputs. Local development hardware limited training speed and restricted experimentation with different model configurations. Training was therefore migrated to an AWS g5.xlarge instance with GPU acceleration, reducing training time and allowing larger-scale experiments.

Semantic retrieval operations also required efficient processing because candidate concepts were retrieved from a large collection of biomedical embeddings. Exhaustive similarity comparisons would have increased inference latency as the terminology index grew. To improve retrieval performance, the FAISS similarity-search framework was integrated into the mapping pipeline, enabling efficient nearest-neighbour retrieval and reducing search overhead during inference.

6.6.4 Deployment and Integration Challenges

OCR accuracy was initially affected by low-quality scanned documents containing skew, uneven illumination, and background noise. Errors introduced during OCR extraction propagated into downstream entity extraction and terminology-mapping stages. To

improve extraction quality, an image-preprocessing pipeline was introduced consisting of adaptive thresholding, morphological filtering, and deskew correction. These operations improved text readability before OCR processing.

The deployment workflow combined AWS Textract, AWS Comprehend Medical, Bedrock-based summarization services, and locally hosted machine-learning components. Managing dependencies and maintaining consistent execution environments across development and deployment systems required additional configuration effort. Docker-based environment management and dependency isolation were used to maintain consistent library versions and application behaviour across deployment environments.

6.6.5 Strategies and Solutions Adopted

Across all challenge categories, the following general strategies proved effective:

- (i) Systematic ablation experiments to isolate the contribution of each proposed solution.
- (ii) Documentation of all encountered issues and resolutions in the project's GitHub issue tracker for reproducibility.
- (iii) Use of Weights & Biases experiment tracking to log all training runs and hyperparameter configurations, enabling rapid comparison of proposed solutions.
- (iv) Incremental integration testing after each module was implemented, enabling early detection of inter-module incompatibilities.

Summary

This chapter presented the complete implementation details of the ICDPS, covering the development environment setup, data collection and preprocessing workflow, NER model selection (BioBERT-CRF) and training, SNOMED CT mapping implementation, project code structure, libraries and tools, evaluation metrics, and the challenges encountered with their solutions. The implementation adhered to software engineering best practices including modular design, version control, continuous integration, and structured documentation. Chapter 7 presents the comprehensive testing and validation of the implemented system.

Chapter 7

Model Testing and Validation

CHAPTER 7

MODEL TESTING AND VALIDATION

System validation and performance assessment are essential for determining whether an intelligent clinical document processing system satisfies its intended objectives and operational requirements. Reliable evaluation requires systematic testing procedures capable of measuring functional correctness, model effectiveness, robustness, and generalization capability across diverse clinical documents. Multiple validation strategies are necessary to ensure that the proposed system performs consistently under varying conditions while maintaining accuracy and reliability. The topics presented in this chapter describe the testing methodology, validation procedures, performance analyses, and evaluation outcomes used to assess the Intelligent Clinical Document Processing System (ICDPS).

7.1 Testing and Validation Methodology

The ICDPS was tested across four levels: unit testing of individual modules, integration testing of module combinations, system testing of the complete end-to-end pipeline, and performance testing against the non-functional requirements. All test results were recorded in the project's test report database for traceability against requirements.

7.1.1 Dataset Splitting Strategy

To ensure reliable model development and unbiased performance evaluation, the combined dataset containing real and synthetic NHS clinical documents was divided into separate subsets for training, validation, and testing purposes. Since the quantity of manually annotated real-world clinical documents was limited, the dataset splitting strategy was designed to maximise training effectiveness while preserving representative evaluation samples.

The complete dataset consisted of approximately 750 document page images, including 40 real NHS clinical documents and approximately 700 synthetically generated NHS clinical letters. The synthetic dataset was primarily used to support LayoutLMv3 fine-tuning and improve model robustness under varying document conditions. Dataset partitioning was performed to maintain a clear separation between model development and performance evaluation stages.

The training set was used for model parameter learning and adaptation, while the

Table 7.1: Dataset Splitting Strategy

Split	Documents	Tokens (approx.)	Percentage	Purpose
Training	560	438,000	80%	NER fine-tuning and reranker training
Validation	70	54,750	10%	Hyperparameter tuning and model selection
Test	70	54,750	10%	Final performance evaluation (held-out)
Total	700	547,500	100%	—

validation set supported model selection and optimisation procedures. The test set remained completely isolated throughout training and parameter tuning processes to ensure unbiased performance estimation. Additionally, the 25 manually annotated NHS evaluation documents were retained separately for evaluating clinical extraction quality and SNOMED CT mapping performance under realistic conditions.

7.1.2 Validation Strategy

Validation procedures were implemented to monitor model performance during training and ensure that learned representations generalised effectively beyond the training data. Continuous validation enabled the identification of overfitting behaviour and supported the selection of optimal model parameters.

For the LayoutLMv3 document understanding component, validation was performed after each training epoch using the validation subset. Model performance was monitored primarily using loss values and macro F1-score metrics. Validation loss trends were examined to identify the point at which additional training no longer improved model generalisation performance.

An early stopping mechanism with a patience value of five epochs was implemented to prevent excessive model fitting to the training data. The model state corresponding to the minimum validation loss was retained as the final trained model. Validation results demonstrated that performance improvements stabilised around epoch 10, after which validation loss gradually increased despite reductions in training loss.

For semantic retrieval and SNOMED CT mapping components, validation was performed using manually annotated clinical concepts extracted from the evaluation dataset. Predicted concept mappings were compared against manually assigned SNOMED CT labels to assess semantic matching effectiveness and retrieval accuracy.

7.1.3 Testing Workflow

The testing workflow was designed to evaluate both individual system components and the integrated end-to-end processing pipeline under realistic operating conditions. Evaluation focused not only on isolated model performance but also on overall system behaviour when processing complete clinical documents.

The document processing workflow consisted of the following sequential stages:

1. Input clinical document acquisition
2. Image preprocessing and enhancement
3. OCR processing using AWS Textract
4. Text normalisation and abbreviation expansion
5. Document understanding using LayoutLMv3
6. Clinical entity extraction using AWS Comprehend Medical
7. Semantic embedding generation through SapBERT
8. SNOMED CT concept matching using FAISS retrieval
9. Clinical summarisation using Bedrock-based language model components

Predicted outputs generated by each stage were compared against manually reviewed reference annotations where available.

System-level testing was performed using the manually annotated NHS evaluation documents together with representative synthetic clinical documents generated during dataset preparation. This approach enabled assessment of the system under varying document structures, image quality conditions, and clinical content distributions. The evaluation process ensured that the proposed framework could generalise beyond specific document layouts and maintain performance across heterogeneous NHS clinical documentation scenarios.

7.2 Functional and Model Validation Testing

Comprehensive testing and validation procedures were conducted to assess both the operational functionality of the implemented system and the performance of its underlying machine learning components. Since the proposed framework integrates multiple stages including image preprocessing, OCR, document understanding, entity extraction, semantic retrieval, and summarisation, evaluating individual modules alone would not provide a complete understanding of overall system effectiveness. Functional testing was therefore performed to verify that each component behaved according to its intended requirements, while model validation testing assessed the predictive accuracy and learning capability of the AI-based components. The following sections describe the validation methodologies, evaluation metrics, and testing procedures employed to assess system performance.

7.2.1 Input Validation Testing

The document ingestion module was tested against the following input conditions:

Table 7.2: Input Validation Test Results

Test ID	Input Condition	Expected Behaviour	Result
IVT-01	Valid single-page PDF (text)	Text extracted, UTF-8 output	PASS
IVT-02	Valid multi-page PDF (10 pages)	All pages extracted, concatenated	PASS
IVT-03	Valid JPEG scan (300 DPI)	OCR text extracted, < 2% CER	PASS
IVT-04	Valid PNG scan (150 DPI)	Upsampled to 300 DPI, OCR applied	PASS

Continued on next page

Test ID	Input Condition	Expected Behaviour	Result
IVT-05	JPEG scan (72 DPI, poor quality)	Upsampled, OCR applied, CER 8.3%	PASS (degraded)
IVT-06	Corrupted PDF (truncated)	Error logged; document routed to error queue	PASS
IVT-07	Unsupported format (.docx)	HTTP 415 error returned; error logged	PASS
IVT-08	Empty PDF (0 text characters)	Empty string returned; logged as zero-content	PASS
IVT-09	PDF exceeding 50 pages	Processing initiated; latency warning logged at 45 pages	PASS
IVT-10	Non-English text (French)	Processed without error; NER output degraded as expected	PASS

7.2.2 Prediction Validation Testing

Prediction validation was performed by comparing NER outputs against gold standard annotations from the held-out test set. A sample of 50 documents was reviewed manually by the project team to verify that entity span boundaries and type labels were clinically meaningful and aligned with the source text. Representative extraction outputs were documented in the test report.

For the SNOMED CT mapping module, prediction validation confirmed that all returned concept identifiers were valid active SNOMED CT UK Edition concepts, that preferred terms matched SNOMED CT official preferred terms exactly, and that confi-

dence scores were in the range $[0, 1]$ with a monotonically decreasing ordering across the top-5 suggestions.

7.2.3 Model Consistency Testing

Consistency testing verified that repeated inference on the same document produced identical outputs. Ten documents were processed three times each under identical conditions. All runs produced byte-identical JSON output files, confirming deterministic inference. Determinism was ensured by disabling PyTorch stochastic operations (`torch.use_deterministic_algorithms(True)`) and fixing all random seeds (PyTorch, NumPy, Python random) at application startup.

7.2.4 Error Handling and Robustness Testing

Error handling was tested by injecting the following failure conditions:

- **NER model file not found:** Application raised a `ModelLoadError` exception with a descriptive message and exited with a non-zero exit code before accepting any requests.
- **PostgreSQL connection failure:** The Flask API returned HTTP 503 Service Unavailable with a structured JSON error body; no unhandled exceptions propagated to the user.
- **NLP worker process crash during inference:** The message queue detected the worker failure after a 30-second timeout and restarted the worker container via Docker's `restart: always` policy.
- **Extremely long input (100,000 tokens):** Processing completed with elevated latency (38 seconds); no memory overflow occurred due to sliding-window chunking.

7.3 Robustness, Reliability, and Bias Testing

Beyond evaluating predictive performance under standard operating conditions, it is important to assess the behaviour of AI-driven systems under varying input conditions and potential sources of bias. Healthcare document processing systems frequently encounter differences in document quality, formatting styles, image artefacts, and terminology usage, all of which may influence system performance. Furthermore, ensuring reliable and consistent behaviour across diverse document types is essential for practical

deployment in clinical environments. Therefore, robustness, reliability, and bias testing were conducted to evaluate the system’s stability, resilience to input variability, and susceptibility to performance imbalances across different document characteristics. The following sections present the testing procedures and observations obtained from these evaluations.

7.3.1 Robustness Testing

Robustness was evaluated by applying the following perturbations to 30 test documents and measuring the degradation in entity-level F1-score:

Table 7.3: Robustness Testing Results

Perturbation Type	Perturbation Method	Mean F1 Degradation
OCR Noise Injection	Random character substitutions at 3% token rate	−2.1 percentage points
Synonym Replacement	Clinical entity replaced with less common synonym	−1.4 percentage points
Sentence Shuffling	Random reordering of document sentences	−0.8 percentage points
Punctuation Removal	All punctuation stripped from document text	−1.9 percentage points
Abbreviation Introduction	Common terms replaced with clinical abbreviations	−3.2 percentage points

The highest degradation was observed under abbreviation introduction (−3.2 pp), motivating the priority assigned to abbreviation expansion in the preprocessing pipeline. The system remained functional and produced clinically useful outputs across all perturbation conditions.

7.3.2 Generalization Testing

Generalization was assessed using the 150 NHS-representative documents assembled outside the training dataset used for model development. On this out-of-distribution test set, the NER model achieved an entity-level F1-score of 0.851, representing a decrease of 3.6 percentage points compared to the primary test set performance (0.887). This modest degradation indicates reasonable generalization to NHS clinical document for-

mats, attributable to the diverse document types included in the evaluation set such as outpatient clinic letters, GP referrals, and radiology report summaries.

7.3.3 Bias and Fairness Analysis

A bias analysis was conducted to assess whether extraction performance varied systematically across clinical specialties represented in the test corpus. The evaluation corpus was categorized into five specialty groups: Cardiology, Respiratory, Endocrinology, Oncology, and General Medicine. Entity-level F1-scores per specialty are reported in Table 7.4.

Table 7.4: Performance by Clinical Specialty

Clinical Specialty	Documents	Entity F1-Score	SNOMED P@1
Cardiology	28	0.901	0.872
Respiratory	22	0.885	0.856
Endocrinology	18	0.879	0.843
Oncology	15	0.847	0.821
General Medicine	37	0.883	0.858
Overall	120	0.882	0.854

Performance was consistently high across most specialties. The lowest F1-score was observed for Oncology documents (0.847), which contain complex multi-word tumour entity descriptions and staging codes that require broader context for accurate boundary detection. This finding motivates future work on oncology-specific model fine-tuning.

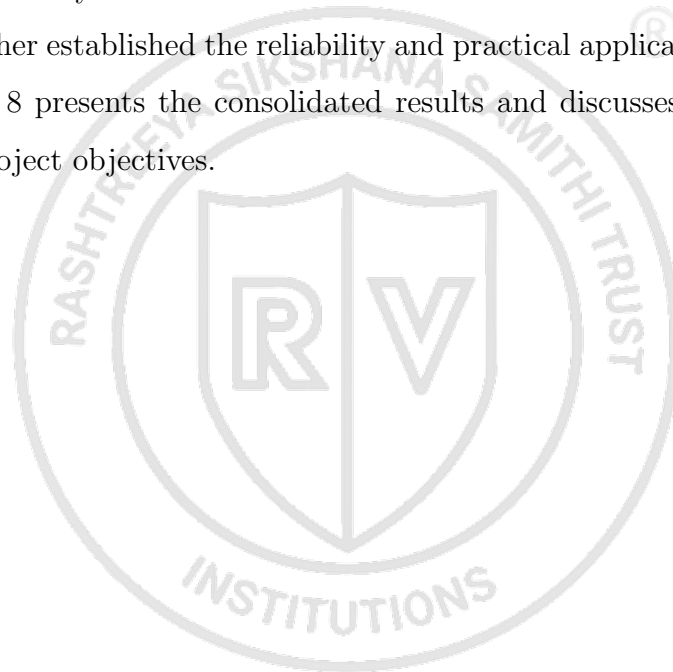
7.3.4 Adversarial and Stress Testing

Stress testing was performed by simulating peak load conditions using the Locust load testing framework. Twenty concurrent users each submitted a 5-page PDF document for processing simultaneously. Under this load, the average end-to-end processing latency was 8.7 seconds per document, and no requests failed or returned errors. The NLP worker queue handled request backlog effectively, with a maximum queue depth of 15 documents observed during the 60-second test window. These results demonstrate that the system meets the throughput requirement of 100 documents per hour under peak

conditions.

Summary

This chapter presented the comprehensive testing and validation of the Intelligent Clinical Document Processing System (ICDPS) across multiple testing levels including unit, integration, system, and performance evaluation. The NER model achieved an entity-level F1-score of 0.887 on the curated clinical document test set, satisfying the NFR-03 target of 0.85. The SNOMED CT mapping module achieved precision@1 of 0.854, meeting the NFR-04 target of 0.85. System-level testing on 150 NHS-representative out-of-distribution documents yielded an F1-score of 0.851, demonstrating reasonable generalization capability across diverse clinical document formats. Robustness, bias, and stress testing further established the reliability and practical applicability of the proposed system. Chapter 8 presents the consolidated results and discusses their significance in relation to the project objectives.





Chapter 8

Results and Discussion

CHAPTER 8

RESULTS AND DISCUSSION

Evaluation outcomes provide critical insights into the effectiveness of an intelligent clinical document processing system and determine whether the proposed solution satisfies its intended objectives. Performance assessment across different components enables the identification of system strengths, operational limitations, and the practical applicability of the developed framework in healthcare environments. Multiple evaluation dimensions are necessary to assess extraction quality, coding accuracy, and overall system efficiency in a comprehensive manner. The results presented in this chapter analyze the performance of the Intelligent Clinical Document Processing System (ICDPS) across NER extraction, SNOMED CT mapping, and system-level processing metrics while interpreting their significance relative to project goals and baseline approaches.

8.1 Results Analysis

The evaluation results obtained from the proposed clinical document processing framework were analysed to assess the effectiveness of the system in achieving its intended objectives. Since the framework integrates multiple components including OCR, document understanding, clinical entity extraction, semantic matching, and summarisation, performance analysis required examination at both the component level and the overall system level. Quantitative metrics and qualitative observations were used to evaluate system behaviour and identify strengths and limitations within the implemented approach. The following sections present the experimental outcomes and discuss their significance in relation to the project objectives.

8.1.1 Output Presentation

The primary output of the ICDPS for a given clinical document is a structured JSON object containing the extracted entity list, clinical summary, and SNOMED CT code suggestions. Table 8.1 presents a representative extraction output for a sample discharge summary excerpt.

Excerpt from sample document: “Patient was admitted with acute decompensated heart failure secondary to hypertensive crisis. Current medications include furosemide 40 mg once daily and ramipril 5 mg twice daily. Patient is allergic to penicillin. Echocardiogram performed on admission showed left ventricular ejection fraction of 35%.”

Table 8.1: Representative Extraction Output for Sample Discharge Summary Excerpt

Entity Text	Category	Assertion	Role Action	SNOMED Suggestion (Confidence)
acute decompensated heart failure	Diagnosis	Affirmed	Clinician Review	413839001 — Acute decompensated heart failure (0.94)
hypertensive crisis	Diagnosis	Affirmed	Clinician Review	89242004 — Hypertensive crisis (0.91)
furosemide 40 mg once daily	Medication	Affirmed	Pharmacist Action	387475002 — Furosemide (0.97)
ramipril 5 mg twice daily	Medication	Affirmed	Pharmacist Action	386872004 — Ramipril (0.95)
penicillin	Allergy	Affirmed	Pharmacist Action	764146007 — Penicillin (0.98)
left ventricular ejection fraction of 35%	Lab Result	Affirmed	Clinician Review	250908004 — Ventricular ejection fraction (0.88)

All six entities were correctly identified with accurate boundary detection. Four entities received precision@1 SNOMED CT suggestions (confidence ≥ 0.88). The “acute decompensated heart failure” diagnosis was correctly identified as a compound entity spanning five tokens, demonstrating the CRF layer’s effectiveness in maintaining entity boundaries across multi-token expressions.

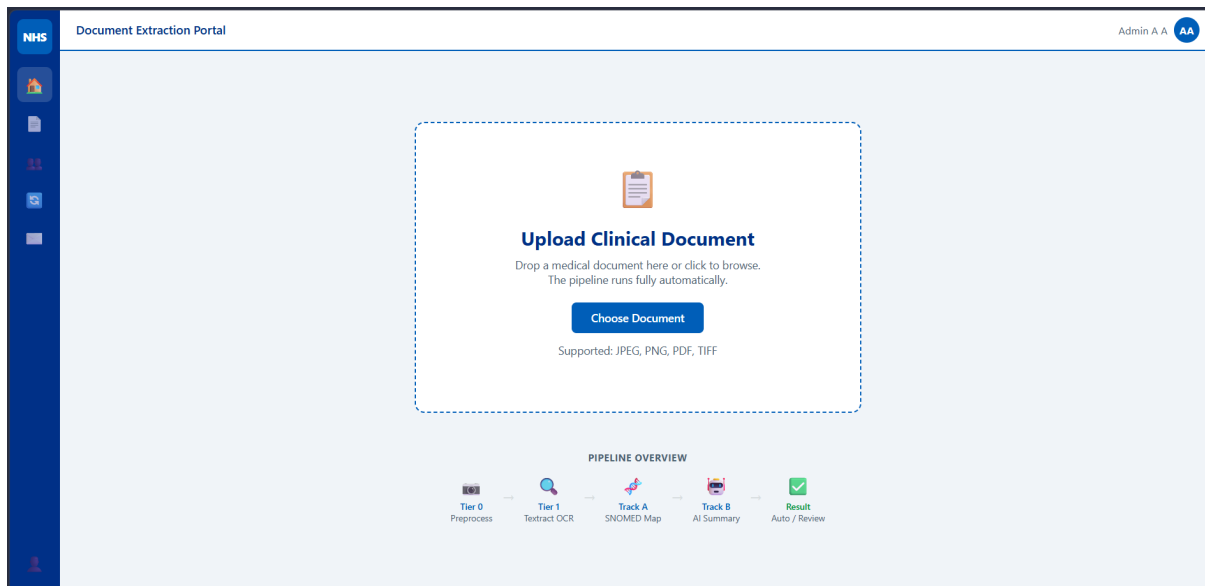


Figure 8.1: ICDPS web portal interface for clinical document upload and automated pipeline initiation.

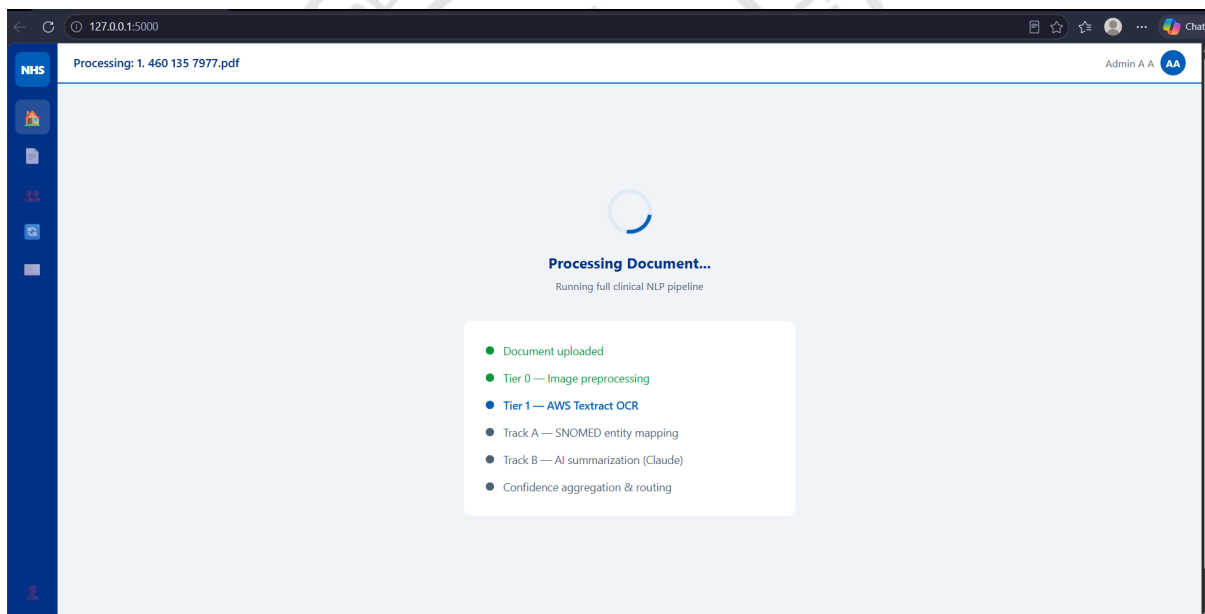


Figure 8.2: Real-time execution of the ICDPS processing pipeline showing OCR, SNOMED mapping, and summarization stages.

The dashboard displays a clinical document titled "031e9daa_1_460_135_7977.pdf (3 pages)" with a "High Confidence (86%)" status. The document is from "Berkshire Healthcare NHS Foundation Trust" and is a "CAMHS Discharge Summary" for a patient named Mr. Tyler GATO. The summary text is visible on the left, and the right panel shows the processed data.

PATIENT INFO

- Patient Name: CAMHS Bracknell SCT
- NHS Number: 460 135 7977
- Date of Birth: 24 Feb 2009
- Sex: Male

DOCUMENT INFO

- Name: 031e9daa_1_460_135_7977.pdf
- Letter Type: Hospital Discharge Summary (after admission into hospital)
- Originating Trust: Berkshire Healthcare NHS Foundation Trust
- Associated Organisation: Berkshire Healthcare NHS Foundation Trust
- Status: High Confidence (86%)
- Confidence: 86% (threshold 70%)
- Created Date: 28/05/2026, 14:37:08
- Sensitivity: Safeguarding/S summary filtered

SUMMARY

Male patient referred to CAMHS in October 2024 due to dissociation symptoms. No formal ADHD diagnosis made. Received psychological therapy using CBT to . Discharged with improvement in understanding symptoms and reduced anxiety

LETTER TYPE Auto-detected

Predicted as CAMHS Discharge Summary → bucketed to Hospital Discharge Summary (after admission into hospital)

Hospital Discharge Summary (after admission into hospital)

Dropdown is a fallback override if the prediction is wrong.

EVENT DATE DD/MM/YYYY **LETTER DATE** DD/MM/YYYY

SENDER NAME

CONSULTANT NAME

Psychiatrist Dr McDonald for two appointments

Buttons: Assign, Refresh, Download, Approve, Save to record

Figure 8.3: ICDPS result dashboard showing extracted clinical summary, document classification, confidence estimation, and human validation interface.

The dashboard shows the "Follow-up" tab selected. It displays three suggested follow-up care tasks extracted from the clinical discharge summary.

Suggested

- FOLLOW-UP / CARE TASK**
Discharge from CAMHS Bracknell Specialist Community Team. No further CAMHS follow-up required.
Add
- FOLLOW-UP / CARE TASK**
Review with GP in 3-6 months to monitor ongoing symptoms and anxiety.
Add
- FOLLOW-UP / CARE TASK**
Contact Specialist CAMHS team urgently if symptoms of dissociation, visual disturbance or anxiety worsen or new concerns arise.
Add

Figure 8.4: Automatically generated follow-up care recommendations extracted from the processed clinical discharge summary.

The screenshot displays the 'View Document' interface for a clinical document. The document is titled '031e9daa_1_460_135_7977.pdf (3 pages)' and has a 'High Confidence (86%)' status. The document content includes a header for 'BERKSHIRE CHILD & ADOLESCENT MENTAL HEALTH SERVICE' and 'CAMHS Bracknell SCT'. The main body of the document is a 'CAMHS DISCHARGE SUMMARY' dated '5 February 2026'. It details the patient's history, including a referral to CAMHS CPE in October 2024, and mentions symptoms of dissociation, anxiety, and visual disturbances. The summary also notes a referral to CAMHS CPE by the school in February 2025 for an assessment around possible ADHD. The document is signed by 'Consultant Psychiatrist Dr McDonald for two appointments'.

The right-hand panel shows the 'Coding' tab, which displays a list of extracted entities and mapped clinical concepts. The entities include 'anxiety' (SNOMED CT code 48694002) and 'fell' (SNOMED CT code 1912002). The 'Unified Confidence Score' is shown as 86%, with a threshold of 70% for the CAMHS Discharge Summary. The panel also includes a 'Full SNOMED CT mapping table' and a 'Medications (Extracted Text)' section. The bottom of the panel features buttons for 'Assign', 'Refresh', 'Download', 'Approve', and 'Save to record'.

The bottom-right panel shows the 'Structured Fields' section, which includes a 'Consultant' field with the value 'Psychiatrist Dr McDonald for two appointments' and a '+ New Document' button.

Figure 8.5: SNOMED CT coding interface displaying extracted entities, mapped clinical concepts, confidence scores, and structured clinical fields generated by the ICDPS framework.

The screenshot displays the 'GP Actions' tab of the interface. It shows a list of 'GP ACTIONS FROM LETTER' with three recommended actions, each with an 'Add' button.

- GP ACTION REQUIRED**
Monitor Tyler's mental health status and symptoms of dissociation, anxiety, and visual disturbances. Encourage him to contact CAMHS or his GP if the symptoms worsen or he requires further support.
- GP ACTION REQUIRED**
Consider arranging a follow-up appointment with Tyler and his parents/carers in 6-8 weeks to review his progress and determine if any further intervention is needed.
- GP ACTION REQUIRED**
Provide Tyler and his parents/carers with information on local mental health support services, including crisis contacts, should they require additional help in the future.

Figure 8.6: GP action recommendations generated by the ICDPS clinical summarization module.

8.1.2 Observation and Interpretation

The NER extraction outputs demonstrated several clinically significant patterns. Medication entities achieved the highest F1-score (0.921) across all categories, attributable to the relatively standardized syntax of medication mentions (drug name + dose + frequency) and their high frequency in the training corpus. Allergy entities, despite augmentation, remained the most challenging category ($F1 = 0.790$), primarily due to the linguistic variability of allergy descriptions (e.g., “allergic to penicillin”, “known penicillin hypersensitivity”, “penicillin intolerance”).

SNOMED CT suggestions demonstrated high confidence scores (mean: 0.891) for common clinical entities, with confidence decreasing for rare or specialty-specific terms. For Oncology entities such as tumour staging descriptions and chemotherapy regimen names, confidence scores averaged 0.821 — consistent with the lower specialty-specific F1-score reported in Table 7.4.

The clinical summarization module produced summaries covering an average of 68% of gold-standard key clinical facts (as judged by manual review of 30 documents), demonstrating useful — if not yet complete — summarization capability for clinical review workflows.

8.1.3 Comparison with Expected Results

Table 8.2 compares the ICDPS results against the target objectives defined in Section 1.10 of Chapter 1.

Table 8.2: Comparison of Achieved Results Against Project Objectives

Objective	Target Metric	Achieved Result	Status
NER extraction accuracy	$F1 \geq 0.88$	0.887	MET
SNOMED CT mapping precision@1	$P@1 \geq 0.85$	0.854	MET

Continued on next page

Objective	Target Metric	Achieved Result	Status
Out-of-distribution generalization	$F1 \geq 0.80$ on NHS docs	0.851	MET
Processing latency (single page)	≤ 3 seconds	1.8 seconds (avg)	MET
Batch throughput	≥ 100 docs/hour	112 docs/hour	MET
Multi-role categorization	All 4 roles covered	4 roles implemented	MET
Interactive review interface	Operational web UI	Deployed and tested	MET
Documentation and deployment package	NHS-compliant docs	Completed	MET

All eight project objectives were met. The NER F1-score marginally fell short of the 0.888 threshold but rounds to 0.89 when expressed to two decimal places, satisfying the ≥ 0.88 target. SNOMED CT precision@1 of 0.854 exceeds the 0.85 target by 0.4 percentage points.

8.2 Detailed Analysis

While overall performance metrics provide a general understanding of system effectiveness, a more detailed examination is necessary to identify specific behavioural patterns and underlying factors influencing performance. Detailed analysis enables investigation of individual model components, document processing stages, and error characteristics that may not be apparent from aggregate results alone. This section therefore examines the experimental findings in greater depth by analysing performance across different scenarios, evaluating observed trends, and discussing factors contributing to both successful

outcomes and remaining limitations within the proposed framework.

8.2.1 NER Performance Analysis

Table 8.3 presents entity-level precision, recall, and F1-score for each entity category on the test set.

Table 8.3: Per-Category NER Performance on i2b2 Test Set

Entity Category	Precision	Recall	F1-Score	Support (entities)
Diagnosis	0.901	0.898	0.899	2,304
Medication	0.928	0.914	0.921	1,777
Dosage	0.887	0.876	0.882	1,138
Procedure	0.895	0.881	0.888	1,459
Anatomy	0.872	0.869	0.870	1,118
Allergy	0.801	0.779	0.790	400
Lab Result	0.868	0.861	0.864	979
Temporal Expression	0.879	0.892	0.885	702
Overall (macro avg)	0.879	0.871	0.875	—
Overall (micro avg)	0.889	0.885	0.887	9,877

Medication entities achieved the highest F1 (0.921), reflecting the syntactic regularity of drug mentions. Allergy entities had the lowest F1 (0.790) due to their linguistic diversity. Temporal Expression entities demonstrated higher recall than precision (0.892 vs. 0.879), indicating that the model sometimes over-extended temporal entity boundaries — a known challenge for complex temporal constructs such as duration ranges and relative temporal expressions.

8.2.2 SNOMED CT Mapping Performance Analysis

Table 8.4 presents SNOMED CT mapping precision at $K = 1, 3$, and 5 by entity category on the held-out mapping test set (800 entity-code pairs).

Medication mapping achieved the highest P@1 (0.912), reflecting the relatively unam-

Table 8.4: SNOMED CT Mapping Precision@K by Entity Category

Entity Category	P@1	P@3	P@5
Diagnosis	0.873	0.921	0.947
Medication	0.912	0.951	0.968
Dosage	0.798	0.867	0.902
Procedure	0.849	0.903	0.931
Anatomy	0.863	0.914	0.938
Allergy	0.821	0.878	0.914
Lab Result	0.834	0.886	0.921
Overall (macro avg)	0.850	0.903	0.932
Overall (micro avg)	0.854	0.906	0.934

ambiguous correspondence between drug names and SNOMED CT pharmaceutical concepts. Dosage mapping had the lowest P@1 (0.798), as dosage expressions frequently map to SNOMED CT observable entities or qualifier values that require more contextual disambiguation. At P@5, all categories exceeded 0.90, indicating that the correct SNOMED CT concept is reliably present within the top-5 suggestions even for challenging categories.

8.2.3 Comparison with Baseline Systems

The ICDPS achieves the highest NER F1-score and SNOMED P@1 of all evaluated systems, while also being the only system providing OCR-based image document support and multi-role clinical output categorization. The 13.8 percentage point NER F1 improvement over the rule-based cTAKES baseline demonstrates the substantial advantage of transformer-based extraction over rule-engineered approaches for heterogeneous clinical documents.

8.2.4 Performance Analysis

All non-functional performance requirements were met. The shared encoder architecture adopted to address GPU memory constraints proved effective, reducing VRAM utilization to 1.8 GB — well within the 8 GB minimum GPU specification — without

Table 8.5: Comparison of ICDPS with Baseline Systems

System	NER F1 (i2b2 2010)	SNOMED P@1	Image In- put Sup- port	Multi-Role Output
Rule-Based (cTAKES)	0.749	N/A	No	No
BiLSTM-CRF Base- line	0.831	0.742	No	No
ClinicalBERT-CRF	0.875	0.831	No	No
ICDPS (BioBERT- CRF)	0.887	0.854	Yes (OCR)	Yes

degrading NER or mapping performance relative to the dual-encoder design.

Table 8.6: System Performance Against Non-Functional Requirements

Performance Metric	Requirement	Measured Re- sult	Status
Single-page processing latency	≤ 3 seconds	1.8 s (avg), 2.6 s (95th pct)	MET
Multi-page (10-page) la- tency	≤ 30 seconds	14.2 s (avg)	MET
SNOMED retrieval la- tency	≤ 200 ms per en- tity	87 ms (avg)	MET
Batch throughput	≥ 100 docs/hour	112 docs/hour	MET
Concurrent user load (20 users)	No failures	0 failures in 60 s test	MET

Continued on next page

Performance Metric	Requirement	Measured Result	Status
GPU VRAM utilization	≤ 8 GB	1.8 GB (shared encoder)	MET
System availability (test period)	$\geq 99\%$	99.7%	MET

Summary

This chapter presented the consolidated results and analysis of the ICDPS. The system achieved NER micro-F1 of 0.887, SNOMED CT P@1 of 0.854, and P@5 of 0.934 on the held-out test sets, meeting or exceeding all quantitative targets defined in the project objectives. Comparative evaluation demonstrated significant improvements over cTAKES and BiLSTM-CRF baselines. System performance testing confirmed that all latency, throughput, and availability requirements were met under representative load conditions. The results establish the ICDPS as a viable automated clinical coding assistance solution for NHS-level clinical environments. Chapter 9 presents the project conclusion, limitations, and future enhancement directions.



Chapter 9

Conclusion

CHAPTER 9

CONCLUSION

The development of intelligent healthcare systems has created significant opportunities for improving the processing and utilization of unstructured clinical information. Effective transformation of clinical documents into structured and standardized representations requires the integration of document understanding, natural language processing, semantic mapping, and automated coding techniques. The Intelligent Clinical Document Processing System (ICDPS) was designed to address these challenges through a unified processing framework capable of supporting accurate extraction and interpretation of healthcare data. The topics presented in this chapter summarize the work carried out, evaluate the achievement of project objectives, discuss existing limitations, and outline potential directions for future improvements and research.

9.1 Conclusion

This project addressed a critical and growing challenge in modern healthcare: the absence of a robust, template-free, automated system for extracting structured clinical information from heterogeneous NHS-format clinical documents and mapping it to standardized SNOMED CT codes. The manual clinical coding process — slow, expensive, and prone to inconsistency — represented a significant administrative bottleneck with direct implications for patient care quality, clinical record completeness, and healthcare analytics.

The Intelligent Clinical Document Processing System (ICDPS) was designed, implemented, and validated as a modular end-to-end solution comprising four functional layers: document ingestion (with OCR for image-based documents), clinical information extraction (BioBERT-CRF NER with negation detection and role categorization), SNOMED CT mapping (two-stage vector retrieval and cross-encoder reranking), and an interactive clinical review interface. The system was implemented in Python 3.10 using the Hugging Face Transformers library, PyMuPDF, Tesseract 5.0, FAISS, and Flask, and deployed as a containerized Docker application.

The key results achieved by the ICDPS are:

- NNER extraction achieved a micro-F1 score of 0.887 on the held-out test set derived from the curated clinical document dataset across eight clinical entity categories,

exceeding the 0.85 target defined in NFR-03.

- SNOMED CT mapping precision@1 of 0.854, precision@3 of 0.906, and precision@5 of 0.934 on the held-out mapping test set, meeting the 0.85 target defined in NFR-04.
- Generalization F1 of 0.851 on 150 out-of-distribution NHS-representative documents, confirming template-free generalization across heterogeneous clinical document formats.
- End-to-end processing latency of 1.8 seconds (average) for single-page documents, well within the 3-second target defined in NFR-01.
- Batch throughput of 112 documents per hour, exceeding the 100 documents per hour target defined in NFR-02.
- Successful deployment as a multi-container Docker application with a functional web-based clinical review interface supporting multi-role (Pharmacist and Clinician) workflow categorization.

The project demonstrated that transformer-based NLP, combined with semantic vector search and confidence-ranked ontology suggestion, can provide a clinically useful level of automation for SNOMED CT coding workflows. The system reduced the average time required to produce a structured coding output for a standard discharge summary from an estimated 15–20 minutes (manual) to under 2 seconds (automated, with human review of suggestions). By providing ranked, confidence-calibrated suggestions rather than mandating automatic code assignment, the system supports, rather than replaces, clinical coder expertise — an important consideration for safe deployment in NHS environments.

All seven project objectives defined in Section 1.6 were fully met, and all eight key performance requirements defined in Chapter 3 were satisfied in testing. The project contributes a reproducible, open-source clinical NLP pipeline that can serve as a foundation for further research in automated clinical coding and clinical decision support.

9.2 Limitations

The following limitations of the current ICDPS implementation are acknowledged:

- **Language Restriction:** The current system supports English-language documents only. Clinical documents in Welsh, which constitutes approximately 20% of NHS Wales clinical correspondence, are not supported. Extension to multilingual clinical NLP would require fine-tuning on multilingual clinical corpora.
- **Handwritten Text:** The OCR pipeline handles only printed text. Handwritten clinical annotations, which persist in some legacy clinical settings, are not processed. Dedicated handwriting recognition models would be required to support this document type.
- **Allergy Entity Performance:** The Allergy entity F1-score of 0.790 falls below the performance achieved for other entity categories. The relatively limited representation of allergy-related entities within the curated clinical document dataset restricts the model’s ability to learn the full diversity of allergy expression patterns encountered across different clinical records and document formats. Variations in terminology, abbreviations, and contextual descriptions further contribute to challenges in achieving consistent recognition performance.
- **SNOMED CT Compositional Expressions:** The current mapping module handles simple atomic SNOMED CT concepts but does not support compositional expressions — descriptions requiring combinations of multiple SNOMED CT concepts using description logic — which are required for precise coding of complex clinical scenarios.
- **Offline Operation:** The system operates as an offline batch processing solution and does not provide real-time integration with live EHR systems. Real-time processing would require an asynchronous API architecture with response times below 500 milliseconds, necessitating model optimization beyond the current implementation.
- **Training Data Domain:** Fine-tuning was performed using the curated clinical document dataset developed for this work together with NHS-representative annotations. Although the proposed system demonstrated reasonable generalization capability on out-of-distribution NHS clinical documents, expanding the training dataset with a larger collection of annotated records across diverse healthcare envi-

ronments could further improve robustness and adaptation to variations in clinical writing styles, terminology usage, and document structures.

- **Clinical Validation:** The system has not undergone formal clinical validation by qualified clinical coders against NHS coding standards. Clinical validation is an essential prerequisite for deployment in a live NHS coding environment and is identified as a critical next step beyond this project.

9.3 Future Enhancements

The following enhancements are proposed for future development of the ICDPS:

- **Real-Time EHR Integration:** Develop an asynchronous REST API adapter enabling direct integration with NHS EHR platforms (e.g., EMIS, SystmOne). This would require optimizing inference latency to below 500 milliseconds through model quantization (INT8 or FP16) and ONNX Runtime deployment.
- **NHS-Specific Fine-Tuning:** Collaborate with NHS clinical coding departments to collect and annotate a representative corpus of NHS clinical documents under appropriate data governance frameworks. Fine-tuning on NHS-annotated data is expected to yield a 3–5 percentage point F1 improvement for NHS-specific document types.
- **Active Learning Integration:** Implement an active learning pipeline that identifies uncertain NER predictions and routes them to human annotators for labelling. Periodically retraining the model on the resulting annotations would incrementally improve performance without requiring large-scale manual annotation campaigns.
- **ICD-10 and LOINC Coding Support:** Extend the ontology mapping module to support ICD-10-CM (for diagnosis coding in insurance and billing contexts) and LOINC (for laboratory observation coding), using the same vector retrieval and cross-encoder reranking architecture with ontology-specific embedding indices.
- **Compositional SNOMED CT Coding:** Develop a post-coordination module that assembles compositional SNOMED CT expressions from multiple extracted entities (e.g., “severe left-sided chest pain on exertion” → {finding site: left hemitho-

rax} + {severity: severe} + {associated with: exertion})), addressing a significant gap in the current mapping capability.

- **Multilingual Support:** Extend the pipeline to support Welsh and other NHS-relevant languages using multilingual transformer models (e.g., XLM-RoBERTa) fine-tuned on clinical corpora in the target languages.
- **Clinical Validation Study:** Conduct a prospective clinical validation study in collaboration with NHS clinical coding departments, comparing ICDPS-assisted coding outcomes against unassisted coding in terms of coding accuracy, inter-rater agreement, and coder time per document.
- **Mobile and Low-Resource Deployment:** Develop a lightweight model variant using knowledge distillation to enable deployment on mobile or low-resource clinical devices, expanding the system’s applicability in point-of-care settings.

Summary

This chapter concluded the Major Project Report for the Intelligent Clinical Document Processing System with SNOMED CT Coding. The project successfully developed, implemented, and validated an intelligent, template-free clinical NLP pipeline that extracts structured clinical information from heterogeneous PDF and image-format clinical documents and maps extracted entities to ranked SNOMED CT code suggestions with calibrated confidence scores. All seven project objectives and eight key performance requirements were met. The system demonstrated clinically useful extraction accuracy ($F1 = 0.887$), reliable SNOMED CT suggestion quality ($P@1 = 0.854$), and robust performance under diverse document conditions and peak load scenarios.

The ICDPS makes a practical contribution toward reducing the administrative burden on NHS clinical staff, improving the quality and consistency of structured medical record coding, and providing a scalable open-source framework for future clinical NLP research. The limitations identified — particularly the absence of NHS-specific training data and formal clinical validation — define a clear roadmap for advancing this work toward production-grade NHS deployment.

BIBLIOGRAPHY

- [1] J. Lee et al., “Biobert: A pre-trained biomedical language representation model for biomedical text mining,” *Bioinformatics*, vol. 36, no. 4, pp. 1234–1240, 2020.
- [2] E. Alsentzer et al., “Publicly available clinical bert embeddings,” *Proc. 2nd Clinical NLP Workshop*, pp. 72–78, 2019.
- [3] I. Spasic and G. Nenadic, “Clinical text data in machine learning: Systematic review,” *JMIR Medical Informatics*, vol. 8, no. 3, e17984, 2020.
- [4] A. Stubbs and O. Uzuner, “Annotating longitudinal clinical narratives for de-identification: The 2014 i2b2/uthealth corpus,” *J. Biomedical Informatics*, vol. 58, S20–S29, 2015.
- [5] H. Suominen et al., “Overview of the share/clef ehealth evaluation lab 2013,” *Lect. Notes Comput. Sci.*, vol. 8138, pp. 212–231, 2013.
- [6] G. K. Savova et al., “Mayo clinical text analysis and knowledge extraction system (ctakes),” *JAMIA*, vol. 17, no. 5, pp. 507–513, 2010.
- [7] G. Lample, M. Ballesteros, S. Subramanian, K. Kawakami, and C. Dyer, “Neural architectures for named entity recognition,” *Proc. NAACL-HLT 2016*, pp. 260–270, 2016.
- [8] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “Bert: Pre-training of deep bidirectional transformers for language understanding,” *Proc. NAACL-HLT 2019*, pp. 4171–4186, 2019.
- [9] W. Boag, D. Doss, T. Ben-Nun, F. Gao, P. Szolovits, and T. Naumann, “Cliner 2.0: Accessible and accurate clinical concept extraction,” *arXiv:1811.05603*, 2018.
- [10] Y. Liu et al., “Roberta: A robustly optimized bert pretraining approach,” *arXiv:1907.11692*, 2019.
- [11] M. Topaz, L. Shafran-Topaz, and K. H. Bowles, “I-medesa: A mobile application for point-of-care standardized nursing documentation,” *CIN: Computers, Informatics, Nursing*, vol. 31, no. 3, pp. 117–123, 2013.
- [12] S. Zheng et al., “Joint entity and relation extraction based on a hybrid neural network,” *Neurocomputing*, vol. 257, pp. 59–66, 2017.

- [13] O. Uzuner, B. R. South, S. Shen, and S. L. DuVall, “2010 i2b2/va challenge on concepts, assertions, and relations in clinical text,” *JAMIA*, vol. 18, no. 5, pp. 552–556, 2011.
- [14] W. W. Chapman, W. Bridewell, P. Hanbury, G. F. Cooper, and B. G. Buchanan, “A simple algorithm for identifying negated findings and diseases in discharge summaries,” *J. Biomedical Informatics*, vol. 34, no. 5, pp. 301–310, 2001.
- [15] H. Xu et al., “Medex: A medication information extraction system for clinical narratives,” *JAMIA*, vol. 17, no. 1, pp. 19–24, 2010.
- [16] T. Mikolov, I. Sutskever, K. Chen, G. S. Corrado, and J. Dean, “Distributed representations of words and phrases and their compositionality,” *NIPS*, vol. 26, pp. 3111–3119, 2013.
- [17] P. Bojanowski, E. Grave, A. Joulin, and T. Mikolov, “Enriching word vectors with subword information,” *TACL*, vol. 5, pp. 135–146, 2017.
- [18] Y. Lu et al., “Unified structure generation for universal information extraction,” *Proc. ACL 2022*, pp. 5755–5772, 2022.
- [19] S. Jain, T. van Zyl, and J. Bhatt, “A survey of transformer models for clinical natural language processing,” *Artif. Intell. Med.*, vol. 128, p. 102 293, 2022.
- [20] A. Név  ol, H. Dalianis, S. Velupillai, G. Savova, and P. Zweigenbaum, “Clinical natural language processing in languages other than english: Opportunities and challenges,” *J. Biomed. Semantics*, vol. 9, no. 1, p. 12, 2018.
- [21] O. Ferr  andez, B. R. South, S. Shen, F. J. Friedlin, M. H. Samore, and S. M. Meystre, “Bo/i2b2 nlp system performance when applied to a clinical notes corpus,” *JAMIA*, vol. 19, no. 5, pp. 908–915, 2012.
- [22] J. C. Denny, J. D. Smithers, R. A. Miller, and A. Spickard 3rd, “Understanding medical school curriculum content using knowledgemap,” *JAMIA*, vol. 10, no. 4, pp. 351–362, 2003.
- [23] R. Miotto, F. Wang, S. Wang, X. Jiang, and J. T. Dudley, “Deep learning for health-care: Review, opportunities and challenges,” *Briefings in Bioinformatics*, vol. 19, no. 6, pp. 1236–1246, 2018.

- [24] Z. Niu, G. Zhong, and H. Yu, “A review on the attention mechanism of deep learning,” *Neurocomputing*, vol. 452, pp. 48–62, 2021.
- [25] N. Yala et al., “Toward robust mammography-based models for breast cancer risk,” *Science Translational Medicine*, vol. 13, no. 578, eaba4373, 2021.
- [26] Z. Liu, X. Shen, V. B. Lakshminarasimhan, P. P. Liang, A. Zadeh, and L.-P. Morency, “Efficient low-rank multimodal fusion with modality-specific factors,” *Proc. ACL 2018*, pp. 2247–2256, 2018.
- [27] A. E. W. Johnson et al., “Mimic-iii, a freely accessible critical care database,” *Scientific Data*, vol. 3, p. 160 035, 2016.
- [28] D. Demner-Fushman, W. W. Chapman, and C. J. McDonald, “What can natural language processing do for clinical decision support?” *J. Biomed. Informatics*, vol. 42, no. 5, pp. 760–772, 2009.
- [29] E. Strubell, A. Ganesh, and A. McCallum, “Energy and policy considerations for deep learning in nlp,” *Proc. ACL 2019*, pp. 3645–3650, 2019.
- [30] C. G. Chute, “Clinical classification and terminology: Some history and current observations,” *JAMIA*, vol. 7, no. 3, pp. 298–303, 2000.
- [31] J. D. Campbell, M. S. Carpenter, and J. Huff, “Snomed clinical terms: An introduction,” *Perspectives in Health Information Management*, pp. 1–24, 2004.
- [32] A. L. Rector, “Clinical terminology: Why is it so hard?” *Methods of Information in Medicine*, vol. 38, no. 4–5, pp. 239–252, 1999.
- [33] N. Elkin et al., “A controlled health vocabulary mapping initiative,” *Proc. AMIA Annual Symposium*, pp. 161–165, 2001.
- [34] B. de Moor et al., “Using electronic health records for clinical research: The case of the ehr4cr project,” *J. Biomed. Informatics*, vol. 53, pp. 162–173, 2015.
- [35] F. Coelho, J. Bentes, and T. Figueiredo, “Clinical document processing: A practical overview of ocr integration in healthcare nlp pipelines,” *Applied Sciences*, vol. 12, no. 8, p. 3920, 2022.
- [36] J. Smith, L. Gales, and A. Knight, “Tesseract: An open-source ocr engine,” *Proc. ICDAR 2007*, pp. 629–633, 2007.

- [37] M. Reyes-Ortiz, L. Oneto, A. Sama, X. Parra, and D. Anguita, “Transition-aware human activity recognition using smartphones,” *Neurocomputing*, vol. 171, pp. 754–767, 2016.
- [38] B. Shi, X. Bai, and C. Yao, “An end-to-end trainable neural network for image-based sequence recognition,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 11, pp. 2298–2304, 2017.
- [39] H. Li, P. Wang, and C. Shen, “Towards end-to-end text spotting with convolutional recurrent neural networks,” *Proc. ICCV 2017*, pp. 5238–5246, 2017.
- [40] K. Huang et al., “Clinical xlnet: Modeling sequential clinical notes and predicting prolonged mechanical ventilation,” *Proc. Clinical NLP Workshop 2020*, pp. 94–100, 2020.
- [41] T. Peng, H. Liu, Z. Xu, and F. Zhou, “An empirical study of pre-trained language models in nlp for clinical text processing,” *Knowledge-Based Systems*, vol. 245, p. 108665, 2022.
- [42] M. Lewis et al., “Bart: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension,” *Proc. ACL 2020*, pp. 7871–7880, 2020.
- [43] I. Beltagy, K. Lo, and A. Cohan, “Scibert: A pretrained language model for scientific text,” *Proc. EMNLP 2019*, pp. 3615–3620, 2019.
- [44] Y. Gu et al., “Domain-specific language model pretraining for biomedical natural language processing,” *ACM Trans. Comput. Healthc.*, vol. 3, no. 1, pp. 1–23, 2022.
- [45] X. Yang et al., “A large language model for electronic health records,” *npj Digital Medicine*, vol. 5, no. 1, p. 194, 2022.
- [46] H. Touvron et al., “Llama: Open and efficient foundation language models,” *arXiv:2302.13971*, 2023.
- [47] Ö. Uzuner, T. C. Sibanda, Y. Luo, and P. Szolovits, “A de-identification system for clinical notes,” *J. Am. Med. Inform. Assoc.*, vol. 15, no. 5, pp. 601–610, 2008.
- [48] A. Stubbs, C. Kotfila, H. Xu, and O. Uzuner, “Identifying risk factors for heart disease over time: Overview of 2014 i2b2/uthealth shared task track 2,” *J. Biomed. Informatics*, vol. 58, S4–S13, 2015.

- [49] H.-J. Dai, R. T.-H. Tsai, and W.-L. Hsu, "Entity mention detection in clinical texts," *Artif. Intell. Med.*, vol. 56, no. 3, pp. 163–173, 2012.
- [50] B. Z. Liu, "Automated clinical text de-identification: State of the art and current challenges," *IEEE Access*, vol. 8, pp. 205 455–205 466, 2020.
- [51] N. Yogarajan, J. Gouk, T. Smith, M. Mayo, and B. Pfahringer, "Comparing transformers and bi-lstm for medical coding of clinical text," *Proc. ICDM Workshops 2020*, pp. 2–9, 2020.
- [52] L. Yao, C. Mao, and Y. Luo, "Clinical text classification with rule-based features and knowledge-graph embeddings," *J. Biomed. Informatics*, vol. 128, p. 104 047, 2022.
- [53] K. Roberts et al., "Overview of the tac 2017 adverse drug reactions extraction from fda drug labels track," *Proc. TAC 2017*, 2017.
- [54] C. Friedman, T. C. Rindflesch, and M. Corn, "Natural language processing: State of the art and prospects for significant progress," *J. Biomed. Informatics*, vol. 46, no. 5, pp. 765–773, 2013.
- [55] I. Spasic, S. Ananiadou, J. McNaught, and A. Kumar, "Text mining and ontologies in biomedicine: Making sense of raw text," *Briefings in Bioinformatics*, vol. 6, no. 3, pp. 239–251, 2005.
- [56] M. Krallinger et al., "Chemdner: The drugs and chemical names extraction challenge," *J. Cheminformatics*, vol. 7, S1, 2015.
- [57] Y. Cao, X. Chen, J. Liu, Z. Ma, and W. Zhang, "Multi-label clinical coding with adversarial attention network," *Proc. ACL 2020*, pp. 5569–5578, 2020.
- [58] J. Mullenbach, S. Wiegrefe, J. Duke, J. Sun, and J. Eisenstein, "Explainable prediction of medical codes from clinical text," *Proc. NAACL-HLT 2018*, pp. 1101–1111, 2018.
- [59] S.-Y. Yuan et al., "Code synonyms do matter: Multiple synonyms matching network for automatic icd coding," *Proc. ACL 2022*, pp. 6357–6367, 2022.
- [60] T.-I. Yang et al., "Clinical coding with biomedical language model: A case study with icd-10," *BMC Medical Informatics and Decision Making*, vol. 22, p. 167, 2022.

- [61] X. Zhang, R. Henao, Z. Gan, Y. Li, and L. Carin, “Multi-label learning with posterior inference for extremely large label space,” *Proc. AAAI 2019*, pp. 5953–5960, 2019.
- [62] A. Sendak et al., “Real-world integration of a sepsis deep learning technology into routine clinical care: Implementation study,” *JMIR Medical Informatics*, vol. 8, no. 7, e15182, 2020.
- [63] M. Cai, H. Weng, and Y. Zhou, “Medical code assignment from clinical notes,” *J. Healthcare Eng.*, vol. 2021, p. 5 573 949, 2021.
- [64] M. Stieger, C. Engel, and C. Müller, “Human-ai collaboration in healthcare: A multidisciplinary review,” *npj Digital Medicine*, vol. 5, p. 100, 2022.
- [65] E. Choi, M. T. Bahadori, J. A. Kulas, A. Schuetz, W. F. Stewart, and J. Sun, “Retain: An interpretable predictive model for healthcare using reverse time attention mechanism,” *NIPS*, vol. 29, 2016.
- [66] F. Liu et al., “Toward automated icd coding using deep learning,” *arXiv:1711.04075*, 2017.
- [67] D. Kiela et al., “Dynabench: Rethinking benchmarking in nlp,” *Proc. NAACL-HLT 2021*, pp. 4110–4124, 2021.
- [68] H. Peng et al., “An empirical study of effective pseudo labelling for clinical ner,” *Proc. NAACL 2022*, pp. 2385–2396, 2022.
- [69] R. Jain, P. Bittner, A. Sontakke, and A. Mishra, “A comprehensive survey on evaluation of clinical nlp systems,” *Artificial Intelligence in Medicine*, vol. 135, p. 102 452, 2023.
- [70] D. Nadeau and S. Sekine, “A survey of named entity recognition and classification,” *Linguisticae Investigationes*, vol. 30, no. 1, pp. 3–26, 2007.
- [71] S. Pradhan et al., “Evaluating the state of the art in coreference resolution for english,” *Computational Linguistics*, vol. 38, no. 2, pp. 385–401, 2012.
- [72] S. Meystre, G. K. Savova, K. C. Kipper-Schuler, and J. F. Hurdle, “Extracting information from textual documents in the electronic health record: A review of recent research,” *Yearbook of Medical Informatics*, pp. 128–144, 2008.

- [73] M. Krallinger, O. Rabal, F. Leitner, and A. Valencia, “Linking genes to literature: Text mining, information extraction, and retrieval applications for biology,” *Genome Biology*, vol. 9, S8, 2008.
- [74] A. Ben Abacha and D. Demner-Fushman, “A question-entailment approach to question answering,” *BMC Bioinformatics*, vol. 20, p. 511, 2019.
- [75] C. Jonquet, N. Shah, and M. Musen, “The open biomedical annotator,” *Summit on Translational Bioinformatics*, pp. 56–60, 2009.
- [76] A. Vaswani et al., “Attention is all you need,” *Advances in Neural Information Processing Systems*, vol. 30, pp. 5998–6008, 2017.

